
Clinical Study Protocol

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A Phase III, Open-label, Randomised, Multicentre Study of Ceralasertib Plus Durvalumab Versus Docetaxel in Patients With Advanced or Metastatic Non-Small Cell Lung Cancer Without Actionable Genomic Alterations, and Whose Disease Has Progressed On or After Prior Anti-PD-(L)1 Therapy and Platinum-based Chemotherapy: LATIFY

Sponsor Name: AstraZeneca AB

Legal Registered Address: 151 85 Södertälje, Sweden

Local Sponsor in Japan: AstraZeneca K.K, 3-1, Ofuka-cho, Kita-ku, Osaka 530 0011, Japan

Local Sponsor in China: AstraZeneca Global R&D (China) Co., Ltd. 16F, 88 Xizang Road North, Jing'an District, Shanghai, China

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Acronym: LATIFY

Study clinical lead: Name and contact information will be provided separately.

International Co-ordinating Investigator:

Benjamin Besse, MD

Gustave Roussy, 114 Rue Edouard Vaillant, 94805 Villejuif, France

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1 PROTOCOL SUMMARY

1.1 Synopsis

Protocol Title:

A Phase III, Open-label, Randomised, Multicentre Study of Ceralasertib Plus Durvalumab Versus Docetaxel in Patients With Advanced or Metastatic Non-Small Cell Lung Cancer Without Actionable Genomic Alterations, and Whose Disease Has Progressed On or After Prior Anti-PD- (L)1 Therapy and Platinum-based Chemotherapy: LATIFY

Short Title:

A Phase III Study of Ceralasertib plus Durvalumab versus Docetaxel in Patients with Non-Small Cell Lung Cancer Whose Disease Progressed On or After Prior Anti-PD-(L)1 Therapy And Platinum-Based Chemotherapy

Rationale:

Immunotherapies targeting the programmed death-ligand 1 (PD-L1) pathway have revolutionised the management of multiple cancers including non-small cell lung cancer (NSCLC). When administered either as monotherapy, in combination with or following chemotherapy, programmed death protein 1/ligand 1 (“PD-(L)1”) blockade therapy significantly improves the durable response rate and prolongs survival with limited adverse effects as monotherapy or combination therapy for locally advanced or metastatic NSCLC. However, despite these improvements in advanced NSCLC, the majority of patients develop treatment resistance and succumb to their disease.

Following progression on platinum-based chemotherapy and PD-(L)1-containing therapies as first-line or second-line for NSCLC, treatment options for patients are limited. Treatment options include single-agent chemotherapies or a combination of docetaxel and ramucirumab or nintedanib, none of these options have been specifically studied in patients who progress following immunotherapy and platinum-based therapy. Modest efficacy and toxicity with these treatment options may limit the patient population that can be treated and therefore, novel treatments for these patients are urgently needed.

The HUDSON study is an open-label, biomarker-directed, multicentre, Phase II umbrella study in which treatment with various drugs is being investigated in patients with progressive NSCLC. In one of the study modules (Module 3), treatment with durvalumab in combination with ceralasertib is being investigated.

Module 3 of the HUDSON Study (D6185C00001) includes patients with locally advanced or metastatic NSCLC who have progressed on prior PD-(L)1 and platinum-based therapy and are being administered ceralasertib 240 mg twice a day (BID) on Days 1 to 7 followed by durvalumab 1500 mg on Day 8 of a 4 week cycle. A total of 66 patients were included in the

analysis and the median follow-up was 14.2 months. Module 3 is divided into three cohorts: Cohort B.3.PRI includes patients who experienced disease progression ≤ 24 weeks from start of their prior anti-PD-(L)1 treatment (primary resistance); Cohort B.3.ACQ includes patients who experienced disease progression > 24 weeks from the start of their prior anti-PD-(L)1 containing therapy (acquired resistance); Cohort A.3.ATM includes patients who are ATM deficient. The efficacy data from HUDSON Module 3 showed objective response rates (ORR)s of 23.8% (ATM deficient cohort, $n = 21$, [A.3.ATM]), 8% (acquired resistance cohort, $n = 25$, [B.3.ACQ]), and 15% (primary resistance cohort, $n = 20$, [B.3.PRI]). The median progression-free survival (PFS) was 8.4 months (A.3.ATM), 5.6 months (B.3.PRI) and 4.6 months (B.3.ACQ). The median overall survival (OS) was 22.8 months (A.3.ATM), 11.8 months (B.3.PRI) and 19.1 months (B.3.ACQ). When combining the data for all three cohorts of Module 3, the median OS was 15.9 months (95% confidence interval [CI], 15.8-19.1). Overall, 97% of patients experienced an adverse event (AE) and 18% of patients experienced an AE that was considered to be Common Terminology Criteria for Adverse Events (CTCAE) Grade 3 or higher and causally related to ceralasertib. Overall, 3 patients experienced an AE that led to death (data cut-off [DCO] 26 October 2021).

Longitudinal peripheral gene expression data from the HUDSON study indicate increases in gene expression signatures associated with tumour necrosis factor- α (TNF α) signalling and CD56dim Natural Killer (NK) cells upon treatment with ceralasertib, which return to baseline during the off-drug period. T-cell receptor sequencing, assessed using Adaptive Biotechnologies' immunoSEQ assay, revealed increases in peripheral richness following treatment with the combination of ceralasertib and durvalumab. Taken together, these data support a mechanism of immune modulation by ceralasertib in combination with durvalumab in this patient population.

In addition, translational data from Study 4 (D5330C00004) and Study 7 (D5330C00007) provide supportive evidence of the immunomodulation seen with ceralasertib in combination with durvalumab in the HUDSON dataset. Longitudinal peripheral blood analysis from Study 4 and Study 7 demonstrated on-drug suppression of peripheral proliferating cytotoxic T-cells (CD8+/Ki67+) during ceralasertib treatment, followed by off-drug rebound above pre-treatment levels. Similar changes were not seen in total T-cells (proliferating cytotoxic T-cells comprise a subset of the total T-cells), suggesting that these effects are primarily restricted to proliferating cells.

This translational data, along with the clinically meaningful data from Module 3 in the HUDSON study suggest that patients who have progressed following anti-PD-(L)1 therapy and platinum doublet chemotherapy could benefit from the ceralasertib plus durvalumab combination.

Therefore, the aim of this Phase III, randomised, comparator-controlled study is to investigate

the efficacy and safety of the combination of ceralasertib and durvalumab versus standard of care docetaxel in patients with locally advanced and metastatic NSCLC after progression on prior anti-PD-(L)1 therapy and platinum-based chemotherapy.

Objectives and Endpoints

Objectives	Estimands/Endpoints
Primary	
To demonstrate superiority of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of OS in participants with advanced NSCLC after second- or third-line therapy and without actionable genomic alterations	OS is defined as time from randomisation until the date of death due to any cause. The comparison will include all randomised participants, regardless of whether the participant withdraws from randomised therapy or receives another anti-cancer therapy. The measure of interest is the hazard ratio (HR) of OS.
Secondary	
To demonstrate superiority of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of PFS	PFS is defined as time from randomisation until progression per Response Evaluation Criteria in Solid Tumours, Version 1.1 (RECIST 1.1) as assessed by the investigator, or until death due to any cause. The analysis will include all randomised participants, as randomised, regardless of whether the participant withdraws from randomised therapy, receives another anti-cancer therapy or clinically progresses prior to RECIST 1.1 progression. However, if the participant progresses or dies immediately after 2 or more consecutive missed visits, the participant will be censored at the time of the latest evaluable assessment prior to the two missed visits. The measure of interest is the HR of PFS.

Objectives	Estimands/Endpoints
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of ORR	ORR is defined as the proportion of participant who have a complete response (CR) or partial response (PR) per RECIST 1.1 as determined by the investigator. The analysis will include all randomised participants with measurable disease at baseline. Data obtained from randomisation up until progression, or the last evaluable assessment in the absence of progression, will be included in the assessment of ORR, regardless of whether the participant withdraws from therapy. Participants who discontinue treatment without a response or progression, receive a subsequent therapy, and then respond will not be included as responders in the ORR. The measure of interest is the odds ratio (OR) of the ORR.
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of duration of response (DoR)	DoR is defined as the time from the date of first documented response until date of documented progression per RECIST 1.1 as assessed by the investigator, or until death due to any cause. The analysis will include all randomised participants with measurable disease at baseline who have a response, regardless of whether the participant withdraws from therapy, receives another anti-cancer therapy or clinically progresses prior to RECIST 1.1 progression. The measure of interest is the median of DoR.

Objectives	Estimands/Endpoints
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of time to response (TTR)	TTR is defined as the time from randomisation until the date of first documented objective response per RECIST 1.1 as assessed by the investigator. The analysis will include all randomised participants with measurable disease at baseline who have a response, as randomised. Data obtained from randomisation until progression, or the last evaluable assessment in the absence of progression, will be included in the assessment of TTR. Participants who discontinue treatment without a response or progression, receive a subsequent therapy, and then respond will not be included. The measure of interest is number and proportion (%) of participants with TTR, analysed as number of responders by scheduled assessment timepoint at which response was first observed.
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of disease control rate (DCR) at 18 weeks	DCR at 18 weeks is defined as the percentage of participants who have a CR or PR or who have stable disease (SD) for at least 17 weeks per RECIST 1.1 as assessed by the investigator. The analysis will include all randomised participants with measurable disease at baseline. Data obtained from randomisation until progression, or the last evaluable assessment in the absence of progression, will be included in the assessment of DCR. The measure of interest is the OR of DCR at 18 weeks.

Objectives	Estimands/Endpoints
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of time to second progression or death (PFS2)	Time from randomisation to PFS2 will be defined as the time from the randomisation to the earliest of the progression event (following the initial progression), subsequent to first subsequent therapy or death. The date of PFS2 will be recorded by the investigator in the electronic case report form (eCRF) and defined according to local standard clinical practice. The comparison will include all randomised participants regardless of whether the participant withdraws from subsequent therapy and regardless of missed visits. The measure of interest is the HR of PFS2.
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of OS at 12 months (OS12)	OS is defined in the same way as the primary endpoint. The comparison will include all randomised participants, regardless of whether the participant withdraws from randomised therapy or receives another anti-cancer therapy. The measure of interest is the landmark OS at 12 months after randomisation.
To assess participant-reported health-related quality of life (QoL)	Time to deterioration (TTD) is defined as the time from randomisation until the date of first confirmed deterioration. Change from baseline and TTD of health-related QoL as measured by the two global QoL items from the European Organisation for Research and Treatment of Cancer (EORTC) 30-item core quality of life questionnaire (QLQ-C30). The analysis will include all randomised participants. The measure of interest will be mean, mean change from baseline, and HR of TTD of items 29 and 30 from the EORTC QLQ-C30.

Objectives	Estimands/Endpoints
To assess participant-reported physical functioning in participants treated with ceralasertib plus durvalumab combination therapy relative to docetaxel	TTD in physical functioning as measured by the EORTC QLQ-C30 Physical Function subscale of the EORTC QLQ-C30. The analysis will include all randomised participants. The measure of interest will be the HR of TTD of Physical Function subscale score.
To evaluate participant-reported treatment tolerability	Participant-reported tolerability will be described among participants as treated, using the following outcomes: • Symptomatic AEs: Proportion of participants experiencing each level of symptomatic AEs as measured by the patient-reported outcome (PRO)-CTCAE • Overall side effect bother: Proportion of participants reporting overall side effect bother on the Patient Global Impression of Treatment Tolerability (PGI-TT) while on treatment The analysis will include all treated participants (safety analysis set). The measure of interest will be the proportion of participants reporting each level of overall tolerability.
To assess the pharmacokinetic (PK) of ceralasertib when administered in combination with durvalumab	Plasma concentration of ceralasertib and sparse PK parameters such as peak concentration and trough, as data allow.
To assess the safety and tolerability of ceralasertib plus durvalumab combination therapy as compared with docetaxel	Safety and tolerability will be evaluated in terms of treatment-emergent adverse events (TEAEs), vital signs, clinical laboratory results, and electrocardiograms (ECG)s.

For Tertiary/Exploratory objectives and endpoints, see Section 3.

Overall Design

Disclosure Statement: This is a Phase III, randomised, open-label, two-arm, multicentre, international study assessing the efficacy and safety of ceralasertib and durvalumab combination therapy compared with docetaxel in patients with advanced or metastatic NSCLC

that has progressed on prior anti-PD-(L)1 therapies and platinum-based chemotherapy.

Participant Population:

Male and female participants aged ≥ 18 years with histologically or cytologically documented locally advanced or metastatic NSCLC disease (according to version 8 of the International Association for the Study of Lung Cancer [IASLC]) without actionable genomic alterations and whose disease has progressed on prior PD (L)1 therapies and platinum-based chemotherapy.

Number of Participants:

Approximately 760 patients with NSCLC will be screened to achieve 580 participants randomised/assigned to study intervention. The study is planned to be conducted in approximately 200 sites across approximately 21 countries worldwide. A recruitment period of approximately 18 months is anticipated.

Enrolled/screened	Estimated 760 patients
Randomised/assigned	Estimated 580 participants

Note: “Enrolled” means a participant’s, or their legally acceptable representative’s, agreement to participate in a clinical study following completion of the informed consent process. Potential patients who are screened for the purpose of determining eligibility for the study, but are not randomised/assigned in the study, are considered “screen failures”, unless otherwise specified by the protocol.

Intervention Groups and Duration:

Participants will be randomised in a 1:1 ratio to one of the two treatment groups:

- **Group A:** Ceralasertib plus durvalumab combination therapy

Each 28-day cycle will begin with ceralasertib from Day 1 to Day 7 followed by durvalumab on Day 8 (approximately 290 participants planned)

- **Group B:** Docetaxel monotherapy

Each 21-day cycle will begin with docetaxel on Day 1 (approximately 290 participants planned)

Participants in Group A will receive ceralasertib oral tablets 240 mg BID for 7 consecutive days (Day 1 to Day 7) and then 1500 mg durvalumab (on Day 8) as an intravenous (IV) infusion every 28 days (q28d). Participants in Group B will receive docetaxel 75 mg/m² body surface area (BSA) as IV infusion every 21 days (q21d). Study intervention will continue until clinical progression, confirmed RECIST 1.1-defined radiological progression (refer to [Appendix F](#)), unacceptable toxicity, withdrawal of consent, or an intervention discontinuation

criterion is met (for study discontinuation criteria, see Section 7.1). There is no maximum treatment duration defined for either group.

If a participant's weight decreases to ≤ 30 kg, then the dose of durvalumab will be reduced to a weight-based dose equivalent of 20 mg/kg durvalumab q28d, after consultation between the principal investigator and the study physician. This dose will continue unless the participant's weight increases to > 30 kg whereby the original fixed dose of 1500 mg durvalumab q28d will be resumed.

Randomisation will be stratified by histology (squamous versus non-squamous), resistance to prior anti-PD-(L)1 therapy (primary versus acquired), tumour cell PD-L1 expression level ($< 1\%$ versus $\geq 1\%$), and geographical region (North America and Western Europe versus Rest of the World [RoW]). **Treatment Through Progression**

Participants who are clinically stable may continue to receive durvalumab and/or ceralasertib beyond RECIST 1.1-defined progression of disease as long as they are continuing to show clinical benefit, as judged by the investigator; see Section 6.1.1.4 for detailed criteria for continuation beyond progression.

Follow-up of Participants Post-discontinuation of Study Intervention:

After study intervention discontinuation, all participants will be followed up for safety assessments 30, 60, and 90 days after their last dose of study intervention (ie, the safety follow-up visits).

Participants who have discontinued study intervention in the absence of RECIST 1.1-defined radiological progression will be followed up with tumour assessments according to the Schedules of Activities (SoA) until RECIST 1.1-defined progression of disease (PD) plus an additional follow-up tumour assessment or death (whichever comes first) regardless of whether or not the participant started a subsequent anticancer therapy, unless they have withdrawn all consent to study-related assessments.

In addition, all participants will be followed up for survival status and assessment of time to second progression after intervention discontinuation every 3 months from the date of last dose until death, withdrawal of consent, or the end of the study as per the SoA. These participants are not eligible for retreatment at any time.

All participants randomised in the study should be followed up for survival.

See Section 6.7 for a description of assessments following DCO.

Independent Data Monitoring Committee:

An independent data monitoring committee (IDMC) comprised of independent experts will be

convened to review unblinded safety data and make recommendations to continue, amend, or stop the study based on safety findings or futility.

For interim futility and interim superiority analyses, the IDMC will review unblinded efficacy data and inform the sponsor whether the interim boundaries specified in Section 9.6 are met. Full details of the IDMC procedures, processes, and interim analyses can be found in the IDMC charter.

Statistical Methods

The primary analysis of OS will be performed when approximately 406 OS events have occurred across the total study population (70% maturity), which will provide 90% power to demonstrate statistical significance at the overall 5% level (using a 2-sided test). The smallest treatment difference that could be statistically significant at the final analysis is an average HR of 0.82.

Assuming an 18 month recruitment period and a minimum follow-up of 13.5 months, it is anticipated that the final OS analysis will be performed approximately 32 months after the first participant is randomised.

Other assumptions are median OS of 9 months for docetaxel and 12.5 months for ceralasertib plus durvalumab combination treatment with 2 months delay of separation of the OS curves (assumed HR = 0.72). The critical HR is 0.82 which translates to an approximate median OS of 10.9 months in ceralasertib plus durvalumab treatment group compared to 9 months in docetaxel group.

OS will be analysed using a log-rank test stratified by histology (squamous versus non-squamous), resistance to prior anti-PD-(L)1 therapy (primary versus acquired), tumour cell PD-L1 expression level ($<1\%$ versus $\geq 1\%$), and geographical region (North America and Western Europe versus RoW). Kaplan-Meier plots of OS will be presented by treatment group and the same stratification variables, where appropriate.

The effect of ceralasertib plus durvalumab combination treatment versus docetaxel treatment will be estimated based on the HR together with its 95% CI (or appropriate CI according to the significance level in the multiple testing procedure [MTP]) and p-value. The HR and CI will be estimated from a stratified Cox Proportional Hazards model (with ties = Efron and stratification variables histology, resistance to prior anti-PD-(L)1 therapy, region and PD-L1 expression level) and the CI calculated using a profile likelihood approach. An HR < 1 will be considered to favour ceralasertib plus durvalumab treatment over docetaxel treatment.

The key secondary endpoints will be analysed using the same methodology specified for the primary endpoint.

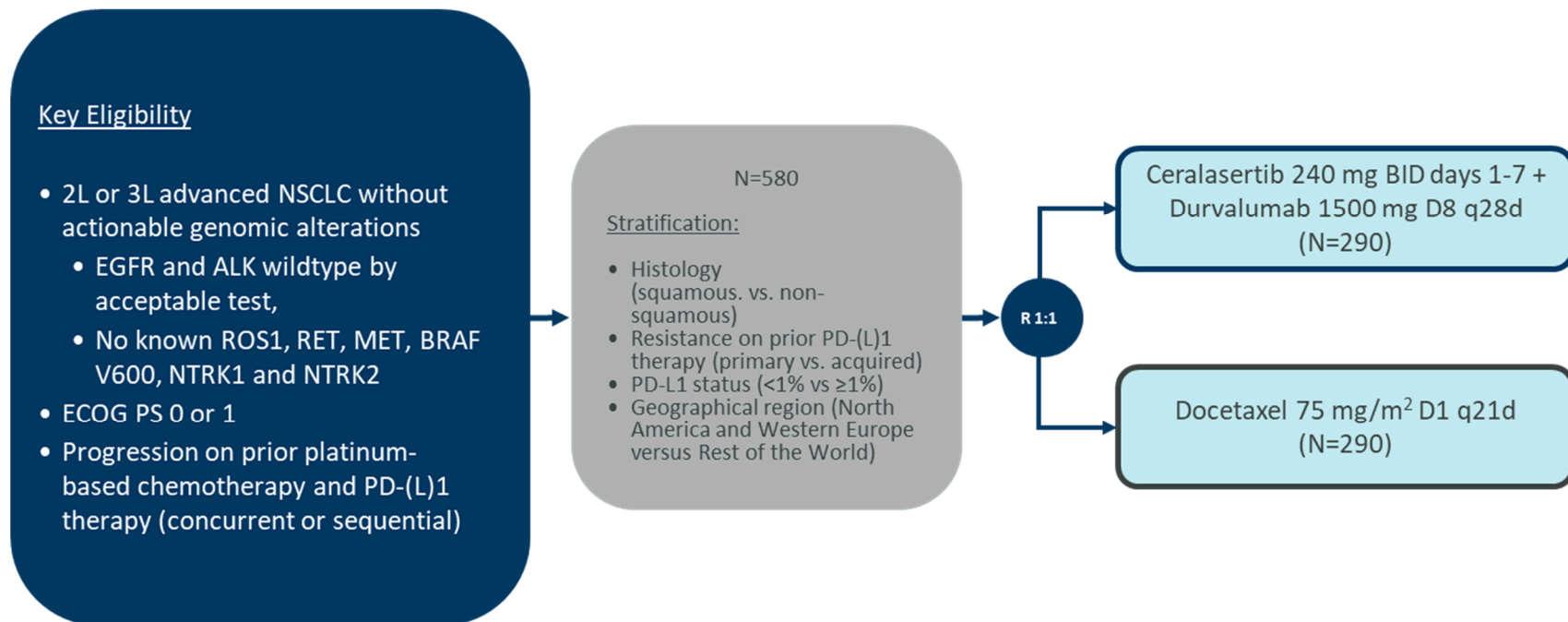
Two interim analyses (IA1 and IA2) will be performed by an IDMC at an information fraction of approximately 32% and 71%, respectively, ie, 130 and 290 deaths, respectively. It is anticipated that these interim analyses will be performed 16 and 23 months, respectively, after the first participant is randomised. IA1 is aimed at assessing the futility of continuing the study. If the HR ≥ 1.05 then the study will be stopped. IA2 is aimed at assessing the superiority of ceralasertib plus durvalumab combination treatment over SoC (critical HR < 0.75). Interim OS will be analysed using a log-rank test stratified as described previously.

Hypotheses will be tested using an MTP with an alpha-exhaustive recycling strategy ([Burman et al 2009](#)). With this approach, hypotheses will be tested in a pre-defined order, where OS for ceralasertib plus durvalumab combination therapy versus docetaxel treatment (FAS) is tested first, followed by test of secondary endpoints of PFS and ORR in the same way. The other hypotheses will then be tested in the MTP using alpha (test mass) splitting and alpha recycling, where the test mass that becomes available after each rejected hypothesis is recycled to secondary hypotheses not yet rejected. This testing procedure stops when the entire test mass is allocated to non-rejected hypotheses. Implementation of this pre-defined ordered testing procedure, including recycling, will strongly control type I error at 5% (2-sided), among all key hypotheses.

Safety data will be summarised descriptively and will not be formally analysed unless otherwise specified. Data will be presented by treatment arm using the safety analysis set.

1.2 Schema

Figure 1. Study Design



1.3 Schedule of Activities

The procedures for this study are presented in the SoA for Group A in [Table 1](#), and for Group B in [Table 2](#).

Table 1 Schedule of Activities for Group A: Ceralasertib plus Durvalumab

Visit	Screening	Intervention period						Post-intervention follow-up period	Reference Section
Cycle (28 days)	-	C1		C2		C3 and onwards			
Day of Cycle		D1	D8	D1	D8	D1	D8		
Study Week	< 4 weeks before C1D1	1	2	5	6	9, 13, etc.	10, 14, etc.		
Window (days)	-	0	± 1	± 1	± 1	± 1	± 1	± 7	
Study procedures									
Informed consent: main study ^a	X								A 3
Informed consent: optional blood sample for Genomic Initiative	X								8.7 and A 3
Mandatory tumour sample ^{b, c}	X								8.6.1
PD-L1 expression status; assessed at the central laboratory ^c	X								8.6.1.1
Inclusion and exclusion criteria ^{d, e}	X								5.1 and 5.2
Demography	X								5.1
Physical examination ^e	X	Targeted physical examination, as clinically indicated.							8.2.1
Height	X								8.2.1
Weight ^e	X		X		X		X		8.2.1
Past and current medical conditions ^e	X								5

Visit	Screening	Intervention period						Post-intervention follow-up period	Reference Section
Cycle (28 days)	-	C1		C2		C3 and onwards			
Day of Cycle		D1	D8	D1	D8	D1	D8		
Study Week	< 4 weeks before C1D1	1	2	5	6	9, 13, etc.	10, 14, etc.		
Window (days)	-	0	± 1	± 1	± 1	± 1	± 1	± 7	
Cancer history including prior anticancer treatment	X								5
ECOG/WHO performance status ^e	X	X		X		X			8.2.5.2
AE/SAE	X	At every visit; and may be conducted by telephone if not tied to a visit.						At 30 ± 3, 60 ± 7, and 90 ± 7 days after last dose	8.3
Concomitant medications	X	At every visit; and may be conducted by telephone if not tied to a visit.							6.5
Pregnancy test (WOCBP only)	Serum pregnancy test	As clinically indicated. Urine pregnancy tests at D1 in every cycle.						At 30 ± 3 days after last dose	8.2.4
Hepatitis B and C and HIV screening ^f	X								8.2.4
Clinical safety laboratory assessments (clinical chemistry including TSH [reflex free T3 or free T4] ^g , haematology, urinalysis ^h) ^{i, k}	X	X	X	X	X	X	X		8.2.4
Coagulation parameters	X	As clinically indicated.							8.2.4
12-lead ECG ^{j, k}	Triplicate ECG	As clinically indicated; single ECGs.							8.2.3
Vital signs ^k	X	X	X	X	X	X	X		8.2.2
Randomisation		X							

Visit	Screening	Intervention period						Post-intervention follow-up period	Reference Section
Cycle (28 days)	-	C1		C2		C3 and onwards			
Day of Cycle		D1	D8	D1	D8	D1	D8		
Study Week	< 4 weeks before C1D1	1	2	5	6	9, 13, etc.	10, 14, etc.		
Window (days)	-	0	± 1	± 1	± 1	± 1	± 1	± 7	
Participant diary ¹		X	X	X	X	X	X		6.4
Subsequent anticancer therapy								X	7.1.1
Survival status								Every 3 months after last dose	8.1.3
PFS2								Every 3 months after PD	8.1.4
Pharmacokinetic assessments									
Plasma sample for ceralasertib PK analysis ^k		X	X	X	X		C4D8, C8D8 and C12D8 only.		8.5.1

Visit	Screening	Intervention period						Post-intervention follow-up period	Reference Section
Cycle (28 days)	-	C1		C2		C3 and onwards			
Day of Cycle		D1	D8	D1	D8	D1	D8		
Study Week	< 4 weeks before C1D1	1	2	5	6	9, 13, etc.	10, 14, etc.		
Window (days)	-	0	± 1	± 1	± 1	± 1	± 1	± 7	
Biomarker analysis									
Whole blood sample for ctDNA and DNA analyses ⁿ		X			X		C4D8, C8D8, and C12D8 only.		8.6.2.1
Whole blood sample for RNA and DNA analyses ⁿ		X	X		X	C3D1 only.			8.6.2.2
Plasma sample for biomarker analyses ⁿ	X	X	X		X	C3D1 only.			8.6.2.3
Optional: Blood sample for Genomic Initiative assessment ^m		X							8.7

Visit	Screening	Intervention period						Post-intervention follow-up period	Reference Section
Cycle (28 days)	-	C1		C2		C3 and onwards			
Day of Cycle		D1	D8	D1	D8	D1	D8		
Study Week	< 4 weeks before C1D1	1	2	5	6	9, 13, etc.	10, 14, etc.		
Window (days)	-	0	± 1	± 1	± 1	± 1	± 1	± 7	
Imaging and Biopsies ^c									Appendix F
Tumour imaging (CT or MRI of chest and abdomen) per RECIST 1.1 ^o	X					First scan at 9 (+1) week; and thereafter every 9 ± 1 weeks until objective PD, with one follow-up scan after 4 ± 2 weeks.			8.1.1
Brain CT/MRI	All participants					For participants with confirmed baseline brain metastases or if clinically indicated. At the same time as tumour evaluations (see above row).			8.1.2
Bone scan or ¹⁸ F-FDG PET/CT or CT/MRI	All participants					For participants with confirmed baseline bone metastases or if clinically indicated. After C1D1, every 6 months ± 2 weeks or as clinically indicated.			8.1.2
Optional: Tumour sample on PD ⁿ		X							8.6.3

Visit	Screening	Intervention period						Post-intervention follow-up period	Reference Section
Cycle (28 days)	-	C1		C2		C3 and onwards			
Day of Cycle		D1	D8	D1	D8	D1	D8		
Study Week	< 4 weeks before C1D1	1	2	5	6	9, 13, etc.	10, 14, etc.		
Window (days)	-	0	± 1	± 1	± 1	± 1	± 1	± 7	
PRO assessments ^P									8.1.5.9
NSCLC-SAQ		pre-dose	Every week for the first 12 weeks and then every 4 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.						8.1.5.1
EORTC IL 171		pre-dose	Every week for the first 12 weeks, then every 4 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.						8.1.5.2
PRO-CTCAE PGI-TT		pre-dose	Every week for the first 12 weeks, then every 4 weeks relative to C1D1 until treatment discontinuation.						8.1.5.4 8.1.5.5
PGI-S		pre-dose	Every 4 weeks relative to C1D1 until PD.						8.1.5.6
PGI-C						Week 12 and Week 24 relative to C1D1.			8.1.5.7
EORTC IL 172		pre-dose	Every 4 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.						8.1.5.3
EQ-5D-5L		pre-dose	Every 8 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.						8.1.5.8
Medical Resource Utilisation									
HOSPAD		X	X	X	X	X	X	X	8.8

Visit	Screening	Intervention period						Post-intervention follow-up period	Reference Section
Cycle (28 days)	-	C1		C2		C3 and onwards			
Day of Cycle		D1	D8	D1	D8	D1	D8		
Study Week	< 4 weeks before C1D1	1	2	5	6	9, 13, etc.	10, 14, etc.		
Window (days)	-	0	± 1	± 1	± 1	± 1	± 1	± 7	
Study intervention ^q									
Ceralasertib (oral) (7 days; D1 to D7; q28d) ^r		X		X		X			6
Durvalumab (IV) (Day 8; q28d)			X		X		X		6
Ceralasertib dispensing ^{r,1}		X		X		X			6
Ceralasertib accountability			X		X		X		6

- a Written informed consent and any locally required privacy act document authorisation must be obtained at screening and prior to performing any protocol-specific procedures, including screening/baseline evaluations.
- b Where available, a tissue sample obtained after progression on prior anti-PD-(L)1 therapy and ≤ 3 months prior to randomisation should be collected. Where no such sample is available a tumour sample taken within the previous 24 months is acceptable. If a suitable archival tumour sample is not available, a fresh tumour sample should be taken via a biopsy. PD-L1 will be evaluated prospectively using the VENTANA PD-L1 (SP263) IHC assay by a central reference laboratory. PD-L1 status must be known prior to randomisation.
- c The mandatory tissue and blood samples will not be collected or analysed until after approval from relevant local authorities (eg, HGR in China).
- d Documented EGFR and ALK wild-type status as determined at a local laboratory using a well-validated, locally approved test.
- e If screening assessments have been performed within 28 days prior to starting study intervention, they do not have to be repeated at C1D1 if the participant's condition has not changed, otherwise the assessments have to be repeated.
- f Participants HCV-Ab positive should test negative for HCV RNA to be eligible.
- g Free T3 and free T4 will only be measured if TSH is abnormal or if there is clinical suspicion of an AE related to the endocrine system. If TSH is measured within 14 days prior to Day 8 (first infusion day), it does not need to be repeated at Day 8.
- h Urinalysis should be performed at baseline (screening) and then as clinically indicated. Serum or plasma clinical chemistry (including LFT monitoring) and haematology may be performed more frequently if clinically indicated from C3D8 onwards.

- i Clinical laboratory assessment parameters are to be assessed on Day 1 or within the previous 3 days (unless all screening laboratory assessments are performed within 3 days prior to Day 1), and as clinically indicated.
- j Triplicate ECG to be done at screening, single ECGs at other visits only if clinically indicated.
- k Whenever ECG, vital signs and blood draws are scheduled on the same day, the assessments should occur in the following order: ECG, vital signs and then blood draws. The timing of the ECG and vital signs assessments must allow for the blood draw (eg, PK blood sample) to occur at the scheduled timepoints.
- l Participants will record the self-administration of ceralasertib tablets in the participant diary.
- m The optional samples for genetic research and Genomic Initiative assessments will be obtained pre-dose on C1D1. If, for any reason, the sample is not drawn on C1D1, it may be taken at any visit until the last study visit. Only one sample should be collected per participant for genetics during the study. This sample will not be collected for participants in China.
- n This sample will not be collected for participants in China.
- o If CT/MRI scan at screening shows disease in any area other than chest and abdomen, then scans of those areas must be continued for all subsequent scans in the study. For participants who continue study intervention following PD (see Section 6.1.1.4), the scans should continue every 9 ± 1 weeks until the end-of-treatment, with a follow-up scan after 4 (+2) weeks.
- p PRO questionnaires will be available on an application for a handheld device and on the web. The device should be fully charged, and checked that it is fully functioning, prior to the participant's first dosing visit. The site should ensure the participant is trained and understands how to use the PRO device prior to the first use. Handheld devices will be provisioned where required. PRO completion windows will be specified during the product design specification process.
- q Every effort should be made to minimise the time between randomisation and starting treatment. First study dose should be administered within 3 business days of randomisation.
- r Ceralasertib tablets for a 7-day treatment (BID) will be dispensed on D1 of each cycle at a site visit.

Note: All assessments on treatment days are to be performed prior to study intervention administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in Section 8.

AE, adverse event; ALK, anaplastic lymphoma kinase; C, cycle; CT, computed tomography; CTCAE, Common Terminology Criteria for Adverse Events; D, study day; ECG, electrocardiogram; ECOG, Eastern Cooperative Oncology Group; EGFR, epidermal growth factor receptor; EORTC, European Organisation for Research and Treatment of Cancer; EQ-5D-5L, EuroQoL 5-dimension, 5-level health state utility index; HIV, human immunodeficiency virus; ICF, informed consent form; LFT, liver function tests; MRI, magnetic resonance imaging; PD, progression of disease; PD-L1, programmed death-ligand 1; PET, positron emission tomography; PFS2, time to second progression or death; PGI-C, Patients' Global Impression of Change; PGI-S, Patients' Global Impression of Severity; PRO, patient-reported outcome; QoL, quality of life; qXd, every X days; QLQ-C30, 30 item core quality of life questionnaire; RECIST 1.1, Response Evaluation Criteria in Solid Tumours, Version 1.1; SoC, standard of care; T3, triiodothyronine; T4, thyroxine; TSH, thyroid-stimulating hormone; WHO, World Health Organisation; WOCBP, women of childbearing potential.

Table 2 **Schedule of Activities for Group B: Docetaxel**

Cycle (21 days)	Screening < 4 weeks before C1D1	Intervention period		Post-intervention period	Reference Section
Days of cycle		C1D1	C2D1, C3D1, etc.		
Study Week		1	4, 7, 10, etc.		
Window (days)		± 1	± 1	± 1	
Study Procedures					
Informed consent: main study ^a	X				A 3
Informed consent: optional blood sample for Genomic Initiative	X				8.7 and A 3
Mandatory tumour sample for PD-L1 ^{b, c}	X				8.6.1
PD-L1 expression status. Assessed at the central laboratory ^c	X				8.6.1
Inclusion and exclusion criteria ^{d, e}	X				5
Demography	X				5.1
Physical examination ^e	X	Targeted physical examination, as clinically indicated.			8.2.1
Height	X				
Weight ^e	X	X	X		
Past and current medical conditions ^e	X				5
Cancer history including prior anticancer treatment	X				5

Cycle (21 days)	Screening < 4 weeks before C1D1	Intervention period		Post-intervention period	Reference Section
Days of cycle		C1D1	C2D1, C3D1, etc.		
Study Week		1	4, 7, 10, etc.		
Window (days)		± 1	± 1	± 1	
ECOG/WHO performance status ^e	X	X	D1 of every cycle		8.2.5.2
AE/SAE	X	At every visit; and may be conducted by telephone if not tied to a visit.		Up to 90 days after the last dose.	8.3
Concomitant medications	X	At every visit; and may be conducted by telephone if not tied to a visit.			6.5
Pregnancy test (for WOCBP)	Serum pregnancy test	As clinically indicated. Urine pregnancy test at D1 in every cycle.		X	8.2.4
Hepatitis B and C and HIV screening ^f	X				8.2.4
Clinical safety laboratory assessments (clinical chemistry including TSH [reflex free T3 or free T4] ^g , haematology, urinalysis ^h) ^{i,k}	X	X	X		8.2.4
Coagulation parameters	X	As clinically indicated.			8.2.4
12-lead ECG ^{j,k}	Triplicate ECG	As clinically indicated; single ECGs.			8.2.3
Vital signs ^k	X	X	X	X	8.2.2
Randomisation		X			
Subsequent anticancer therapy				X	7.1.1

Cycle (21 days)	Screening < 4 weeks before C1D1	Intervention period		Post-intervention period	Reference Section
Days of cycle		C1D1	C2D1, C3D1, etc.		
Study Week		1	4, 7, 10, etc.		
Window (days)		± 1	± 1	± 1	
Survival status				Every 3 months after last dose.	8.1.3
PFS2				Every 3 months after PD.	
Biomarker analysis ^c					
Whole blood sample for ctDNA and DNA assessments ^l		X	C3D1, C5D1, C10D1, and C16D1 only.	X	8.6.2.1
Whole blood for RNA and DNA analysis ^l		X	C3D1 only.		8.6.2.2
Plasma sample for biomarker analysis ^l		X	C3D1 only.		8.6.2.3
Optional: Blood sample for Genomics Initiative assessments ^m		X			8.7

Cycle (21 days)	Screening < 4 weeks before C1D1	Intervention period		Post-intervention period	Reference Section
Days of cycle		C1D1	C2D1, C3D1, etc.		
Study Week		1	4, 7, 10, etc.		
Window (days)		± 1	± 1	± 1	
Imaging and Biopsies ^c					Appendix F
Tumour evaluation (CT or MRI of chest and abdomen) per RECIST 1.1 ⁿ	X		First scan at 9 (+ 1) week and thereafter every 9 ± 1 weeks until objective PD with one follow-up scan after 4 ± 2 weeks after PD.		8.1.1
Brain CT/MRI	All participants		For participants with confirmed baseline brain metastases or if clinically indicated. At the same time as tumour evaluations (see above row).		8.1.2
Bone scan or ¹⁸ F-FDG PET/CT or CT/MRI	All participants		For participants with confirmed baseline bone metastases or if clinically indicated. Every 6 months ± 2 weeks or as clinically indicated.		8.1.2
Optional: Tumour tissue biopsy sample at PD ¹		X			8.6.3

Cycle (21 days)	Screening < 4 weeks before C1D1	Intervention period		Post-intervention period	Reference Section
Days of cycle		C1D1	C2D1, C3D1, etc.		
Study Week		1	4, 7, 10, etc.		
Window (days)		± 1	± 1	± 1	
PRO assessments ^o					8.1.5.9
NSCLC-SAQ		pre-dose	Every week for the first 12 weeks and then every 3 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.		8.1.5.1
EORTC IL 171		pre-dose	Every week for the first 12 weeks, then every 3 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.		8.1.5.2
PRO-CTCAE PGI-TT		pre-dose	Every week for the first 12 weeks, then every 3 weeks relative to C1D1 until treatment discontinuation.		8.1.5.4 8.1.5.5
PGI-S		pre-dose	Every 3 weeks relative to C1D1 until PD.		8.1.5.6
PGI-C			Week 12 and Week 24 relative to C1D1.		8.1.5.7
EORTC IL 172		pre-dose	Every 3 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.		8.1.5.3
EQ-5D-5L		pre-dose	Every 6 weeks relative to C1D1 until PD; then every 12 weeks for 3 timepoints after PD.		8.1.5.8
Medical Resource Utilisation					
HOSPAD	X	X	X	X	8.8

Cycle (21 days)	Screening < 4 weeks before C1D1	Intervention period		Post-intervention period	Reference Section
Days of cycle		C1D1	C2D1, C3D1, etc.		
Study Week		1	4, 7, 10, etc.		
Window (days)		± 1	± 1	± 1	
Study intervention Administration					
Docetaxel (IV) ^P		X	X		6

- a Written informed consent and any locally required privacy act document authorisation must be obtained at screening and prior to performing any protocol-specific procedures, including screening/baseline evaluations.
- b Where available, a tissue sample obtained after progression on prior anti-PD-(L)1 therapy and ≤3 months prior to randomisation should be collected. Where no such sample is available a tumour sample taken within the previous 24 months is acceptable. If a suitable archival tumour sample is not available, a fresh tumour sample should be taken via a biopsy. PD-L1 will be evaluated prospectively using the VENTANA PD-L1 (SP263) IHC assay by a central reference laboratory. PD-L1 status must be known prior to randomisation.
- c The mandatory tissue and blood samples will not be collected or analysed until after approval from relevant local authorities (eg, HGR in China).
- d Documented EGFR and ALK wild-type status as determined at a local laboratory using a well-validated, locally approved test.
- e If screening assessments have been performed within 28 days prior to starting study intervention, they do not have to be repeated at C1D1 if the participant's condition has not changed, otherwise the assessments have to be repeated.
- f Participants HCV-Ab positive should test negative for HCV RNA to be eligible.
- g Free T3 and free T4 will only be measured if TSH is abnormal or if there is clinical suspicion of an AE related to the endocrine system. If TSH is measured within 14 days prior to Day 1 (first infusion day), it does not need to be repeated at Day 1.
- h Urinalysis should be performed at baseline (screening) and then as clinically indicated. Serum or plasma clinical chemistry (including LFT monitoring) and haematology may be performed more frequently if clinically indicated from C3D8 onwards.
- i Clinical laboratory assessment parameters are to be assessed on Day 1 or within the previous 3 days (unless all screening laboratory assessments are performed within 3 days prior to Day 1), and as clinically indicated.
- j Triplicate ECG to be done at screening, single ECGs at other visits only if clinically indicated.
- k Whenever ECG, vital signs and blood draws are scheduled on the same day, the assessments should occur in the following order: ECG, vital signs and then blood draws. The timing of the ECG and vital signs assessments must allow for the blood draw (eg, PK blood sample) to occur at the scheduled timepoints.
- l This sample will not be collected for participants in China.
- m The optional samples for genetic research and Genomic Initiative assessments will be obtained pre-dose on C1D1. If, for any reason, the sample is not drawn on C1D1, it may be taken at any visit until the last study visit. Only one sample should be collected per participant for genetics during the study. This sample will not be collected for patients in China.

- n If CT/MRI scan at screening shows disease in any area other than chest and abdomen, then scans of those areas must be continued for all subsequent scans in the study. For participants who continue study intervention following PD (see Section 6.1.1.4), the scans should continue every 9 ± 1 weeks until the end-of-treatment, with a follow-up scan after 4 (+2) weeks.
- o PRO questionnaires will be available on an application for a handheld device and on the web. The device should be fully charged, and checked that it is fully functioning, prior to the participant's first dosing visit. The site should ensure the participant is trained and understands how to use the PRO device prior to the first use. Handheld devices will be provisioned where required. PRO completion windows will be specified during the product design specification process.
- p Every effort should be made to minimise the time between randomisation and starting treatment. First study dose should be administered within 3 business days of randomisation.

Note: All assessments on treatment days are to be performed prior to study intervention administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in Section 8.

AE, adverse event; ALK, anaplastic lymphoma kinase; C, cycle; CT, computed tomography; CTCAE, Common Terminology Criteria for Adverse Events; D, study day; ECG, electrocardiogram; ECOG, Eastern Cooperative Oncology Group; EGFR, epidermal growth factor receptor; EORTC, European Organisation for Research and Treatment of Cancer; EQ-5D-5L, EuroQoL 5-dimension, 5-level health state utility index; HIV, human immunodeficiency virus; ICF, informed consent form; LFT, liver function tests; MRI, magnetic resonance imaging; PD, progression of disease; PD-L1, programmed death-ligand 1; PET, positron emission tomography; PFS2, time to second progression or death; PGI-C, Patients' Global Impression of Change; PGI-S, Patients' Global Impression of Severity; PRO, patient-reported outcome; QoL, quality of life; qXd, every X days; QLQ-C30, 30 item core quality of life questionnaire; RECIST 1.1, Response Evaluation Criteria in Solid Tumours, Version 1.1; SoC, standard of care; T3, triiodothyronine; T4, thyroxine; TSH, thyroid-stimulating hormone; WHO, World Health Organisation; WOCBP, women of childbearing potential.

2 INTRODUCTION

Durvalumab is a human Ig G1 kappa mAb that blocks the interaction of PD-L1 (but not PDL2) with PD-1 on T-cells and CD80 (B7.1) on immune cells and is engineered to reduce antibody-dependent cell-mediated cytotoxicity and complement activation.

Ceralasertib (AZD6738) is an oral, potent, selective inhibitor of the ATR kinase, with good selectivity against other PIKK family members. ATR is a key component of the DNA damage response, in particular in the cell's response to replication stress. In addition, there is evidence that ATR can modulate the tumour immune environment and inhibition of ATR can potentiate the effects of treatment by modulation of T-cells.

Background on the Disease to be Treated

Lung cancer is the most common cancer globally, with an estimated incidence of 2.21 million new cases in 2020, accounting for 11.4% of all new cancers. It is the leading cause of cancer-related mortality, with 1.80 million deaths in 2020 (18.0% of the total) (GLOBOCON 2020). NSCLC represents approximately 80% to 85% of all lung cancers, of which at least 70% of cases present late with inoperable Stage III or metastatic Stage IV disease. In some Stage III cases chemoradiotherapy ± surgery can provide long-term control while nearly all patients with Stage IV disease are incurable and treated with palliative systemic medical treatment.

Advances over the past 20 years have seen histological and molecular classification of NSCLC such that squamous and non-squamous carcinomas are clearly distinguishable, and subdivision based on molecular aberrations is now possible ([Hirsch et al 2016](#)). Whilst some of these molecular segments have approved therapies, including patients with EGFR-activating and resistance mutations and ALK translocations and ROS1 (a receptor tyrosine kinase) aberrations amongst others, more than 50% of adenocarcinomas and more than 95% of squamous cell carcinomas do not.

Historically, platinum-based combination chemotherapy was the standard of care for first-line therapy in non-squamous and squamous cell carcinoma ([Ettinger et al 2017](#)). More recently, immune checkpoint inhibitors targeting the PD-1 or PD-L1 inhibitory axis, referred to as “PD-(L)1”, have demonstrated efficacy and tolerability when combined with platinum-based combinations or as monotherapy in patients with locally advanced or metastatic NSCLC ([Ettinger et al 2020](#); [Planchard et al 2018](#)). The current first-line standard of care treatment for patients with metastatic NSCLC depends on PD-L1 expression on the tumour cells. In patients with PD-L1 expression $\geq 50\%$, preferred treatment options include pembrolizumab monotherapy, atezolizumab monotherapy, cemiplimab monotherapy and pembrolizumab plus platinum doublet chemotherapy. In patients with tumours with PD-L1 expression $< 50\%$, the preferred treatment option is pembrolizumab plus platinum doublet. For patients who do not receive anti-PD-(L)1 therapy in first-line, nivolumab, pembrolizumab or atezolizumab

monotherapy is preferred as second-line therapy. In patients with unresectable, locally advanced disease, durvalumab is preferred as sequential therapy following chemoradiation therapy ([NCCN 2021](#), [Planchard et al 2018](#)).

Despite the improvement in outcomes with immune checkpoint inhibitors in the first- or second-line NSCLC setting, it is becoming increasingly clear that there is a new and significant patient population emerging that does not respond to anti-PD-(L)1-containing therapy or progresses during anti-PD-(L)1-containing therapy ([Fares et al 2019](#), [Kluger et al 2020](#), [Sharma et al 2017](#), [Walsh and Soo 2020](#)).

Following progression on platinum-based chemotherapy and PD-(L)1 inhibitor therapies as first-line or second-line for NSCLC, treatment options for patients are limited. Treatment options include single-agent chemotherapies or a combination of docetaxel and ramucirumab or nintedanib. However, the studies underpinning the registrational approval of these therapies did not include patients who have progressed on both platinum doublet therapy and immune checkpoint therapy. Modest efficacy and toxicity with these treatment options may limit the number of patients that can be treated and therefore, novel treatments for these patients are urgently needed.

Analysis of real-world treatment patterns in patients from Europe, North America, Asia, and South America suggest that the three most frequently used single-agent chemotherapies were docetaxel, pemetrexed and gemcitabine for second-line and third-line treatment of advanced NSCLC ([Davies et al 2017](#)). Single-agent chemotherapy accounts for about 50% of third-line treatments in general with docetaxel, gemcitabine, vinorelbine, and pemetrexed being the most frequently administered third-line chemotherapies ([NCCN 2021](#), [Planchard et al 2018](#)).

2.1 Study Rationale

Immunotherapies targeting the PD-L1 pathway have revolutionised the management of multiple cancers including NSCLC. When administered either as monotherapy, in combination with or following chemotherapy, PD-(L)1 blockade therapy significantly improves the durable response rate and prolongs survival with limited adverse effects as monotherapy or combination therapy for locally advanced or metastatic NSCLC. However, despite these improvements in advanced NSCLC, the majority of patients develop treatment resistance and succumb to their disease.

Following progression on platinum-based chemotherapy and PD-(L)1-containing therapies as first-line or second-line for NSCLC, treatment options for patients are limited. Treatment options include single-agent chemotherapies or a combination of docetaxel and ramucirumab or nintedanib, of which, the studies underpinning the registrational approval of these therapies did not include patients who have progressed on both platinum doublet therapy and immune checkpoint therapy. Modest efficacy and toxicity with these treatment options may limit the

patient population that can be treated and therefore, novel treatments for these patients are urgently needed.

The HUDSON study is an open-label, biomarker-directed, multicentre, Phase II umbrella study in which treatment with various drugs is being investigated in patients with progressive NSCLC. In one of the study modules (Module 3) treatment with durvalumab in combination with ceralasertib is being investigated.

Module 3 of the HUDSON Study (D6185C00001) includes patients with advanced or metastatic NSCLC who had progressed on prior PD-(L)1 and platinum-based therapy and were administered ceralasertib 240 mg BID on Days 1 to 7 followed by durvalumab 1500 mg on D8 of a 4 week cycle. A total of 66 patients were included in the analysis and the median follow-up was 14.2 months. Module 3 is divided into three cohorts: Cohort B.3.PRI includes patients who experienced disease progression $\leq$ 24 weeks from start of their prior anti-PD-(L)1 treatment (primary resistance); Cohort B.3.ACQ includes patients who experienced disease progression $>$ 24 weeks from the start of their prior anti-PD-(L)1 containing therapy (acquired resistance); Cohort A.3.ATM includes patients who are ATM deficient. The efficacy data from HUDSON Module 3 showed ORRs of 23.8% (ATM deficient cohort, $n = 21$, [A.3.ATM]), 8% (acquired resistance cohort, $n = 25$, [B.3.ACQ]), and 15% (primary resistance cohort, $n = 20$, [B.3.PRI]). The median PFS was 8.4 months (A.3.ATM), 5.6 months (B.3.PRI) and 4.6 months (B.3.ACQ). The median OS was 22.8 months (A.3.ATM), 11.8 months (B.3.PRI) and 19.1 months (B.3.ACQ). When combining the data for all three cohorts of Module 3, the median OS was 15.9 months (95% CI 15.8-19.1). Overall, 97% of patients experienced an AE and 18% of patients experienced an AE that was considered to be CTCAE Grade 3 or higher and causally related to ceralasertib. Overall, 3 patients experienced an AE that led to death (DCO, 26 October 2021).

Longitudinal peripheral gene expression data from the HUDSON study indicate increases in gene expression signatures associated with TNF α signalling and CD56dim NK cells upon treatment with ceralasertib, which return to baseline during the off-drug period. T-cell receptor sequencing, assessed using Adaptive Biotechnologies' immunoSEQ assay, revealed increases in peripheral richness following treatment with the combination of ceralasertib and durvalumab. Taken together, these data support a mechanism of immune modulation by ceralasertib in combination with durvalumab in this patient population.

In addition, translational data from Study 4 (D5330C00004) and Study 7 (D5330C00007) provide supportive evidence of the immunomodulation seen with ceralasertib in combination with durvalumab in the HUDSON dataset. Longitudinal peripheral blood analysis from Study 4 and Study 7 demonstrated on-drug suppression of peripheral proliferating cytotoxic T-cells (CD8+/Ki67+) during ceralasertib treatment, followed by off-drug rebound above

pre-treatment levels. Similar changes were not seen in total T-cells (proliferating cytotoxic T-cells comprise a subset of the total T-cells), suggesting that these effects are primarily restricted to proliferating cells.

This translational data, along with the clinically meaningful data from Module 3 in the HUDSON study suggest that patients who have progressed following anti-PD-(L)1 therapy and platinum doublet chemotherapy could benefit from the ceralasertib plus durvalumab combination.

Therefore, the aim of this Phase III, randomised, comparator-controlled study is to investigate the efficacy and safety of the combination of ceralasertib and durvalumab versus standard of care docetaxel in patients with locally advanced and metastatic NSCLC after progression on prior anti-PD-(L)1 therapy and platinum-based chemotherapy.

2.2 Background

2.2.1 Immunotherapies

It is increasingly understood that cancers are recognised by the immune system, and under some circumstances, the immune system may control or even eliminate tumours ([Dunn et al 2004](#)).

PD-L1 is part of a complex system of receptors and ligands that are involved in controlling T-cell activation. The PD-1 receptor (cluster of differentiation [CD]279) is expressed on the surface of activated T-cells ([Keir et al 2008](#)). It has 2 known ligands: PD-L1 (B7-H1; CD274) and programmed death-ligand 2 (PD-L2) (B7-DC; CD273) ([Okazaki and Honjo 2007](#)). PD-1 and PD-L1/PD-L2 belong to a family of immune checkpoint proteins that act as co-inhibitory factors, which can halt or limit the development of T-cell response. When PD-L1 binds to PD-1, an inhibitory signal is transmitted into the T cell, which reduces cytokine production and suppresses T-cell proliferation. Tumour cells exploit this immune checkpoint pathway as a mechanism to evade detection and inhibit immune response.

PD-L1 is constitutively expressed by B-cells, dendritic cells, and macrophages ([Qin et al 2016](#)). Importantly, PD-L1 is commonly over-expressed on tumour cells or on non-transformed cells in the tumour microenvironment ([Pardoll 2012](#)). PD-L1 expressed on the tumour cells binds to PD-1 receptors on the activated T-cells, leading to the inhibition of cytotoxic T-cells. These deactivated T-cells remain inhibited in the tumour microenvironment. The PD-1/PD-L1 pathway represents an adaptive immune resistance mechanism that is exerted by tumour cells in response to endogenous anti-tumour activity.

The inhibitory mechanism described above is co-opted by tumours that express PD-L1 as a way of evading immune detection and elimination. The binding of an anti-PD-L1 agent to the PD-L1 receptor inhibits the interaction of PD-L1 with the PD-1 and CD80 receptors expressed

on immune cells. This activity overcomes PD-L1-mediated inhibition of anti-tumour immunity. While functional blockade of PD-L1 results in T-cell reactivation, this mechanism of action is different from direct agonism of a stimulatory receptor such as CD28.

PD-L1 is expressed in a broad range of cancers. Based on these findings, an anti-PD-L1 antibody could be used therapeutically to enhance anti-tumour immune responses in participants with cancer. Results of preclinical and clinical studies of monoclonal antibodies (mAbs) targeting the PD-L1/PD-1 pathway have shown evidence of clinical activity and a manageable safety profile, supporting the hypothesis that an anti-PD-L1 antibody could be used to therapeutically enhance anti-tumour immune response in cancer participants (Antonia et al 2014; Brahmer et al 2012; Hirano et al 2005, Iwai et al 2002, Okudaira et al 2009, Topalian et al 2012, Zhang et al 2008) with responses that tend to be more pronounced in participants with tumours that express PD-L1 (Powles et al 2014, Rizvi et al 2015, Segal et al 2015). In addition, high mutational burden (eg, in bladder carcinoma; Alexandrov et al 2013) may contribute to the responses seen with immune therapy.

2.2.2 Ceralasertib (AZD6738)

Ceralasertib is being developed as an oral anti-tumour agent in patients with DDR-deficient tumours, including (but not limited to) BRCA mutant and ATM gene or protein deficiency, or in combination with durvalumab, chemotherapy or poly-adenosine 5'-diphosphate-ribose polymerase (PARP) inhibitors. Activity may not be restricted to these patient populations and there is preclinical evidence of activity as monotherapy or combinations across broader malignant disease types and molecular aberrations.

Ceralasertib is an inhibitor of the serine/threonine protein kinase, ATR, a member of the phosphatidylinositol 3 kinase-related kinase family (Foote et al 2018). ATR is the apical kinase in the replication stress response DNA damage induced checkpoint pathway (Cimprich and Cortez 2008, Forment and O'Connor 2018) and during normal DNA replication is recruited at stalled replication forks, which can progress to DSBs if left unrepaired. Following resection of DSBs, ATR is recruited to single-strand DNA coated with Replication Protein A following single-strand DNA damage. Recruitment and activation of ATR leads to cell cycle arrest in S-phase while the DNA is repaired, and the stalled replication fork resolved. Loss of ATR function leads to the inability to resolve stalled replication forks, the accumulation of DNA damage and rapid cell death exemplified by nuclear fragmentation.

DNA damage, replication stress and ATR have been associated with activation of the innate immune response, leading to the priming of the adaptive immune system which may also drive enhanced anti-tumour responses (Dillon et al 2019, Feng et al 2020, Parkes et al 2017, Pilger et al 2021, Sato et al 2017, Sheng et al 2020, Sun et al 2018, Vendetti et al 2018). Unrepaired DNA lesions and/or chromosome mis-segregation during mitosis contribute to the formation of endogenous cytosolic DNA (often in the form of micronuclei) or RNA. This

leads to the activation of cGAS-STING dependent DNA and/or RIG-I-MAVS dependent RNA sensing pathways to induce type I IFN signalling, resulting in presentation of the cancer cell for CD8⁺ T-cell-mediated immune destruction. Therefore, ATR inhibitors have the potential to work synergistically when combined with immune checkpoint inhibitors such as durvalumab (an anti-PD-L1 therapy) to further enhance anti-tumour immunity potentially through, but not restricted to, tumour intrinsic cytosolic nuclei acid sensing pathways and modulation of T-cells in the tumour microenvironment.

A detailed description of the chemistry, pharmacology, efficacy, and safety of ceralasertib is provided in the IB.

2.2.3 Durvalumab

Durvalumab is a human mAb of the immunoglobulin G 1 kappa subclass that blocks the interaction of PD-L1 (but not PD-L2) with PD-1 on T-cells and CD80 (B7.1) on immune cells. It is being developed by AstraZeneca for use in the treatment of cancer. The proposed mechanism of action for durvalumab is interference in the interaction of PD-L1 with PD-1 and CD80 (B7.1). Blockade of PD-L1/PD-1 and PD-L1/CD80 interactions releases the inhibition of immune responses, including those that may result in tumour elimination. In vitro studies demonstrate that durvalumab antagonises the inhibitory effect of PD-L1 on primary human T-cells resulting in the restored proliferation of IFN- γ ([Stewart et al 2015](#)). In vivo studies have shown that durvalumab inhibits tumour growth in xenograft models via a T-cell-dependent mechanism ([Stewart et al 2015](#)). Based on these data, durvalumab is expected to stimulate the patient's anti-tumour immune response by binding to PD-L1 and shifting the balance toward an anti-tumour response. Durvalumab has been engineered to reduce antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity.

As of the DCO date of the current durvalumab IB, 12 July 2021, an estimated 12,874 patients have received durvalumab in AstraZeneca or MedImmune-sponsored interventional studies in multiple tumour types, stages of disease and lines of therapy. Of these, 5136 patients received durvalumab monotherapy, 3712 patients received durvalumab in combination with tremelimumab, and 4026 patients received durvalumab or durvalumab plus tremelimumab in combination with an investigational and/or an approved product. An estimated 13,113 patients have been randomised and treated to the various treatment/comparator arms in sponsor-blinded and/or double-blinded studies. In addition, > 5000 patients have participated in the durvalumab Early Access Program (Study D4194C00002) for patients with locally advanced, unresectable NSCLC whose disease has not progressed following platinum-based chemoradiation therapy.

As of 12 July 2021, durvalumab is approved in more than 70 countries. Durvalumab has been approved to treat locally advanced or metastatic urothelial carcinoma, unresectable Stage III NSCLC, and extensive stage small cell lung cancer (administered in combination with

chemotherapy). Refer to the package insert (or label) for your specific country, as applicable.

Refer to the current durvalumab IB for a complete summary of preclinical and clinical information including safety, efficacy, and PK.

2.2.4 Ceralasertib in Combination with Durvalumab

Clinical data demonstrating the safety and anti-tumour activity of the combination of ceralasertib and durvalumab in patients with NSCLC comes from two ongoing AZ-sponsored studies, Study 4 (D5330C00004) Module 3 and HUDSON Study (D6185C00001); and one ongoing externally-sponsored scientific research, MATRIX Study (D6180C00001).

As of the latest data cut of the ceralasertib IB, there are a total of 161 patients that have been treated with ceralasertib in combination with durvalumab in NSCLC.

Clinical efficacy data from Study 4 is immature as the study is ongoing. However, there is preliminary evidence of efficacy for the combination of ceralasertib and durvalumab in four patients in the dose-escalation part of the study; these four patients achieved a confirmed RECIST response (n = 1 confirmed CR and n = 3 PRs). Data from the MATRIX study is not available at this time as this study is ongoing, however data from the HUDSON study is available and is presented in Section 2.2.

In addition to clinical data, preclinical data supporting the use of the ceralasertib and durvalumab combination for the treatment of NSCLC can also be derived from literature, in vitro data, and in vivo data. Durvalumab is a human mAb that binds to PD-L1 and blocks the interaction of PD-L1 with PD-1 and CD80. Durvalumab antagonises the immune inhibitory effects of PD-L1 on T-cells, resulting in restored effector T-cell functions and tumour growth inhibition via T-cell dependent mechanisms. Pharmacological inhibition of ATR is expected to show anti-tumour activity by (i) direct growth suppression of tumour cells through exacerbation of replication stress and DNA damage, (ii) subsequent signalling (priming) of innate and adaptive T-cell responses and (iii) remodelling of the tumour microenvironment to a more immune permissive state. Emerging evidence suggests ceralasertib delays tumour growth while reducing terminally differentiated and exhausted T-cells when on treatment yet maintains naïve or effector T-cells and promotes lymphocytic infiltration when off treatment which together would make tumours susceptible (or re-sensitise) to T-cell-mediated immune responses. The addition of immune checkpoint blockade such as durvalumab would promote and fully activate effector T-cell functions to maximise the potential anti-tumour activity and durability of responses.

Preclinically, ceralasertib shows selective growth inhibition activity with enhanced sensitivity enriched in subsets of NSCLC models which contain ATM-DDR defects (~5-10% NSCLC), KRAS/STK11 mutations (~7-13% NSCLC) and/or high DNA replication stress signatures. This is consistent with a tumour-based ‘synthetic lethality’ concept where there is a specific

dependency on ATR in the repair and survival of cancer cells undergoing high DNA replication stress (associated with molecular aberrations noted above), but not in other cancer or normal cells with low DNA replication stress ([Cimprich et al 2008](#), [Foote et al 2018](#), [Forment and O'Connor 2018](#)). In addition, ATR inhibition has been associated more broadly (not limited to high replication stress cancers) with increased or priming of the innate and adaptive immune responses. Preclinical studies have shown ATR inhibition induces micronuclei, activation of cystolic DNA sensing cGAS-STING-type I IFN pathways ([Parkes et al 2017](#), [Sato et al 2017](#), [Sun et al 2018](#)) and remodelling of the tumour immune micro-environment through promotion of CD8+ effector T-cell infiltration while inhibiting immunosuppressive cells ([Dillon et al 2019](#), [Sheng et al 2020](#), [Vendetti et al 2018](#)). Furthermore, ceralasertib plus anti-PD-L1 combination has shown improved anti-tumour efficacy over monotherapies alone in immunocompetent mouse tumour models (Appendix B of ceralasertib IB, data on file). Finally, clinical translational studies with ceralasertib have indicated treatment-related modulation of T-cell populations and other immune effector cells, including, NK cells in patients. Therefore, we hypothesised that ceralasertib has the potential to make tumours more susceptible or (re-)sensitise immunotherapy resistant (primary or acquired) tumours to T-cell-mediated immune responses.

Clinical experience of the combination of ceralasertib and durvalumab is summarised in the ceralasertib IB.

2.3 Benefit/Risk Assessment

2.3.1 Risk Assessment

Data from non-clinical studies and emerging data from the clinical development programme of ceralasertib shows that ceralasertib as monotherapy or in combination with durvalumab has a manageable safety profile in an advanced cancer population. There were no risks identified that would preclude investigation of ceralasertib as monotherapy or in combination with durvalumab in advanced cancers.

Identified risks for ceralasertib include anaemia, neutropenia (including febrile neutropenia), thrombocytopenia, nausea, vomiting and diarrhoea, which have been reported during the clinical use of ceralasertib as monotherapy or in combination with olaparib or durvalumab. The emerging safety profile of ceralasertib either as monotherapy or in combination with either olaparib or durvalumab is consistent with preclinical findings and/or the safety profile of the respective combination agent. Detailed information on the identified and potential risks, including the most commonly observed AEs, for ceralasertib when used alone or in combination are provided in the ceralasertib IB.

Risks with durvalumab include, but are not limited to, diarrhoea/colitis, pneumonitis/ILD, endocrinopathies (ie, events of hypophysitis/hypopituitarism, adrenal insufficiency, hyper and hypothyroidism, type I diabetes mellitus which may present with diabetic ketoacidosis and

diabetes insipidus), hepatitis/increases in transaminases, nephritis/increases in creatinine, rash/dermatitis (including pemphigoid), myocarditis, myositis/polymyositis, immune thrombocytopenia, infusion-related reactions, pancreatitis, encephalitis, hypersensitivity reactions, serious infections, and other rare or less frequent inflammatory events including neuromuscular toxicities (eg, Guillain-Barré syndrome, myasthenia gravis).

In monotherapy clinical studies with durvalumab, AEs at an incidence $\geq 20\%$ include events, such as fatigue, cough, decreased appetite, dyspnoea, and nausea. Approximately 10% of patients discontinued the drug due to an AE. The majority of treatment-related AEs were manageable with dose delays, symptomatic treatment, and in the case of events suspected to have an immune basis, the use of established treatment guidelines for immune-mediated toxicity.

A detailed summary of durvalumab monotherapy AE data, including AEs, SAEs, and CTCAE Grades 3 to 5 events reported across the durvalumab program can be found in the current version of the durvalumab IB.

Table 3 Risk Assessment

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Ceralasertib		
Currently identified risks for ceralasertib treatment include anaemia, neutropenia thrombocytopenia, nausea, vomiting, and diarrhoea.	For updates on the emerging safety data and a detailed description of all identified and potential risks for ceralasertib please refer to the latest ceralasertib IB.	Routine monitoring of haematology and biochemistry blood counts is planned. Dose interruptions and reductions of ceralasertib will be permitted (Section 6.6.1).
Durvalumab		
Risks with durvalumab include, but are not limited to, diarrhoea/colitis, pneumonitis/interstitial lung disease (ILD), endocrinopathies (ie, events of hypophysitis/hypopituitarism, adrenal insufficiency, hyper- and hypothyroidism, type I diabetes mellitus and diabetes insipidus), hepatitis/increases in transaminases, nephritis/increases in creatinine, rash/dermatitis (including pemphigoid), myocarditis, myositis/polymyositis, immune thrombocytopenia, infusion-related reactions, hypersensitivity reactions, pancreatitis, serious infections, and other rare or less frequent inflammatory events including neuromuscular toxicities (eg, Guillain-Barré syndrome, myasthenia gravis).	In monotherapy clinical studies, AEs at an incidence of $\geq 20\%$ include events such as fatigue and decreased appetite. Approximately 10% of patients discontinued the drug due to an AE. Please see the current version of the IB for a detailed summary of the monotherapy data including AEs, SAEs, and CTCAE Grade 3 to 5 events reported across the durvalumab programme.	Refer to “Dosing Modification and Toxicity Management Guidelines for Durvalumab Monotherapy or in Combination with Other Products” as provided to the investigative site as an Annex document, which is maintained within the Site Master File.

For updates on the emerging safety data and a detailed description of all potential risks for ceralasertib and durvalumab are available in the latest ceralasertib and durvalumab IBs.

2.3.2 Benefit Assessment

As reviewed in Section 2.2, the combination of durvalumab plus ceralasertib has the potential to significantly improve outcomes for patients with advanced or metastatic NSCLC.

Participants in this study will benefit from the monitoring of all AEs arising during the clinical study, related or not related to the study treatment. In addition, they will be contributing to the

development of new therapies for advanced cancer.

The study rationale and dose justification are summarised in Section 2.1 and Section 4.3, respectively.

2.3.3 Overall Benefit: Risk Conclusion

Taking into account the measures taken to minimise risk to participants enrolling in this study, the potential risks identified in association with ceralasertib in combination with durvalumab are justified by the anticipated benefits that may be afforded to patients with advanced NSCLC.

Further details can be found in the most recent versions of the ceralasertib and durvalumab IBs.

3 OBJECTIVES AND ENDPOINTS

Table 4 Objectives and Endpoints

Objectives	Estimands/Endpoints
Primary	
To demonstrate superiority of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of OS in participants with advanced NSCLC after second- or third-line therapy and without actionable genomic alterations	OS is defined as time from randomisation until the date of death due to any cause. The comparison will include all randomised participants, regardless of whether the participant withdraws from randomised therapy or receives another anti-cancer therapy. The measure of interest is the HR of OS.
Secondary	
To demonstrate superiority of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of PFS	PFS is defined as time from randomisation until progression per RECIST 1.1 as assessed by the investigator, or until death due to any cause. The analysis will include all randomised participants, as randomised, regardless of whether the participant withdraws from randomised therapy, receives another anti-cancer therapy or clinically progresses prior to RECIST 1.1 progression. However, if the participant progresses or dies immediately after 2 or more consecutive missed visits, the participant will be censored at the time of the latest evaluable assessment prior to the two missed visits. The measure of interest is the HR of PFS.

Objectives	Estimands/Endpoints
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of ORR	ORR is defined as the proportion of participant who have a CR or PR per RECIST 1.1 as determined by the investigator. The analysis will include all randomised participants with measurable disease at baseline. Data obtained from randomisation up until progression, or the last evaluable assessment in the absence of progression, will be included in the assessment of ORR, regardless of whether the participant withdraws from therapy. Participants who discontinue treatment without a response or progression, receive a subsequent therapy, and then respond will not be included as responders in the ORR. The measure of interest is the OR of the ORR.
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of DoR	DoR is defined as the time from the date of first documented response until date of documented progression per RECIST 1.1 as assessed by the investigator, or until death due to any cause. The analysis will include all randomised participants with measurable disease at baseline who have a response, regardless of whether the participant withdraws from therapy, receives another anti-cancer therapy or clinically progresses prior to RECIST 1.1 progression. The measure of interest is the median of DoR.

Objectives	Estimands/Endpoints
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of TTR	TTR is defined as the time from randomisation until the date of first documented objective response per RECIST 1.1 as assessed by the investigator. The analysis will include all randomised participants with measurable disease at baseline who have a response, as randomised. Data obtained from randomisation until progression, or the last evaluable assessment in the absence of progression, will be included in the assessment of TTR. Participants who discontinue treatment without a response or progression, receive a subsequent therapy, and then respond will not be included. The measure of interest is number and proportion (%) of participants with TTR, analysed as number of responders by scheduled assessment timepoint at which response was first observed.
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of DCR at 18 weeks	DCR at 18 weeks is defined as the percentage of participants who have a CR or PR or who have SD for at least 17 weeks per RECIST 1.1 as assessed by the investigator. The analysis will include all randomised participants with measurable disease at baseline. Data obtained from randomisation until progression, or the last evaluable assessment in the absence of progression, will be included in the assessment of DCR. The measure of interest is the OR of DCR at 18 weeks.

Objectives	Estimands/Endpoints
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of PFS2	Time from randomisation to PFS2 will be defined as the time from the randomisation to the earliest of the progression event (following the initial progression), subsequent to first subsequent therapy or death. The date of PFS2 will be recorded by the investigator in the eCRF and defined according to local standard clinical practice. The comparison will include all randomised participants regardless of whether the participant withdraws from subsequent therapy and regardless of missed visits. The measure of interest is the HR of PFS2.
To estimate the effectiveness of ceralasertib plus durvalumab combination therapy relative to docetaxel by assessment of OS12	OS is defined in the same way as the primary endpoint. The comparison will include all randomised participants, regardless of whether the participant withdraws from randomised therapy or receives another anti-cancer therapy. The measure of interest is the landmark OS at 12 months after randomisation.
To assess participant-reported health-related QoL	TTD is defined as the time from randomisation until the date of first confirmed deterioration. Change from baseline and TTD of health-related QoL as measured by the two global QoL items from the EORTC QLQ-C30. The analysis will include all randomised participants. The measure of interest will be mean, mean change from baseline, and HR of TTD of items 29 and 30 from the EORTC QLQ-C30.

Objectives	Estimands/Endpoints
To assess participant-reported physical functioning in participants treated with ceralasertib plus durvalumab combination therapy relative to docetaxel	TTD in physical functioning as measured by the EORTC QLQ-C30 Physical Function subscale of the EORTC QLQ-C30. The analysis will include all randomised participants. The measure of interest will be the HR of TTD of Physical Function subscale score.
To evaluate participant-reported treatment tolerability	Participant-reported tolerability will be described among participants as treated, using the following outcomes: • Symptomatic AEs: Proportion of participants experiencing each level of symptomatic AEs as measured by the PRO-CTCAE • Overall side effect bother: Proportion of participants reporting overall side effect bother on the PGI-TT while on treatment The analysis will include all treated participants (safety analysis set). The measure of interest will be the proportion of participants reporting each level of overall tolerability.
To assess the PK of ceralasertib when administered in combination with durvalumab	Plasma concentration of ceralasertib and sparse PK parameters such as peak concentration and trough, as data allow.
To assess the safety and tolerability of ceralasertib plus durvalumab combination therapy as compared with docetaxel	Safety and tolerability will be evaluated in terms of TEAEs, vital signs, clinical laboratory results, and ECGs.

Objectives	Estimands/Endpoints
Tertiary	
To assess pulmonary symptoms in participants treated with ceralasertib plus durvalumab combination therapy compared with docetaxel	TTD in pulmonary symptoms as measured by the NSCLC-SAQ. • TTD in three core pulmonary symptoms of dyspnoea, cough, and chest pain • TTD in overall NSCLC-SAQ score The analysis will include all randomised participants. The measure of interest is the HR of TTD in core lung cancer symptoms and NSCLC-SAQ total scores.
To assess participant-reported functioning	Change from baseline and TTD in functioning as measured by the modified EORTC QLQ-C30 (EORTC IL 172) including role functioning, emotional functioning, cognitive functioning, social functioning, and modified physical functioning scale (EORTC IL171). The analysis will include all randomised participants. The measure of interest will be mean, mean change from baseline, and HR of TTD in EORTC IL 172 and IL 171 scores.
To assess participant-reported global overall impression of severity of cancer symptoms and change in condition since starting the study	Descriptive summary of the proportion of participants reporting each severity level of cancer symptoms (PGI-S) and change in condition since study start (PGI-C). The analysis will include all randomised participants. The measure of interest will be mean and mean change from baseline in PGI-S and PGI-C scores.

Objectives	Estimands/Endpoints
To explore the impact of the study intervention and disease on participant-reported health status	VAS mean score and change from baseline and 5-dimension scores as measured by the EQ-5D-5L. The analysis will include all randomised participants. The measure of interest will be mean and mean change from baseline of VAS score and proportion of participants reporting different levels of each 5-dimension score.
To collect blood and tissue samples, or utilise residual samples, and medical imaging for diagnostic development and the analysis of tumoural and peripheral biomarkers that may influence the progression of cancer, identify participants more likely to benefit from durvalumab plus ceralasertib combination treatment, or may be surrogate markers of drug response or activity	Exploratory biomarker analysis that may include (but is not limited to) medical imaging, the presence of or changes in the levels of RNA, DNA, and ctDNA; somatic mutations; tumour mutational burden; gene and protein expression profiles; the number, phenotype, and expression profile of immune cells such as T-cells, HLA allotypes; and diversity of immune cell receptor repertoires.
To explore the impact of treatment and disease on medical resource utilisation	Number and type of hospitalisations, reason for hospitalisation, hospital attendances, procedures undertaken, and length of hospital stay, based on the HOSPAD module.
Optional samples: To collect and store DNA from tissue and/or blood according to each country's local and ethical procedures for future diagnostic development and exploratory research into genes and genetic variation that may influence response (including distribution, efficacy, pharmacodynamics, tolerability, or safety) to study treatments (ceralasertib as monotherapy or in combination with durvalumab) and/or to disease susceptibility	Correlation of polymorphisms with variation in PK, pharmacodynamics, safety, or treatment response parameters and/or susceptibility to disease may also be analysed.

4 STUDY DESIGN

4.1 Overall Design

This is a Phase III, randomised, open-label, two-arm, multicentre, international study assessing the efficacy and safety of ceralasertib plus durvalumab combination therapy compared with docetaxel in patients with advanced or metastatic NSCLC without actionable genomic alterations and whose disease has progressed on prior anti-PD-(L)1 therapies and platinum-based chemotherapy.

Approximately 580 participants will be randomised in a 1:1 ratio to one of the two treatment groups:

- **Group A:** Ceralasertib plus durvalumab combination therapy

Each 28-day cycle will begin with ceralasertib (oral tablets; 240 mg BID) from Day 1 to Day 7 followed by durvalumab (IV; 1500 mg) on Day 8 (approximately 290 participants planned)

- **Group B:** Docetaxel monotherapy

Each 21-day cycle will begin with docetaxel (IV; 75 mg/m² BSA) on Day 1 (approximately 290 participants planned)

The study is planned to be conducted in approximately 200 sites across approximately 21 countries. A recruitment period of approximately 18 months is anticipated.

Randomisation will be stratified by histology (squamous versus non-squamous), resistance to prior anti-PD-(L)1 therapy (primary versus acquired), tumour cell PD-L1 expression level (< 1% versus ≥ 1%), and geographical region (North America and Western Europe versus RoW).

4.1.1 Study Conduct Mitigation During Study Disruptions Due to Cases of Civil Crisis, Natural Disaster, or Public Health Crisis

The guidance given below supersedes instructions provided elsewhere in this CSP and should be implemented only during cases of civil crisis, natural disaster or public health crisis (eg, during quarantines and resulting site closures, regional travel restrictions and considerations if site personnel or study participants become infected with SARS-CoV-2 or similar pandemic infection), which would prevent the conduct of study-related activities at study sites, thereby compromising the study site staff or the participant's ability to conduct the study. The investigator or designee should contact the study sponsor to discuss whether the mitigation plans below should be implemented.

To ensure continuity of the clinical study during a civil crisis, natural disaster, or public health crisis, changes may be implemented to ensure the safety of study participants, maintain

compliance with GCP, and minimise risks to study integrity. Where allowable by local health authorities, ethics committees, health care provider guidelines (eg, hospital policies) or local government, these changes may include the following options:

- Obtaining consent/reconsent for the mitigation procedures (note, in the case of verbal consent/reconsent, the ICF should be signed at the participant's next contact with the study site).
- Rescreening: Additional rescreening for screen failure and to confirm eligibility to participate in the clinical study can be performed once in previously screened participants. The investigator should confirm this with the designated study clinical lead.
- Home or Remote visit: Performed by a site qualified HCP or HCP provided by a TPV.
- Telemedicine visit: Remote contact with the participants using telecommunications technology including phone calls, virtual or video visits, and mobile health devices.

For further details on study conduct during civil crisis, natural disaster, or public health crisis, refer to [Appendix J](#).

4.2 Scientific Rationale for Study Design

There is still a significant unmet medical need for treatment options other than docetaxel for patients with metastatic NSCLC whose disease has progressed on immune checkpoint and platinum doublet therapies.

In this study, the superiority of combination of ceralasertib and durvalumab over docetaxel, (current SoC) will be studied in a population of patients with advanced or metastatic NSCLC refractory to prior anti-PD-(L)1 therapies and platinum-based chemotherapy and without actionable genomic alterations. Current cancer treatments for advanced or metastatic NSCLC, except those provided as part of the study, are prohibited in order to ensure that efficacy and safety observations in this study can be more reliably attributed to the study treatments, or the disease under study, and will not be confounded by other cancer treatments. WHO PS is restricted to 0 or 1 to ensure that participants are of sufficient quality of health to participate in the study and benefit from study treatment.

4.3 Justification for Dose

4.3.1 Ceralasertib and Durvalumab Dose and Treatment Regimen Justification

The ceralasertib plus durvalumab combination therapy doses and regimen selected for this study are based on the goal of selecting an optimal combination dose that would demonstrate promising efficacy and have an acceptable safety profile.

The durvalumab dose and regimen selected for this study is the currently approved dosing

regimen.

Ceralasertib combination with durvalumab 1500 mg q28d is being tested in Module 3 of the ongoing Study 4 (Study D5330C00004) where patients with NSCLC and squamous cell cancer of the head and neck have been treated. The combination is also being tested in the HUDSON study (Study D6185C00001) in patients with NSCLC (clinical data presented in Section 2.1).

The RP2D for ceralasertib and durvalumab combination was declared as 240 mg BID ceralasertib 14-day schedule and 1500 mg durvalumab, q28d (Study D5330C00004). The platform, Phase 2 study, HUDSON explored the option of treating patients with ceralasertib on a 14-day schedule with durvalumab as per the RP2D and at an alternative schedule with ceralasertib 240 mg BID for 7 days followed by durvalumab. The latter schedule demonstrated a clear efficacy benefit (survival benefit) and a tolerable safety profile (refer to Section 2.1 for further details).

Therefore, the proposed dose and schedule for this study is ceralasertib 240 mg BID, Days 1 to 7 and durvalumab 1500 mg, Day 8, q28d.

4.3.2 Justification for Ceralasertib Dose

The dose of ceralasertib selected for this study (240 mg BID for 7 days in a 28 day cycle) is primarily based on the clinical results observed in the HUDSON study (refer to Section 2.2), with supportive data on preclinical, translational and clinical pharmacology data/modelling from Study 4 (preliminary PK-PD report). Preliminary PK observations from Study 4 showed less than dose proportional increase in exposure, where the plasma concentrations following a single dose at 320 mg OD were very similar to that observed at 240 mg OD (> 90% overlap in AUC), suggesting the need to change to BID schedules in order to increase ceralasertib exposure beyond the dose of 240 mg OD. This data also indicated that doses above 240 mg may not provide additional clinical benefit as exposure of ceralasertib in plasma is expected to be similar at doses above 240 mg.

Ceralasertib 240 mg BID for 7 days in 28-day cycles is based on clinical safety and tolerability (mainly bone marrow safety outcomes) and clinical PK/pharmacodynamic data. The dose level of 240 mg BID is predicted to maintain ceralasertib concentrations above the estimated concentration of an inhibitor where ATR catalytic activity is reduced by 90% (IC₉₀) threshold for 24 hours in > 95% of the patients based on simulations from a preliminary population PK model, and hence, it achieves maximum IC₉₀ daily coverage. The dosing schedule of 240 mg BID for 7 days is predicted to maximize the percentage of patients (70%) having an optimal cycle cover of 182 hours, defined as the time in which plasma concentrations of ceralasertib are maintained above the in vitro IC₉₀ per cycle. The target of 182 hours is based on preclinical data and is considered optimal to achieve maximum tumour

regression under ceralasertib monotherapy. A lower dose of 160 mg BID 7 day is predicted to reduce the optimal cover over IC90 per cycle in 10% of patients compared to 240 mg BID 7-day dosing (preliminary PK-PD report). In addition, 1 week ceralasertib dosing maximises cytotoxic T-cell (CD8+/Ki67+) proliferative burst (by minimising the time in which proliferative T-cells are depleted) during the three-week off period, potentially promoting upregulation of PD-L1 which can be subsequently blocked with durvalumab to enhance anti-tumour T-cell responses.

Considering emerging PK data from ongoing studies, there is no evidence to indicate any ethnic sensitivity in Asian patients compared to Western patients.

The choice of ceralasertib schedule is supported by preclinical data in syngeneic mouse tumour models in which the combination treatment showed improved anti-tumour efficacy compared to monotherapy, and an intermittent dosing schedule for ceralasertib is most efficacious in terms of balancing the direct anti-tumour effects, the potential for priming an IFN-mediated immune response and dynamic modulation of the tumour immune microenvironment (see ceralasertib IB). An intermittent schedule of ceralasertib is also supported by a population PK/PD model describing ceralasertib-induced thrombocytopenia and neutropenia which showed that a two-week off treatment period is needed for platelet/neutrophil recovery. This same model also demonstrated that haematological toxicity is reduced with a 7-day schedule compared to 14-day schedule of ceralasertib (preliminary PK-PD report).

The immunomodulatory effects of ceralasertib have been evaluated in multiple clinical studies. Longitudinal peripheral blood analysis from Study 4 and Study 7 (D5330C00007) demonstrated on-drug suppression of peripheral proliferating cytotoxic T-cells (CD8+/Ki67+) during ceralasertib treatment, followed by off-drug rebound above pre-treatment levels. Similar changes were not seen in total T-cells (proliferating cytotoxic T-cells comprise a subset of the total T-cells), suggesting that these effects are primarily restricted to proliferating cells.

Refer to the current ceralasertib IB for a complete summary of clinical information including safety, efficacy, and PK at the oral dose of 240 mg BID for D1-7 of q28d regimen.

4.3.3 Justification for Durvalumab Dose

A population PK model was developed for durvalumab using monotherapy data Study 1108 (N = 292; doses = 0.1 to 10 mg/kg q2w or 15 mg/kg q3w; solid tumours). Population PK analysis indicated only minor impact of body weight on the PK of durvalumab (coefficient of ≤ 0.5). The impact of body weight-based (10 mg/kg q2w) and fixed dosing (750 mg q2w) of durvalumab was evaluated by comparing predicted steady state PK concentrations (5th, median and 95th percentiles) using the population PK model. A fixed dose of 750 mg was

selected to approximate 10 mg/kg (based on median body weight of ~75 kg). A total of 1000 patients were simulated using body weight distribution of 40 to 120 kg. Simulation results demonstrate that body weight-based and fixed dosing regimens yield similar median steady state PK concentrations with slightly less overall between-patient variability with fixed dosing regimen.

Similar findings have been reported by others ([Narwal et al 2013](#); [Ng et al 2006](#), [Wang et al 2009](#), [Zhang et al 2012](#)). Wang and colleagues investigated 12 mAbs and found that fixed and body size-based dosing perform similarly, with fixed dosing being better for 7 of 12 antibodies (Wang et al 2009). In addition, they investigated 18 therapeutic proteins and peptides and showed that fixed dosing performed better for 12 of 18 in terms of reducing the between-patient variability in PK/pharmacodynamic parameters ([Zhang et al 2012](#)).

A fixed dosing approach is preferred by the prescribing community due to ease of use and reduced dosing errors. Given the expectation of similar PK exposure and variability, AstraZeneca considered it feasible to switch to fixed dosing regimens.

A fixed dose of 1500 mg durvalumab administered q28d is to be used for all patients with a body weight greater than 30 kg. Currently, the use of a fixed dose of 1500 mg durvalumab, administered both in combination with chemotherapy and as monotherapy, is approved for treatment of extensive stage small cell lung cancer; additionally, the 1500 mg fixed dose is approved in some countries as monotherapy for unresectable Stage III NSCLC.

Refer to the current durvalumab IB for a complete summary of clinical information including safety, efficacy, and PK at the 20 mg/kg q28d IV regimen.

4.4 End of Study Definition

A participant is considered to have completed the study if he/she has completed all phases of the study, including the last visit or the last scheduled procedure shown in the SoA (including OS determination).

The study may be stopped if, in the judgement of AstraZeneca, study participants are placed at undue risk because of clinically significant findings.

The end of the study is defined as the date of the last visit of the last participant in the study or last scheduled procedure shown in the SoA for the last participant in the study globally (including OS determination).

See Section [6.7](#) for details on participant management following the final DCO as well as following study completion.

5 STUDY POPULATION

The target population of interest in this study is adult patients with NSCLC without actionable genomic alterations and who have progressed after anti-PD-(L)1 therapy and platinum-based chemotherapy.

Prospective approval of protocol deviations to recruitment and enrolment criteria, also known as protocol waivers or exemptions, is not permitted.

Patients who do not meet the eligibility criteria requirements are screen failures; refer to Section 5.4.

5.1 Inclusion Criteria

Patients are eligible to be included in the study only if all of the following criteria apply.

Age

- 1 Participant must be ≥ 18 years at the time of screening.

Type of Participant and Disease Characteristics

- 2 Histologically or cytologically documented NSCLC that is locally advanced or metastatic according to Version 8 of the IASLC Staging Manual in Thoracic Oncology, at the time of study enrolment. Participants with unknown status are not allowed onto the study.
Note: A computed tomography or magnetic resonance imaging scan of the brain at baseline is required for all subjects.
- 3 Where available, a tissue sample obtained after progression on prior anti-PD-(L)1 therapy and ≤ 3 months prior to randomisation should be provided. Where no such sample is available a tumour sample taken ≤ 24 months prior to screening is acceptable. Samples should be of sufficient quality to enable assessment of tumour cell PD-L1 expression. Tumour sample must be FFPE, with sufficient material for sectioning preferably 16 slides (5 micron thickness). If FFPE blocks cannot be provided, then a set of newly cut unstained slides that enable necessary testing should be provided (preferably a minimum of 16 slides). Refer to Section 8.6.1 and the Laboratory Manual for details.
- 4 Documented tumour cell PD-L1 status as assessed by a central laboratory using the VENTANA SP263 PD-L1 IHC assay prior to randomisation. Participants with unknown PD-L1 status are not eligible for study.
- 5 Documented EGFR and ALK wild-type status as determined at a local laboratory using a well-validated, locally approved test. Participants with sensitising EGFR mutations (eg, exon 19 deletion; exon 21 L858R or L861Q; exon 18 G719X; or exon 20 insertion or S768I mutation) or ALK rearrangements are excluded from the study.

- 6 Tumours harbouring mutations in any of the following genes, if known, as determined by existing local test results: ROS1, RET, MET, BRAF V600, NTRK1 and NTRK2 are excluded.
- 7 Documented radiological PD whilst on or after receiving the most recent treatment regimen.
- 8 Eligible for second- or third-line therapy and must have received an anti-PD-(L)1 therapy and a platinum doublet containing therapy for locally advanced or metastatic NSCLC either separately or in combination. Prior durvalumab is acceptable. Refer to exclusion criterion 16 for the requirements for prior anti-PD-(L)1 therapy.
- 9 Received a minimum of 8 weeks of an anti-PD-(L)1 and at least 2 cycles of platinum-doublet containing regimen for locally advanced or metastatic NSCLC (either separately or in combination).
- 10 Participant must have had a treatment-free interval of ≥ 4 weeks from any prior therapy before the start of study intervention. In addition, the following intervals between the end of the prior treatment and first dose of study intervention must be observed:
 - (a) Minor surgical procedures (as defined by the investigator): 7 post-operative days
 - (b) Major surgery (as defined by the investigator): ≥ 4 weeks
 - (c) Radiotherapy: ≥ 4 weeks (participants who receive palliative radiation for non-target tumour lesions need not be subjected to this washout period and can be enrolled immediately).
- 11 ECOG/WHO performance status of 0 or 1 with no deterioration over the previous 2 weeks prior to baseline or day of first dosing.

At least 1 lesion, not previously irradiated, that qualifies as a RECIST 1.1 TL at baseline and can be accurately measured at baseline as ≥ 10 mm in the longest diameter (except lymph nodes, which must have short axis ≥ 15 mm) with CT or MRI and is suitable for accurate repeated measurements. A previously irradiated lesion can be considered a TL if the lesion has clearly progressed.
- 12 Adequate organ function and marrow reserve (within 7 days of C1D1) as follows:
 - (a) Haemoglobin ≥ 9.0 g/dL, with no blood transfusions (packed red blood cells) in the past 28 days
 - (b) Absolute neutrophil count $\geq 1.5 \times 10^9/\text{L}$ with no G-CSF administration in the past 28 days
 - (c) Platelet count $\geq 100 \times 10^9/\text{L}$, with no platelet transfusions in the past 28 days
 - (d) Total bilirubin (TBL) $\leq 1.5 \times \text{ULN}$ or TBL $\leq 3 \times \text{ULN}$ in the presence of documented Gilbert's syndrome (unconjugated hyperbilirubinemia). Note: Local practice guidelines and/or the local docetaxel label should be used to assess eligibility for the study.
 - (e) ALT and AST $\leq 2.5 \times \text{ULN}$; and ALT and AST $\leq 5 \times \text{ULN}$ in case of hepatic metastases
 - (f) Serum albumin > 30 g/L

- (g) Calculated CrCL > 40 mL/min as determined by Cockcroft-Gault (using actual body weight)

Creatinine clearance will be determined using the Cockcroft-Gault formula:

Males:

$$\text{CrCL (mL/min)} = \frac{\text{Body weight (kg)} \times (140 - \text{Age})}{72 \times \text{serum creatinine (mg/dL)}}$$

Females:

$$\text{CrCL (mL/min)} = \frac{\text{Body weight (kg)} \times (140 - \text{Age}) \times 0.85}{72 \times \text{serum creatinine (mg/dL)}}$$

- 13 Minimum life expectancy of 12 weeks.

Weight

- 14 Body weight > 30 kg and no cancer-associated cachexia, eg, CTCAE Grade 2 or worse weight loss over the past 3 months.

Sex

- 15 Male or female.

Contraceptive use by men or women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies.

Reproduction

- 16 Negative pregnancy test (serum test) for WOCBP.
- 17 Female participants must be 1 year post-menopausal, surgically sterile, or using one highly effective form of birth control (a highly effective method of contraception is defined as one that can achieve a failure rate of less than 1% per year when used consistently and correctly). Women of childbearing potential must agree to use one highly effective method of birth control. They should have been stable on their chosen method of birth control for a minimum of 3 months before entering the study and remain stable throughout the total duration of the study and for 3 months (Group A) or 6 months (Group B) after the last dose of study intervention. Non-sterilised male partners of a WOCBP must use a male condom plus spermicide (condom alone in countries where spermicides are not approved) throughout this period. Refer to [Appendix G](#) for details.
- 18 Male participants in Group A must use a condom (with spermicide) with all sexual partners throughout the total duration of the study and for 1 week after the last dose of study intervention. Male participants who intend to be sexually active with a WOCBP partner must be surgically sterile or agree to use one highly effective methods of contraception from the time of screening, throughout the total duration of the study and

for 6 months after the last dose of study intervention to prevent pregnancy in a partner. Male participants must not donate or bank sperm during this same time period. Refer to [Appendix G](#) for details.

Informed Consent

- 19 Capable of giving signed informed consent which includes compliance with the requirements and restrictions listed in the ICF and in this protocol.
- 20 Provision of signed and dated, written Optional Genetic Research Information informed consent prior to collection of sample for optional genetic research that supports Genomic Initiative.

5.2 Exclusion Criteria

Participants are excluded from the study if any of the following criteria apply.

Medical Conditions

- 1 Participant with mixed SCLC and NSCLC histology.
- 2 As judged by the investigator, any evidence of diseases (such as severe or uncontrolled systemic diseases, including uncontrolled hypertension, active bleeding diseases, active infection, active interstitial lung disease/pneumonitis, serious chronic gastrointestinal conditions associated with diarrhoea, psychiatric illness/social situations), history of allogenic organ transplant, which, in the investigator's opinion, makes it undesirable for the participant to participate in the study or that would jeopardise compliance with the protocol.
- 3 Refractory nausea and vomiting, chronic gastrointestinal disease, inability to swallow a formulated product, or previous significant bowel resection that would preclude adequate absorption, distribution, metabolism, or excretion of ceralasertib.
- 4 History of another primary malignancy except for malignancy treated with curative intent with no known active disease ≥ 5 years before the first dose of study intervention and of low potential risk for recurrence, basal cell carcinoma of the skin, squamous cell carcinoma of the skin or lentigo maligna that has undergone potentially curative therapy or adequately treated carcinoma in situ without evidence of disease.
- 5 Brain metastases or spinal cord compression unless the participant is stable (asymptomatic, no evidence of new or emerging brain metastases) and off steroids for at least 14 days prior to start of study treatment. Following radiotherapy and/or surgery, participants with brain metastases must wait 4 weeks following the intervention and must confirm stability with imaging before randomisation.
- 6 Persistent toxicities (CTCAE Grade > 2) caused by previous anticancer therapy; alopecia and vitiligo are excluded toxicities. Participants with Grade ≥ 2 neuropathy will be

evaluated on a case by-case basis after consultation with the study clinical lead.

Participants with irreversible toxicity that is not reasonably expected to be exacerbated by study interventions may be included (eg, hearing loss) after consultation with the AstraZeneca study clinical lead.

- 7 History of ILD/pneumonitis (non-infectious) that required management with steroids.
- 8 Active or prior documented autoimmune or inflammatory disorders (including inflammatory bowel disease [eg, colitis or Crohn's disease], diverticulitis [with the exception of diverticulosis], systemic lupus erythematosus, sarcoidosis, granulomatosis with polyangiitis, Graves' disease, rheumatoid arthritis, hypophysitis, uveitis, etc), autoimmune pneumonitis and autoimmune myocarditis). The following are exceptions to this criterion:
 - (a) Participants with vitiligo or alopecia
 - (b) Participants with hypothyroidism (eg, following Hashimoto syndrome) stable on hormone replacement
 - (c) Any chronic skin condition that does not require systemic therapy
 - (d) Participants without active disease in the last 5 years may be included but only after consultation with the study physician
 - (e) Participants with celiac disease controlled by diet alone
- 9 History of leptomeningeal carcinomatosis.
- 10 Known active hepatitis infection, positive HCV antibody, HBsAg or anti-HBc at screening. Participants with a past or resolved HBV infection (defined as the presence of anti-HBc and absence of HBsAg) are eligible. Participants positive for HCV antibody are eligible only if polymerase chain reaction is negative for HCV RNA.
- 11 Known to have tested positive for human immunodeficiency virus (HIV) (positive anti-HIV 1 or anti-HIV 2 antibodies) or active tuberculosis infection (clinical evaluation that may include clinical history, physical examination and radiographic findings, or tuberculosis testing in line with local practice).
- 12 Investigator judgement of one or more of the following:
 - (a) Mean resting corrected QT interval > 470 ms, obtained from triplicate ECGs performed at screening.
 - (b) History of QT prolongation associated with other medications that required discontinuation of that medication, or any current concomitant medication known to prolong the QT interval and cause Torsades de Pointes (TdP).
 - (c) Congenital long QT syndrome, family history of long QT syndrome, or unexplained sudden cardiac death under 40 years of age in first-degree relatives.
- 13 History of symptomatic congestive heart failure, unstable angina pectoris, uncontrolled cardiac arrhythmia (multifocal premature ventricular contractions, bigeminy, trigeminy, ventricular tachycardia), which is symptomatic or requires treatment (CTCAE Grade 3),

symptomatic or uncontrolled atrial fibrillation despite treatment, or asymptomatic sustained ventricular tachycardia. Participants with atrial fibrillation controlled by medication or arrhythmias controlled by pacemakers may be permitted upon discussion with the study clinical lead.

- 14 Known history of drug or alcohol abuse.
- 15 Diagnosis of ataxia telangiectasia.

Prior/Concomitant Therapy

- 16 Participants who have received more than one line of prior anti-PD-(L)1, either alone or in any combination, ie, patients who have received two or more prior lines of anti-PD-(L)1 therapy with a different anti-PD-(L)1 agent in the metastatic setting are excluded with the following exceptions:
 - (a) Patients who completed treatment with an anti-PD-(L)1 therapy for locally advanced disease, and then were subsequently treated with a different anti-PD-(L)1 therapy for metastatic disease, may be included if they had disease progression on the treatment.
 - (b) Patients who discontinued treatment with an anti-PD-(L)1 therapy for a reason other than disease progression or significant toxicity and have been retreated with the same anti-PD-(L)1 agent again, may be included if they had disease progression on the treatment.
- 17 Participants:
 - (a) Must not have experienced a toxicity that led to permanent discontinuation of the prior anti-PD(L)1 therapy.
 - (b) All AEs while receiving prior anti-PD(L)1 therapy must have completely resolved or resolved to baseline prior to screening for this study.
 - (c) Must not have experienced a Grade ≥ 3 imAE or an immune-related neurologic or ocular AE of any grade while receiving prior anti-PD(L)1 therapy. Note: Participants with an endocrine AE of Grade ≤ 2 are permitted to enrol if they are stably maintained on appropriate replacement therapy and are asymptomatic.
 - (d) Must not have required the use of additional immunosuppression other than corticosteroids for the management of an AE, not have experienced recurrence of an AE if re-challenged, and not currently require maintenance doses of > 10 mg prednisone or equivalent per day.
- 18 Participants who have received more than one prior line of platinum-based chemotherapy in metastatic setting.
- 19 Participants who have received a prior ATR inhibitor.
- 20 Participants who have received prior docetaxel.

- 21 Current or prior use of immunosuppressive medication within 14 days before the first dose of durvalumab. The following are exceptions to this criterion:
 - (a) Intranasal, inhaled, topical steroids, or local steroid injections (eg, intra-articular injection)
 - (b) Systemic corticosteroids at physiologic doses not to exceed 10 mg/day of prednisone or its equivalent
 - (c) Steroids as premedication for hypersensitivity reactions (eg, CT scan premedication)
- 22 Receipt of live attenuated vaccine within 30 days prior to the first dose of study intervention. Note: Participants, if enrolled, should not receive live vaccine whilst receiving study intervention and up to 180 days after the last dose of study intervention.

Prior/Concurrent Clinical Study Experience

- 23 Previous randomisation in the present study.
- 24 Participants with a known hypersensitivity to durvalumab, ceralasertib or docetaxel or any of the excipients of these products.
- 25 Involvement in the planning and/or conduct of the study (applies to both AstraZeneca staff and/or staff at the study site).
- 26 Judgement by the investigator that the participant should not participate in the study if the participant is unlikely to comply with study procedures, restrictions and requirements.

5.3 Lifestyle Considerations

The following restrictions apply while the participant is receiving study intervention and for the specified times before and after:

- 1 Participants must follow the contraception requirements outlined in [Appendix G](#).
- 2 Participants should not donate blood or blood components while participating in this study and through 90 days after receipt of the last dose of the study intervention or until alternate anti-cancer therapy is started.
- 3 During study intervention and for 4 weeks after the last dose of study intervention, participants should be advised to avoid prolonged exposure to the sun, wear protective clothing including a hat, and seek shade from the sun as much as possible; in addition, SP30+ sunscreen should be used. Exposure to other sources of ultraviolet light including sun beds and tanning booths, etc, should be avoided.

Restrictions relating to concomitant therapies are described in [Appendix I 2](#).

5.3.1 Meals and Dietary Restrictions

Participants must fast (water permitted) from at least 2 hours prior to taking a dose of

ceralasertib to at least 1-hour post-dose for all doses.

Participants should avoid concomitant drugs, herbal supplements and/or ingestion of foods known to modulate CYP3A4 activity ([Appendix I](#)).

5.4 Screen Failures

Screen failures are defined as participants who consent to participate in the clinical study but are not subsequently randomised to study intervention. A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the Consolidated Standards of Reporting Trials (CONSORT) publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, eligibility criteria and any SAE.

Individuals who do not meet the criteria for participation in this study (screen failure) may be rescreened a single time, but they must not be re-randomised. Rescreened participants should be assigned the same identification number (ie, E-code) as for initial screening. However, rescreening should be documented so that its effect on study results, if any, can be assessed. All assessments must be repeated for rescreening unless these are within 28 days of randomisation.

These participants should have the reason for study withdrawal recorded in eCRF as “eligibility criteria not fulfilled” (ie, participant does not meet the required inclusion/exclusion criteria). This reason for study withdrawal is only valid for screen failures (ie, participants who are not randomised in the study).

Participant enrolment and randomisation is described in [Section 6.3](#).

6 STUDY INTERVENTION

Study intervention is defined as any investigational intervention(s), marketed product(s), placebo, or medical device(s) intended to be administered to a study participant according to the CSP.

6.1 Study Interventions Administered

6.1.1 Investigational Products

AstraZeneca will supply ceralasertib and durvalumab as study interventions. Docetaxel (SoC agent) will either be locally sourced by the study site or centrally supplied by AstraZeneca.

Table 5 **Investigational Products**

Intervention name	Ceralasertib	Durvalumab	Docetaxel
Type	Synthetic	Biologic	Semi-synthetic
Dosage form	Tablets for oral administration	Concentrate solution for infusion	Concentrate solution for infusion
Unit strength	Tablets in either 80 or 120 mg	500 mg (50 mg/mL)	Either 20 mg (0.5 mL) or 80 mg (2 mL) or 160mg (4 mL)
Dosage regimen	q28d; 7 consecutive days; 240 mg BID	q28d; 1500 mg; or 20 mg/kg in participants who weigh ≤ 30 kg	q21d; 75 mg/m ² BSA
Route of administration	Oral	IV infusion	IV infusion
Use	Experimental	Experimental	Active comparator
IMP or NIMP	IMP	IMP	IMP
Sourcing	Provided centrally by the sponsor.	Provided centrally by the sponsor.	Provided centrally by the sponsor or locally by the study site, subsidiary, or designee.
Packaging and labelling	High-density polyethylene bottles with child-resistant closures. Labels will be prepared in accordance with GMP and local regulatory guidelines. The labels will fulfil GMP Annex 13 requirements for labelling. Label text will be translated into local language, as required.	Vials. Each vial will be labelled in accordance with GMP Annex 13 and per country regulatory requirement. Labels will be prepared in accordance with GMP and local regulatory guidelines. The labels will fulfil GMP Annex 13 requirements for labelling. Label text will be translated into local language, as required.	Glass vials in a commercial package. Where centrally provided, the labels will fulfil GMP Annex 13 requirements for labelling. Label text will be translated into local language, as required.

The treatment schedule for the two treatment groups will be as shown in [Table 6](#).

Table 6 Treatment Schedule

Treatment Group	Days of study intervention administration	
	D1 to D7	D8
Group A		
Durvalumab (IV), 1500 mg, q28d		X
Ceralasertib (oral, BID), 240 mg, BID	X	
Group B	D1	
Docetaxel (IV), 75 mg/m ² BSA; q21d	X	

6.1.1.1 Ceralasertib

Ceralasertib tablets will be supplied by AstraZeneca.

Participants in Group A will receive ceralasertib as per the schedule in [Table 1](#) and as described in [Table 6](#). Ceralasertib tablets should be taken with a glass of water. When ceralasertib is administered, participants must fast (water permitted) for at least 2 hours prior to taking a dose, to at least 1 hour post-dose for all doses.

Ceralasertib will be administered orally, 240 mg twice daily, approximately 12 ± 2 hours apart. If the interval between two doses is more than 14 hours, then the missed dose should not be taken, and participant should continue with the next dose at the allotted time. If participant wishes to bring forward the time of their scheduled dose, the dose can be taken up to a maximum of 2 hours prior to the scheduled time, ie, ± 2 hour window.

6.1.1.2 Durvalumab

Durvalumab will be supplied by AstraZeneca as a concentrate for solution for infusion. Durvalumab 500 mg will be supplied in a vial containing 50 mg/mL durvalumab, 26 mM histidine/histidine hydrochloride, 275 mM trehalose dihydrate and 0.02% weight/volume (w/v) polysorbate 80; the solution has a pH of 6.0 and density of 1.05 g/mL. The label-claim volume for durvalumab is 10 mL.

Durvalumab is a sterile, clear to opalescent, colourless to slightly yellow solution, free from visible particles.

Participants in Group A will receive durvalumab as per the schedule in [Table 1](#) and as described in [Table 6](#).

If a participant's weight falls to 30 kg or below (≤ 30 kg) the participant should receive weight-based dosing equivalent to 20 mg/kg of durvalumab q28d after consultation between investigator and study clinical lead, until the weight improves to > 30 kg, at which point the

participant should start receiving the fixed dosing of durvalumab 1500 mg q28d.

6.1.1.3 Docetaxel: Standard of Care (Comparator)

Participants in Group B will receive docetaxel per the schedule in [Table 2](#) and as described in [Table 6](#).

For docetaxel, refer to the local Prescribing Information with regard to warnings, precautions, and contraindications.

If injection-related symptoms occur, participants should be treated according to best medical practice and participants will be monitored until adequate resolution of signs and symptoms.

Dose reductions for docetaxel will be as per local guidelines and site practice. However, in general docetaxel should be administered when the neutrophil count is $\geq 1,500$ cells/mm³. In addition, patients who experienced either febrile neutropenia, neutrophil count < 500 cells/mm³ for more than one week, severe or cumulative cutaneous reactions or severe peripheral neuropathy during docetaxel therapy, the dose of docetaxel should be reduced from 75 to 60 mg/m². If the patient continues to experience these reactions at 60 mg/m², the treatment should be discontinued.

Premedication consisting of an oral corticosteroid, such as dexamethasone 16 mg per day (eg, 8 mg BID) for 3 days starting 1 day prior to docetaxel administration, unless contraindicated, can be used to reduce the incidence and severity of fluid retention as well as the severity of hypersensitivity reactions.

6.1.1.4 Duration of Treatment

All participants will receive the study interventions as described in [Table 6](#) until clinical progression, confirmed RECIST 1.1-defined radiological progression (refer to [Appendix F](#)), unacceptable toxicity, withdrawal of consent, or an intervention discontinuation criterion is met (for study discontinuation criteria, see [Section 7.1](#)). There is no maximum treatment duration defined.

Participants who are clinically stable may continue to receive durvalumab and/or ceralasertib beyond RECIST 1.1-defined PD as long as they are continuing to show clinical benefit, as judged by the investigator and sponsor, and in the absence of discontinuation criteria and if the following criteria are met:

- Absence of clinical symptoms or signs indicating clinically significant disease progression.
- No decline in performance status.
- Absence of rapid disease progression or threat to vital organs or critical anatomical sites (eg, central nervous system metastasis, respiratory failure due to tumour, tumour

compression, or spinal cord compression) requiring urgent alternative medical intervention.

- No significant, unacceptable, or irreversible toxicities related to study intervention.

6.2 Preparation/Handling/Storage/Accountability of Interventions

- 1 The investigator or designee (eg, pharmacist) must confirm appropriate temperature conditions have been maintained during transit for all study intervention received at the site and throughout the entire study duration until authorisation is provided for on-site destruction or removal of the study intervention, reflecting completion of the study. In the event of a temperature excursion detected at any time during the study, sites will follow the reporting procedures for notifying the sponsor (or designated party); release of study intervention for clinical use can only occur once the event has been reviewed and approval is provided by the sponsor (or designated party).
- 2 Only authorised site staff may prepare, dispense and administer study intervention. All study intervention must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labelled storage conditions with access limited to authorised site staff (and investigator, where applicable).
- 3 The investigator is responsible for study intervention accountability, reconciliation, and record maintenance (ie, receipt, reconciliation, and final disposition records reflecting destruction or return of all unused study intervention); this task may be delegated to study staff members identified on the site delegation log.
- 4 Further guidance and information for the final disposition of unused study interventions are provided in the Site Manual.

6.2.1 Storage of Study Interventions

6.2.1.1 Storage of Ceralasertib

Ceralasertib should be stored in the pack provided and used according to the product label instructions specified on the ceralasertib bottle.

6.2.1.2 Storage of Durvalumab

Durvalumab vials are stored at 2°C to 8°C (36°F to 46°F) and must not be frozen. The investigator, or an approved representative (eg, pharmacist), will ensure that all study intervention is stored in a secured area, in refrigerated temperatures (2°C to 8°C; 36°F to 46°F) and in accordance with applicable regulatory requirements. A temperature log will be used to record the temperature of the storage area. Temperature excursions outside the permissible range listed in the clinical supply packaging are to be reported to the monitor upon detection. A calibrated temperature-monitoring device will be used to record the temperature conditions in the drug storage facility. Storage conditions stated in the IB may be superseded by the label storage.

Study intervention must be kept in original packaging until time of preparation to prevent prolonged light exposure.

The dose of durvalumab must be prepared by the investigator's or site's designated study intervention manager using aseptic technique. Total time from needle puncture of the study intervention vial to the start of administration must not exceed:

- 24 hours at 2°C to 8°C (36°F to 46°F).
- 4 hours at room temperature.

If the final product is stored at both refrigerated and ambient temperatures, the total time must not exceed 24 hours otherwise a new dose must be prepared from new vials. Durvalumab and docetaxel vials do not contain preservatives; any unused portion of the vial must be discarded immediately after use.

6.2.1.3 Storage of Docetaxel

Docetaxel should be stored according to the local Prescribing Information.

6.2.2 Durvalumab Infusion Preparation and Administration

6.2.2.1 Preparation

A dose of 1500 mg for participants > 30 kg in weight will be prepared using an IV bag containing 0.9% sodium chloride for injection or 5% dextrose for injection, with a final durvalumab concentration ranging from 1 to 15 mg/mL. Add 30 mL (ie, 1500 mg) of durvalumab to the IV bag. The IV bag size should be selected such that the final concentration is within 1 to 15 mg/mL. Mix the bag by gently inverting to ensure homogeneity of the dose in the bag.

If participant weight falls to ≤ 30 kg, weight-based dosing at 20 mg/kg will be prepared using an IV bag size selected such that the final concentration is within 1 to 15 mg/mL.

6.2.2.2 Administration

Durvalumab infusions are to be administered through an IV administration set with a 0.2 or 0.22 μ m filter; acceptable configurations include an IV set containing an in-line filter or the attachment of a separate filter to the distal end of the IV tubing.

The durvalumab infusion time is 1 hour $\pm$ 10 minutes; however, if there are interruptions, the total allowed time must not exceed 8 hours with the infusion bag maintained at room temperature, otherwise a new dose must be prepared from new vials.

Other drugs should not be co-administered through the same infusion line.

The IV line will be flushed according to local practices to ensure the full dose is administered.

Infusion time does not include the final flush time.

6.2.3 Docetaxel Infusion Preparation and Administration

Docetaxel is administered as a one-hour infusion every three weeks.

Sites should consult the locally approved Prescribing Information or local treatment guidelines for docetaxel for complete Prescribing Information (including warnings, precautions, contraindications, and adverse reactions) and follow institutional procedures for the administration of docetaxel.

The SoC comparator, docetaxel, will either be locally sourced by the study site or centrally supplied by AstraZeneca. Under certain circumstances when local sourcing by the study site is not feasible, AstraZeneca will centrally supply the study intervention, which will be labelled and accompanied by Prescribing Information with local language translated text in accordance with regulatory guidelines.

Accountability and storage requirements, as specified in Sections 6.2.1 apply for any study intervention supplied by AstraZeneca.

6.3 Measures to Minimise Bias: Randomisation and Blinding

6.3.1 Participant Enrolment and Randomisation

This is an open-label study. Potential bias will be reduced by use of the following.

All participants will be centrally assigned to randomised study intervention using an IXRS/IRT. Before the study is initiated, the call/log-in directions and user guides for the IXRS/IRT will be provided to each site.

If a participant withdraws from participation in the study, then his/her enrolment code cannot be reused. Withdrawn participants will not be replaced.

Investigators should keep a record (ie, the participant screening log) of participants who entered screening.

Study intervention will be dispensed at the study visits as summarised in the SoAs. Returned study intervention should not be re-dispensed to the participants.

At screening/baseline (Days -28 to -1), the investigators or suitably trained delegate will:

- Obtain signed informed consent before any study-specific procedures are performed. If laboratory or imaging procedures were performed for alternate reasons prior to signing consent, these can be used for screening purposes with consent of the participant.

However, all screening laboratory and imaging results must have been obtained within 28 days of the first dose of study intervention.

- Participants will be identified to the IXRS/IRT per country regulations. Obtain a unique 7-digit enrolment number (E-code), through the IXRS/IRT in the following format (ECCNNXXX: CC being the country code, NN being the centre number, and XXX being the participant enrolment code at the centre). This number is the participant's unique identifier and is used to identify the participant on the eCRFs.
- Obtain sample and send for centralised PD-L1 testing. (Obtaining the sample should be given the highest priority and, as such, the sample may be obtained and sent for central testing prior to the 28-day screening window (after obtaining signed informed consent) in order to permit analysis prior to randomisation study intervention). Screening procedures may be obtained while PD-L1 status/expression is being assessed.
- Determine participant eligibility (see Sections 5.1 and 5.2).
- Obtain signed informed consent for genetic research study (optional). Participants who decide not to sign the specific genetic ICF but the general study ICF are eligible for study enrolment and all other study procedures.

At randomisation, once the participant is confirmed to be eligible, the investigator or suitably trained delegate will:

- Assign a randomised treatment group via the IXRS/IRT. The system will randomise the eligible participant to one of the two treatment groups. Randomisation codes will be assigned strictly sequentially within each stratum as participants become eligible for randomisation.

If the participant is ineligible and not randomised, the IXRS/IRT should be accessed to terminate the participant in the system.

Participants will begin treatment on Day 1. Treatment should start no more than 3 working days after being randomised. Participants must not be randomised and treated unless all eligibility criteria have been met.

6.3.2 Procedures for Handling Incorrectly Enrolled or Randomised Participants

Participants who fail to meet the eligibility criteria should not, under any circumstances, be enrolled or receive study medication. There can be no exceptions to this rule. Participants who are enrolled but subsequently found not to meet all the eligibility criteria must not be randomised or started on study intervention and must be withdrawn from the study.

Where a participant does not meet all the eligibility criteria but is randomised in error, or incorrectly started on treatment, the investigator should inform the AstraZeneca study clinical

lead immediately, and a discussion should occur between the AstraZeneca study clinical lead and the investigator regarding whether to continue or discontinue the participant from treatment. The AstraZeneca study clinical lead must ensure all decisions are appropriately documented and that the potential benefit/risk profile remains positive for the participant.

6.3.3 Methods for Assigning Treatment Groups

The actual treatment given to participants will be determined by the randomisation scheme in the IXRS/IRT. The randomisation scheme will be produced by a computer software programme that incorporates a standard procedure for generating randomisation numbers. One randomisation list will be produced for each of the randomisation strata. A blocked randomisation will be generated, and randomisation will be balanced within the IXRS/IRT at the stratum level. All centres will use the same list in order to minimise any imbalance in the number of participants assigned to each treatment group.

Randomisation codes will be assigned strictly sequentially, within each stratum as participants become eligible for randomisation. The IXRS/IRT will provide the kit identification number to be allocated to the participant at the randomisation visit and subsequent treatment visits.

6.3.4 Methods for Ensuring Blinding

This is an open-label study; however, the specific intervention to be taken by a participant will be assigned using an IXRS/IRT. The site will contact the IXRS/IRT prior to the start of study intervention administration for each participant. The site will record the intervention assignment on the applicable eCRF, if required.

In order to reduce bias in study conduct, data analysis and data interpretation, the study will be sponsor-blind. Sponsor personnel directly involved in the study conduct will not undertake or have access to aggregated data by treatment group until conclusion of the OS analysis. Detailed procedures for maintenance of the blind will be described in the study integrity plan.

6.4 Study Intervention Compliance

Sites will follow local practices for verification of study intervention doses, prepared and administered to study participants. The study interventions, durvalumab or docetaxel, will be prepared, dispensed and administered by the investigator or designee, under medical supervision. The date and time, of the administered study intervention will be recorded in the source documents and recorded in the eCRF. The dose of study intervention and study participant identification will be confirmed at the time of dosing in accordance with local treatment verification practices.

Ceralasertib will be self-administered by the participants at home. Participants will record their ceralasertib dosing in a participant diary which should be returned to the site at the next available visit. Additionally, compliance will be assessed at each visit by direct questioning,

counting returned tablets during the site visits, and documented in the source documents and eCRF. A record of the number of ceralasertib tablets dispensed to and taken by each participant must be maintained and reconciled with study intervention and compliance records. Intervention start and stop dates, including dates for intervention delays and/or dose reductions will also be recorded in the eCRF. Deviations from the prescribed dosage regimen should be recorded in the eCRF. The investigator is responsible for ensuring that the participant has returned all unused study intervention.

6.5 Concomitant Therapy

Any concomitant treatment, procedure, or other medication considered necessary by the investigator for the participant's safety and wellbeing, or vaccine (including over-the-counter or prescription medicines, vitamins, and/or herbal supplements) or other specific categories of interest) that the participant is receiving from the time of screening or receives during the study including the 90-day follow-up period following the last dose of study intervention must be recorded in the eCRF along with:

- Reason for use
- Dates of administration including start and end dates
- Dosage information including dose and frequency

The study clinical lead should be contacted if there are any questions regarding concomitant or prior therapy.

If any concomitant therapy is administered due to new or unresolved AE, it should be recorded.

Participants must be instructed not to take any medications, including over-the-counter products, without first consulting with the investigator.

Restricted, prohibited, and permitted concomitant medications/therapies are described in more detail in Appendix [I 2](#).

Guidance regarding potential interactions with concomitant medications is provided in Appendix [I 1](#).

6.5.1 Ceralasertib and Durvalumab Drug-drug Interactions

There is no information to date on drug-drug interactions of ceralasertib with durvalumab either preclinically or in participants.

As durvalumab is a mAb and therefore a protein, it will be degraded to small peptides and amino acids and will be eliminated by renal and reticuloendothelial clearance (CL). It is

therefore not expected that durvalumab will induce or inhibit the major drug metabolising CYP pathways. As a result, there are no expected PK drug-drug interactions. The mechanism of action of durvalumab involves binding to PD-L1, and therefore significant pharmacodynamic drug interactions with commonly administered concomitant medications are not expected.

Guidance regarding potential interactions with concomitant medications is provided in Appendix I 1.

6.5.2 Effect of Ceralasertib on Other Drugs

Ceralasertib is an investigational drug for which no data on in vivo interactions are currently available. Potential interaction and guidelines below are considered on the basis of preclinical in vitro data only.

Avoid concomitant medications, herbal supplements and/or ingestion of foods that significantly modulate CYP3A4 or P-gp activity (see guidelines below). Note: These include common azole antifungals, macrolide antibiotics, etc. In the absence of discontinuation criteria, if the investigator feels that concomitant administration of medications, herbal supplements or foods that significantly modulate CYP3A4 activity is necessary based upon medical judgement, such products may be administered with caution following discussion between the investigator and the study physician.

Concomitant medication may be given as medically indicated with the following exceptions:

- The principal enzyme for metabolising ceralasertib is CYP3A4. Participants should avoid concomitant drugs, herbal supplements, and/or ingestion of foods known to modulate CYP3A4 activity from the time they enter the screening period until 28 days after the last dose of study medication.
- Ceralasertib is a potential inducer of CYP3A4. Caution should be applied with co-administration of drugs that are either completely metabolised by CYP3A4, or that are CYP3A4 substrates and also have a narrow therapeutic index. Investigators should be aware that the exposure of other drugs metabolised by CYP3A4 may be reduced.
- Strong CYP3A4 inducers. For participants taking any of these drugs the required washout periods prior to starting ceralasertib is 2 weeks, except for St. John's Wort, which is 3 weeks.
- Prior to study medication, use of potent inducers or inhibitors of CYP3A are not permitted. For subjects taking any of these drugs, the required washout periods before starting ceralasertib is five half-lives; except for St. John's wort, which is 3 weeks.
- On study medication, if there is no suitable alternative concomitant medication other than a potent inhibitor of CYP3A, the investigator must interrupt ceralasertib for the duration

of the potent CYP3A inhibitor and wait for the required washout period (five half-lives) before dosing ceralasertib again. If potent CYP3A inducers are considered necessary for the participant's safety and welfare, this may diminish the clinical efficacy of ceralasertib, and the participant should be monitored carefully for any change in the efficacy of study treatment.

- If the use of any strong inducers or inhibitors of CYP3A4 are considered necessary for the participant's safety and welfare, the investigator must contact the study office and a decision to allow the participant to continue in the study will be made on a case-by-case basis. In the absence of discontinuation criteria, if the investigator feels that concomitant administration of medications, herbal supplements or foods that significantly modulate CYP3A4 activity is necessary based upon medical judgement, such products may be administered with caution following discussion between the investigator and the study physician.
- Ceralasertib is also a P-gp substrate. Co-administration of P-gp inhibitors or inducers may affect exposure to ceralasertib and therefore should not be co-administered with ceralasertib. If the use of any inhibitors or inducers of P-gp are considered necessary for the participant's safety and welfare, the investigator must contact the study office and a decision to allow the participant to continue in the study will be made on a case-by-case basis.
- Ceralasertib is a substrate of BCRP. Co-administration of BCRP inhibitors or inducers may affect exposure to ceralasertib; therefore, it is recommended that the investigators must interrupt ceralasertib for the duration of the BCRP inhibitor or inducer and wait for the required washout period of the BCRP modulator (five half-lives) before dosing ceralasertib again.
- Ceralasertib is an inhibitor of OATP1B1 and BCRP. Co-administration of substrates of OATP1B1 and/or BCRP may affect exposure to ceralasertib; therefore, it is recommended that caution should be applied when such drugs are to be administered with ceralasertib.
- The use of any natural/herbal products or other 'folk remedies' should be discouraged.
- Anticoagulation therapy: participants on warfarin may participate in this study but it is recommended that their INR is monitored more frequently.

6.5.3 Rescue Medicine

The study site will supply the required rescue medication and will be obtained locally. The following rescue medications are required to be available at the site:

- 1 Infliximab/infliximab biosimilar (eg, for colitis).
- 2 Mycophenolate (eg, for hepatitis).

Under certain circumstances when local sourcing by the study site is not feasible or local regulations prevent the use of infliximab or mycophenolate for this use (as they are considered off-label for management of immunotherapy related toxicities), AstraZeneca will centrally supply the required rescue medications, which will be labelled and accompanied by Prescribing Information with local language translated text in accordance with regulatory guidelines. Accountability and storage requirements, as specified in Section 6.2, apply for any study intervention supplied by AstraZeneca. If required to manage an imAE, then the IXRS/IRT will allocate the specific medication either by a kit or another identification number, to the pharmacist for a specific participant at the time of the event.

The date of rescue medication administration as well as the name and dosage regimen of the rescue medication must be recorded.

As a result of imAEs that could potentially be experienced by participants on durvalumab, appropriate treatment (eg, steroids and specific immunosuppressant rescue medications) must be made readily available to this participant population.

6.6 Dose Modification

6.6.1 Ceralasertib

If a participant experiences a clinically significant and/or unacceptable toxicity, dosing will be interrupted, and supportive therapy administered as required.

If the toxicity resolves or reverts to CTCAE Grade ≤ 1 or 2 (depending on the toxicity; see [Table 7](#)), treatment with ceralasertib may be restarted using the rules in [Table 7](#) and [Table 8](#) for dose modifications.

If the toxicity does not resolve to CTCAE Grade ≤ 1 or 2 or the participant is not showing clinical benefit, then the participant should be discontinued from treatment and observed until resolution of the toxicity.

Repeat dose interruptions are allowed as required for a maximum of 28 days on each occasion as recommended in [Table 7](#). If the duration of ceralasertib dose interruption is longer than 28 days, the case should be discussed with the study clinical lead.

In the instance ceralasertib dosing needs to be withheld while continuing durvalumab, the durvalumab dose should be administered at the scheduled time without delay, apart from the following situation: if ceralasertib toxicity has resolved by D8 of a cycle, in which case ceralasertib should be administered from D8 onwards for 7 days followed by durvalumab on D15.

Table 7 Ceralasertib Dose Interruption and Stopping Criteria

Event	Action
Grade 1 neutropenia and/or thrombocytopenia	No change to ceralasertib dosing is advised.
Grades 1 or 2 toxicities (except Grade 2 neutropenia and thrombocytopenia)	Ceralasertib dosing may continue with supportive treatment (as required) or investigator decision whether to interrupt ceralasertib (maximum 28 days). Following interruption, ceralasertib may be resumed at the same dose level.
Grade 2 neutropenia or Grade 3 anaemia	Blood counts may recover during the non-dosing days of the 28 day cycle. Ceralasertib dosing may continue with supportive treatment as required (eg, transfusion) or investigator may decide to interrupt ceralasertib (maximum 28 days). Following interruption, ceralasertib may be resumed at the same dose level or dose reduced by 1 level.
Grades 2 or 3 thrombocytopenia	First occurrence Interrupt ceralasertib only (maximum 28 days) and give appropriate supportive treatment, as needed, until platelet count improves to at least $\geq 75000/\text{mm}^3$. Following interruption, ceralasertib may be resumed at the same dose level, as blood counts may recover during the “off period” on the intermittent schedule. If blood counts do not recover by the start of the next dosing period, ceralasertib should be restarted with a dose reduction by 1 level. Subsequent occurrences Interrupt ceralasertib (maximum 28 days) and give appropriate supportive treatment, as needed, until platelet count improves to at least $\geq 75000/\text{mm}^3$. Treatment may be restarted with a dose reduction by 1 level.
Grade 4 thrombocytopenia	Interrupt ceralasertib (maximum 28 days) and give appropriate supportive treatment until the platelet count has recovered to $\geq 75000/\text{mm}^3$. Following interruption, ceralasertib should be restarted with a dose reduction by 1 level.
Grades 3 or 4 toxicity (except Grade 3 anaemia or Grades 3 or 4 thrombocytopenia)	First occurrence Interrupt ceralasertib (maximum 28 days) and give appropriate supportive treatment. When toxicity has resolved to Grade 1, ceralasertib should be restarted with a dose reduction by 1 level. Subsequent occurrences Investigator discretion on whether to interrupt ceralasertib (maximum 28 days) or to stop treatment. Restart treatment with a ceralasertib dose reduction of 1 or 2 levels.
Vomiting	If vomiting occurs shortly after ceralasertib is swallowed, the dose should only be replaced if all of the intact tablets can be counted. Resume with the following scheduled dose.
Missed dose	Allowed to take the scheduled dose up to 2 hours after the scheduled dose time. If greater than 2 hours, the missed dose should not be taken, and the participant should continue with next dose at the allotted time.

This table is for guidance. Therefore, for example, it may be deemed appropriate by the investigator to reduce the dose by more than 1 dose level depending on the individual participant circumstances.

Table 8 Ceralasertib Dose Reduction Levels

Dose level	Ceralasertib (Days 1-7 in q28d)
Initial dose	240 mg BID
Level 1 dose reduction	160 mg BID
Level 2 dose reduction	160 mg OD
Level 3 dose reduction	Stop treatment

Ceralasertib dose interruption for conditions other than toxicity resolution should be kept as short as possible. If a participant cannot restart study treatment within 28 days for resolution of intercurrent conditions not related to disease progression or toxicity, the case should be discussed with the Study clinical lead.

Ceralasertib should be stopped at least 3 days prior to planned surgery. After surgery study treatment can be restarted when the wound has healed. No stoppage of study treatment is required for any biopsy procedure.

Ceralasertib dose must not be adjusted under any other circumstances unless prior agreement is given by the sponsor.

All dose modifications and interruptions (including any missed doses) and the reasons for the modifications/interruptions are to be recorded in the CRF.

Individual Stopping Criteria:

Hepatic

Elevations observed after initiation of study treatment and which cannot be explained by alternative etiology ie, baseline liver metastases.

- ALT or AST or ALP* $> 5 \times$ ULN
- ALT or AST or ALP* $> 3 \times$ ULN with the appearance of symptoms associated with a clinical diagnosis of hepatitis including right upper quadrant pain or tenderness, fever, rash or eosinophilia ($> 5\%$)
- ALT or AST $> 3 \times$ ULN and TBL $> 2 \times$ ULN or INR⁺ > 1.5 or other evidence of impairment to the synthesis function of the liver

* In the presence of bone metastasis, assess bone specific isoform of raised ALP in the presence of a raised gamma-GT (to ensure the ALP change is specific to the liver).

+ Unless participant is receiving warfarin.

Please refer to [Appendix E](#) for further details.

Cardiovascular

Clinically significant hypotension defined as an asymptomatic decrease of more than

20 mmHg in systolic blood pressure to below 70 mmHg persisting for at least 10 minutes.

Symptomatic orthostatic fall in systolic blood pressure of more than 20 mmHg compared to resting supine systolic blood pressure.

6.6.2 Durvalumab

Dose delays (interruptions) are permitted for durvalumab treatment; however, dose reduction is **not permitted** (see Dosing Modification and Toxicity Management Guidelines in the Annex document to this CSP).

Participants may delay dosing under certain circumstances.

- Dosing may be delayed per the Dosing Modification and Toxicity Management Guidelines, due to either an immune or a non-immune-related AE.
- If dosing must be delayed for reasons other than treatment-related toxicity, dosing will resume as soon as feasible.
- Dosing intervals of subsequent cycles may be shortened as clinically feasible in order to gradually align treatment cycles with the schedule of tumour efficacy (RECIST 1.1) and PRO assessments. Subsequent time between 2 consecutive doses cannot be less than 21 days, based on the half-life of durvalumab (see current durvalumab IB).

6.6.3 Docetaxel (SoC Comparator)

Please refer to locally approved Prescribing Information for guidance on docetaxel monitoring and management of toxicities. Dose adjustments of docetaxel should be made based on the specific types of toxicities observed, graded using NCI CTCAE v 5.0.

It is not recommended to administer docetaxel when the neutrophil count is $< 1,500$ cells/mL. In order to monitor the occurrence of neutropenia, which may be severe and result in infection, frequent blood cell counts should be performed on all participants receiving docetaxel.

It is not recommended to administer docetaxel to participants with serum bilirubin $> \text{ULN}$ and/or ALT and AST $> 3.5 \times \text{ULN}$ associated with ALP $> 6 \times \text{ULN}$. Bilirubin, AST, or ALT values should be obtained prior to each cycle of docetaxel.

In participants who experienced either febrile neutropenia, neutrophil count < 500 cells/mL for more than 1 week, severe or cumulative cutaneous reactions, or severe peripheral neuropathy attributed to docetaxel therapy, the dose of docetaxel should be reduced from 75 to 60 mg/m² BSA. If the participant continues to experience these reactions at 60 mg/m² BSA, the docetaxel treatment should be discontinued.

6.6.4 Management of Toxicities

The following general guidance should be followed for management of toxicities.

- Treat each of the toxicities with maximum supportive care (including holding the agent suspected of causing the toxicity if required).
- If the symptoms promptly resolve with supportive care, consideration should be given to continuing the same dose of the assigned study intervention along with appropriate continuing supportive care. If medically appropriate, dose modifications are permitted.
- All dose modifications should be documented with clear reasoning and documentation of the approach taken.

All toxicities will be graded according to NCI CTCAE, version 5.

Guidelines for the management of immune-mediated reactions, infusion-related reactions, and non-immune-mediated reactions for durvalumab monotherapy and durvalumab combination therapies are provided in the Dosing Modifications and Toxicity Management Guidelines. The most current version of these guidelines is to be maintained within the Site Master File.

6.6.4.1 Ceralasertib-related Toxicities

Refer to Section [6.6.1](#).

6.6.4.2 Durvalumab-related Toxicities

Comprehensive toxicity management guidelines have been developed to assist investigators with the recognition and management of toxicities associated with the use of the immune checkpoint inhibitor durvalumab (a PD-L1 inhibitor). These guidelines are applicable to the management of participants receiving durvalumab as monotherapy or in combination with other anticancer drugs (ie, antineoplastic chemotherapy, targeted agents) administered concurrently or sequentially as part of a protocol-specific treatment regimen. The toxicity management guidelines provide information for the management of immune-mediated reactions, infusion-related reactions, and non-immune-mediated reactions that may be observed with checkpoint inhibitor monotherapy or combination checkpoint inhibitor regimens, with specific instructions for dose modifications (including discontinuations) and treatment interventions. Investigators are advised however to use local practice guidelines and consult local references for the management of toxicities observed with other cancer treatment. The most current version of the toxicity management guidelines entitled “Dosing Modification and TMGs for Durvalumab Monotherapy or in Combination with Other Products” is provided to the investigative site as an Annex document and is maintained within the Site Master File.

Participants should be thoroughly evaluated, and appropriate efforts should be made to rule out neoplastic, infectious, metabolic, toxin, or other etiologic causes of the imAE. Serological,

immunological, and histological (biopsy) data, as appropriate, should be used to support an imAE diagnosis. In the absence of a clear alternative aetiology, events should be considered potentially immune-related. In addition, there are certain circumstances in which durvalumab plus ceralasertib should be permanently discontinued (see Section 7.1 of this protocol and the Dosing Modification and TMGs). These guidelines have been prepared by the sponsor to assist the investigator in the exercise of his/her clinical judgement in treating these types of toxicities. These guidelines apply to AEs considered causally related to durvalumab plus ceralasertib regimen by the reporting investigator.

6.6.4.3 Docetaxel-related Toxicities

For management of toxicities due to docetaxel, including dose and treatment modifications, refer to the locally approved Prescribing Information or manage in accordance with institutional guidelines.

6.7 Intervention After the End of the Study

As described in Section 4.4, the study will remain open until all participants have discontinued study intervention and completed their last expected visit/contact.

After the final DCO for this study, AstraZeneca will continue to supply open-label ceralasertib plus durvalumab combination therapy or monotherapies to participants until PD as judged by the investigator or until meeting any other discontinuation criteria as defined in Section 7.1.

Participants who continue to receive benefit from their assigned treatment at the final DCO and database closure may continue to receive their assigned treatment for as long as they and their physician considers they are gaining clinical benefit. For participants continuing to receive durvalumab and ceralasertib combination treatment following the final DCO and database closure, it is recommended that the participants continue the scheduled site visits and investigators monitor the participants' safety laboratory results prior to and periodically during the treatment in order to manage AEs in accordance with the durvalumab and ceralasertib Dosing Modification and TMGs (see the Annex document to this CSP).

In the event that a roll-over or safety extension study is available at the time of the final DCO and database closure, participants without confirmed PD during the period of treatment, and who are potentially eligible for retreatment with ceralasertib and durvalumab combination therapy may be transitioned to such a study, and the current study would reach its end. The roll-over or extension study would ensure treatment continuation with visit assessments per its protocol, as applicable. Any participant who would be proposed to move to such a study would be given a new informed consent, as applicable.

7 DISCONTINUATION OF STUDY INTERVENTION AND PARTICIPANT DISCONTINUATION/WITHDRAWAL

7.1 Discontinuation of Study Intervention

It may be necessary for a participant to permanently discontinue (definitive discontinuation) study intervention. If study intervention is permanently discontinued, the participant will remain in the study to be evaluated for safety follow-up. The investigator should instruct the participant to contact the site before or at the time if study intervention is stopped. A participant that decides to discontinue study intervention will always be asked about the reason(s) and the presence of any AEs. The reason for discontinuation should be documented in the source document and the appropriate section of the eCRF.

Participants who have permanently discontinued from further receipt of study intervention will need to be discontinued from the IXRS/IRT. All study intervention (ceralasertib) and study devices should be returned by the participant at their next on-site study visit or unscheduled visit.

Participants will be discontinued from study intervention in the following situations.

- Clinical or RECIST 1.1-defined radiological progression (refer to Section [8.1.1](#) and [Appendix F](#)).
- Investigator determination that the participant is no longer benefiting from study intervention.
- An AE that, in the opinion of the investigator or AstraZeneca, contraindicates further dosing.
- Any AE that meets criteria for discontinuation defined in the dose modification guidelines for management of study intervention-related toxicities (see the TMGs that is an Annex document to this CSP) or as defined in the local Prescribing Information for docetaxel.
- Participant decision. The participant is at any time free to discontinue treatment, without prejudice to further treatment. Participants who discontinue treatment are normally expected to continue to participate in the study (eg, for safety and survival follow-up) unless they specifically withdraw their consent to all further participation in any study procedures and assessments (see Section [7.2](#)).
- Severe non-compliance with the CSP as judged by the investigator or AstraZeneca.
- Pregnancy or intent to become pregnant. Initiation of subsequent anti-cancer therapy, including another investigational agent.

Note that discontinuation from study intervention is NOT the same thing as a withdrawal from the study.

Crossover within the study will not be permitted.

If participants in Group A discontinue treatment with either one of the combination agents due to toxicity, they may continue on monotherapy with the other drug as long as they are continuing to show clinical benefit, as judged by the investigator and sponsor, and in the absence of discontinuation criteria.

Participants who have discontinued study intervention prior to objective RECIST 1.1-defined radiological progression, regardless of whether or not they have commenced subsequent anti-cancer therapy, will be followed up with tumour assessments as indicated in the SoA until RECIST 1.1-defined PD or death regardless of whether or not the participant started a subsequent anti-cancer therapy, unless they have withdrawn all consent to study-related assessments.

See the SoA for data to be collected at the time of intervention discontinuation and follow-up and for any further evaluations that need to be completed.

7.1.1 Follow-up of Participants Post-discontinuation of Study Intervention

All participants who discontinue the study intervention will be followed up for safety assessments up to 90 days after their last dose of study intervention, or until initiation of another anti-cancer therapy, whichever is earlier. Additional assessments to be performed are detailed in the SoA.

Participants who have discontinued study intervention prior to objective RECIST 1.1-defined radiological progression, regardless of whether or not they have commenced subsequent anti-cancer therapy, will be followed up with tumour assessments as indicated in the SoA until RECIST 1.1-defined PD or death regardless of whether or not the participant started a subsequent anti-cancer therapy, unless they have withdrawn all consent to study-related assessments.

7.1.2 Follow-up of Participants to PFS2

Following objective progression, participants will have their subsequent progression status recorded every 3 months after PD per local standard clinical practice to assess PFS2. Assessments will be performed according to the local practice, and formal RECIST 1.1 measurements will not be collected for assessment of PFS2. See Section [8.1.4](#) for additional information.

7.1.3 Follow-up for Survival

Participants will be followed up for survival status as indicated in the SoA until death, withdrawal of consent, or the end of the study. Survival information may be obtained via telephone contact with the participant or the participant's family, or by contact with the participant's current physician. Additional assessments to be performed at the time of survival follow-up are detailed in the SoA.

7.2 Participant Withdrawal from the Study

- A participant may withdraw from the study at any time at his/her own request or may be withdrawn at any time at the discretion of the investigator for safety, behavioural, compliance, or administrative reasons. This is expected to be uncommon.
- A participant who considers withdrawing from the study must be informed by the investigator about modified follow-up options to ensure the collection of endpoints and safety information including new AEs and follow-up on any ongoing AEs and concomitant medications (eg, telephone contact at 30 days (+7 days) after study intervention is discontinued, a contact with a relative or treating physician, or information from medical records).
- If the participant withdraws consent for disclosure of future information, the sponsor may retain and continue to use any data collected before such a withdrawal of consent.
- If a participant withdraws from the study, it should be confirmed if he/she is still agrees for existing samples to be used in line with the original consent. If he/she requests withdrawal of consent for use of samples, destruction of any samples taken and not tested should be carried in line with what was stated in the informed consent and local regulation. The investigator must document the decision on use of existing samples in the site study records and inform the Global Study Team.

7.3 Lost to Follow-up

A participant will be considered lost to follow-up if he or she repeatedly fails to return for scheduled visits and no contact has been established by the time the study is completed (see Section 4.4), such that there is insufficient information to determine the participant's status at that time.

Participants who decline to continue participation in the study, including telephone contact, should be documented as “withdrawal of consent” rather than “lost to follow-up.”

Investigators should document attempts to re-establish contact with missing participants throughout the study period. If contact with a missing participant is re-established, the participant should not be considered lost to follow-up and evaluations should resume according to the protocol.

The following actions must be taken if a participant fails to return to the clinic for a required study visit:

- The site must attempt to contact the participant and reschedule the missed visit as soon as possible and counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether or not the participant wishes to and/or should continue in the study.

- Before a participant is deemed lost to follow-up, the investigator or designee must make every effort to regain contact with the participant (where possible, 3 telephone calls and, if necessary, a certified letter to the participant's last known mailing address or local equivalent methods). These contact attempts should be documented in the participant's medical record.
- Should the participant continue to be unreachable, he/she will be considered to have been lost to follow-up from the study.
- Site personnel, or an independent third-party, will attempt to collect the vital status of the participant during survival follow-up within legal and ethical boundaries for all participants randomised, including those who did not get study intervention. Public sources may be searched for vital status information. If vital status is determined as deceased, this will be documented, and the participant will not be considered lost to follow-up. Sponsor personnel will not be involved in any attempts to collect vital status information.

Discontinuation of specific sites or of the study as a whole are handled as part of [Appendix A](#).

In order to support key efficacy endpoints of OS and PFS analyses, the survival status of all participants in the FAS and the Safety Analysis Set should be re-checked; this includes those participants who withdrew consent or are classified as "lost to follow-up."

- Lost to follow-up – Site personnel should check hospital records and a publicly available death registry (if available), as well as checking with the participants' current physician, to obtain a current survival status (the applicable eCRF modules will be updated).
- In the event that the participant has actively withdrawn consent to the processing of their personal data, the survival status of the participant can be obtained by site personnel from publicly available death registries (if available) where it is possible to do so under applicable local laws to obtain a current survival status.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are summarised in the SoA. Data collection following study analysis until the end of the study is described below.

- Protocol waivers or exemptions are not allowed.
- Immediate safety concerns should be discussed with the sponsor immediately upon occurrence or awareness to determine if the participant should continue or discontinue study intervention.
- Adherence to the study design requirements, including those specified in the SoA, is essential and required for study conduct.
- All screening evaluations must be completed and reviewed to confirm that potential participants meet all eligibility criteria. The investigator will maintain a screening log to record details of all participants screened and to confirm eligibility or record reasons for screening failure, as applicable.
- Procedures conducted as part of the participant's routine clinical management (eg, blood count) and obtained before signing of the ICF may be utilised for screening or baseline purposes provided the procedures met the protocol-specified criteria and were performed within the time frame defined in the SoA.

Data Collection Following Study Analysis until the End of the Study

After the final DCO and database closure, only SAEs will be reported for the purposes of this study (see Section 8.3.11). Following the DCO for the final analysis, all participants who remain in the study will continue the scheduled “progression/survival follow-up” site visits indicated in the SoA.

8.1 Efficacy Assessments

8.1.1 Imaging Tumour Assessments

Tumour assessments will use images from IV contrast CT (preferred) or MRI (if CT is contraindicated), of the chest, abdomen, and pelvis (including liver and adrenal glands). Any other areas of disease involvement should be additionally imaged at screening based on known metastasis sites or by the signs and symptoms of individual participants. The main screening tumour assessments will be used as the baseline assessment. The imaging modality used for baseline tumour assessment, CT/MRI for chest and abdomen and MRI for brain, should be kept the same consistently at each subsequent assessment throughout the study, if possible.

It is important to follow the tumour assessment schedule as closely as possible (refer to the SoA). Screening/baseline imaging should be performed no more than 28 days before start of

study intervention and ideally should be performed as close as possible to and prior to the start of study intervention. Scans obtained as part of standard clinical practice, prior to informed consent, but within the 28-day period are acceptable.

Tumour assessments will continue throughout treatment until RECIST 1.1-defined radiological progression (see [Appendix F](#)). If an unscheduled assessment is performed (eg, to investigate clinical signs/symptoms of progression) and the participant has not progressed, every attempt should be made to perform the subsequent assessments at the next scheduled visit.

Tumour assessments should be performed every 9 ($\pm$ 1) weeks starting at C3D1 and continuing until objective PD as defined by RECIST 1.1 and confirmed with a subsequent scan (no earlier than 4 weeks after initial progressive disease assessment) in the absence of clinically significant deterioration as determined by the investigator. Scans at Week 27 and Week 36 should be done before durvalumab infusion. On PD, in case the study intervention is continued at the principal investigator's discretion (see Section [6.1.1.4](#)), then the scans should continue at the same frequency until the end-of-treatment with a follow-up scan after 6 ($\pm$ 2) weeks.

The RECIST 1.1 assessments of baseline images should identify TLs (defined as measurable) and NTLs. On-study images will evaluate TLs and NTLs chosen at baseline, and NLs when they appear. This allows determination of follow-up TL response, NTL lesion response, the presence of unequivocal NLs, and overall timepoint responses (CR, PR, SD, PD, or NE).

8.1.2 Central Reading of Scans

Images, including unscheduled visit scans, will be collected on an ongoing basis and sent to an AstraZeneca-appointed iCRO for quality control and storage. Guidelines for image acquisition, de-identification, storage of digital copies at the investigative site (as source documents), and transfer to the iCRO will be provided in the imaging acquisition guideline handling document. Digital copies of all original scans should be stored at the investigator site as source documents. Electronic image transfer from the sites to the iCRO is strongly encouraged.

8.1.3 Overall Survival

Assessments for survival (dead or alive; date of death) will be conducted every 3 months ($\pm$ 1 week) following objective PD or treatment discontinuation. Survival information may be obtained via telephone contact with the participant, participant's family, by contact with the participant's current physician, or local death registries as described in Section [7.2](#).

8.1.4 Time to Second Progression or Death

Following objective progression, participants will have their subsequent progression status

recorded every 3 months per local standard clinical practice to assess PFS2. A participant's PFS2 status is defined according to the local practice and may involve any of: objective radiological progression (preferred), symptomatic progression, or death. Scans will be performed according to the local practice and formal RECIST 1.1 measurements will not be collected for assessment of PFS2. The second progression event must have occurred during or after treatment with a subsequent treatment after the progression event used for PFS or death. The date of PFS2 assessment and investigator opinion of progression status (progressed or non-progressed) at each assessment will be recorded in the source documents and the eCRF.

8.1.5 Clinical Outcome Assessments

A COA is any assessment that may be influenced by human choices, judgement, or motivation and may support either direct or indirect evidence of treatment benefit.

The PRO assessment is one type of COA and is a general term referring to all outcomes and symptoms that are directly reported by the participant. PROs have become important in evaluating the efficacy and tolerability of study interventions in clinical studies and will aid in understanding of the benefit/risk evaluation ([Kluetz et al 2018](#)). The following PRO instruments will be administered in this study:

- NSCLC-SAQ
- EORTC IL 171 (Physical Function)
- EORTC IL 172 (modified EORTC QLQ C30)
- PRO-CTCAE
- Patient Global Impression of Treatment Tolerability (PGI-TT)
- Patient Global Impression of Severity (PGI-S)
- Patient Global Impression of Change (PGI-C)
- EuroQol 5 dimensions, 5 levels (EQ-5D-5L)

PRO questionnaires will be administered according to the SoA (see [Appendix H](#) for sample copies of each questionnaire). The PRO questionnaires will be completed by participants if a linguistically validated version is available in their language for the country in which they live.

8.1.5.1 NSCLC-SAQ

The NSCLC-SAQ ([McCarrier et al 2016](#)) is a 7-item PRO assessment developed to measure the following symptom domains: cough, pain, dyspnoea, fatigue, and appetite. All items are rated on a 5-level verbal rating scale using a recall period of 7 days. It was developed as part of the pre-competitive C-PATH Consortium NSCLC working group, a partnership between multiple sponsors to develop a fit-for-purpose PRO symptom assessment for use as a

secondary endpoint. The NSCLC-SAQ is currently pre-qualified for use as an exploratory endpoint and the Center for Drug Evaluation and Research has signalled their intent to qualify this instrument as a secondary endpoint once longitudinal data are available to inform meaningful change.

8.1.5.2 Physical Function Scale (EORTC IL 171)

The EORTC item library is an online platform comprised of individual items from multiple questionnaires. As the static questionnaires might not always be sufficient to meet the demands of quickly evolving treatment modalities, selecting items from the item library offers new opportunities to leverage existing items for capturing relevant concepts for a given study. Items from these libraries will be used in this study to assess physical function.

A total of 8 pre-selected items from the EORTC IL will be used, five of which make up the physical functioning scale of the EORTC QLQ-C30 ([Aronson et al 1993](#)).

The three additional EORTC IL items are “Do you have any trouble walking up a flight of stairs?”, “Do you have any trouble running a short distance, such as to catch the bus?”, and “Have you felt slowed down?”.

Participants can choose among four response options ranging from “Not at All to “Very Much”.

8.1.5.3 Modified EORTC QLQ-C30 (EORTC IL 172)

The EORTC QLQ-C30 was developed by the EORTC QoL Group. It consists of 30 items and measures symptoms, functioning, and global health status/QoL ([Aronson et al 1993](#)) for all cancer types.

In this trial, only relevant subscales of the EORTC QLQ-C30 will be used. Relevant subscales include role functioning (2 items), emotional functioning (4 items), cognitive functioning (2 items), and global health status (2 items). This modular approach is to reduce redundancy across PROs, therefore reducing participant burden. Subscales will be used in their entirety in order to maintain scoring conventions.

Financial difficulties will not be included due to the lack of relevance in a trial setting. Single items assessing symptoms will also not be assessed due to irrelevance or duplicate information being collected in the PRO-CTCAE and NSCLC-SAQ.

8.1.5.4 PRO-CTCAE

The PRO-CTCAE is included to address tolerability from the participants’ perspective. It was developed by the NCI. It was developed in recognition that collecting side effect data directly from participants using PRO tools can improve the accuracy and efficiency of symptomatic AE data collection. This was based on findings from multiple studies demonstrating that

physicians and nurses underestimate side effect onset, frequency, and severity in comparison with participant ratings ([Antonia et al 2014](#); [Litwin et al 1998](#); [Sprangers and Aaronson 1992](#)). These side effects have been converted to participant terms (eg, the CTCAE term “myalgia” has been converted to “aching muscles”). For several symptoms, like fatigue and pain, additional questions are asked about the frequency, severity, and interference with usual activities. The items included in the PRO-CTCAE have undergone extensive qualitative review among experts and participants. These items have been extensively evaluated by cancer participants to be clear, comprehensible, and measure the symptom of interest. In this study, only items that are considered relevant for the trial, site of cancer, and cancer treatment are selected.

The items pre-selected for this study include mouth or throat sores, nausea, vomiting, constipation, diarrhoea, rash, hair loss, and numbness or tingling in the hands or feet. These items are based on a review of the core symptom set from NCI, as well as expected treatment-related symptoms. The free text item in the PRO-CTCAE instrument is not included in the study, as the utility of this information and the analysis method have not been established.

8.1.5.5 PGI-TT

The PGI-TT item is included to assess how a participant perceives the overall burden of treatment-related side effects of cancer treatment over the past 7 days. Participants will be asked to choose the response that best describes the level of burden by the side effect of their cancer treatment over the past week. The response options are: “not at all”, “a little bit”, “somewhat”, “quite a bit”, and “very much”. This item is included to aid in the interpretation of other PRO measures and to evaluate the overall impact of treatment-related side effects.

8.1.5.6 PGI-S

The PGI-S item is included to assess how a participant perceives the overall severity of cancer symptoms over the past 7 days. Participants will be asked to choose the response that best describes the severity of their overall cancer symptoms over the past week. The response options are: “none”, “mild”, “moderate”, and “severe”. This item is included to aid in the interpretation of other PRO measures and to evaluate the overall impact of treatment on the global severity of cancer symptoms.

8.1.5.7 PGI-C

The PGI-C scale assesses how a participant perceives his/her overall change in health status since the start of study intervention. This is a single item questionnaire, and participants will choose among seven response options ranging from “Much Better” to “Much Worse”.

8.1.5.8 EQ-5D-5L

The EQ-5D-5L will be used to explore the impact of treatment and disease state on health state utility.

The EQ-5D-5L, developed by the EuroQoL Group, is a generic questionnaire that provides a simple descriptive profile of health and a single index value for health status for economic appraisal ([EuroQol 2019](#)). The EQ-5D-5L questionnaire comprises 6 questions that cover 5 dimensions of health (eg, mobility, self-care, usual activities, pain/discomfort, and anxiety/depression). Respondents also assess their health today using the EQ-VAS, which ranges from 0 (worst imaginable health) to 100 (best imaginable health).

8.1.5.9 Administration of Electronic PRO (ePRO) Questionnaires

Baseline PRO questions will be completed on site at C1D1, after informed consent and before dosing.

After C1D1, PRO questionnaires will be self-administered electronically at home by the participants using an application on a personal or provisioned handheld device at the timepoints indicated in the SoA. Participants should complete the ePROs prior to or at the site if the assessment timepoint coincides with a scheduled site visit. Alerts will remind participants when ePRO assessments should be completed.

Participants must be instructed to bring the device to all visits.

Each site must allocate the responsibility for the administration of the ePRO instruments to a specific individual (eg, a research nurse or study coordinator) and, if possible, assign a back-up person to cover if that individual is absent.

A web back-up may be available to answer the questionnaires if there are technical problems with the device.

Approximately 10 to 15 minutes is required for participants to complete all questionnaires administered remotely while on treatment; however, participants will complete assessments at their own pace. The amount of time necessary at each timepoint will vary based on the specific questionnaires administered.

The below instructions should be followed when collecting PRO data:

- The research nurse or appointed site staff should explain to participants the value and relevance of these data so they are motivated to comply with questionnaire completion. Inform the participant that these questions are being asked to find out, directly from them, how they feel.
- It is vital that the ePRO reporting is initiated at C1D1 (baseline) as specified in the SoA and prior to the first dose of study intervention and any other study procedures to capture the effect of study intervention. The ePRO device must be charged and fully functional at the beginning of the baseline visit to ensure that the PROs can be completed at the start of the visit.

- The participant should bring the ePRO device to each site visit so the research nurse or appointed site staff can check if there are available PRO questionnaires to be completed and that the device is functioning properly.
- PRO questionnaires completed at sites must be completed prior to treatment administration performed at the site and ideally before any discussions of health status to avoid biasing the participant's responses to the questions. As feasible, site staff should also ensure PRO questionnaires are completed prior to other study procedures, such as collection of laboratory samples, to further minimise bias.
- On completion of the questionnaire at the site, the device should be handed back to the research nurse or appointed staff nurse, who should check that all questionnaires were completed.
- Study sites must ensure that a participant completes baseline questionnaires before revealing the treatment arm allocated to the participant.
- When each instrument is due to be completed, the following order must be observed:
 - NSCLC-SAQ
 - EORTC IL 171 (physical function)
 - EORTC IL 172
 - PRO-CTCAE
 - PGI-TT
 - PGI-S
 - PGI-C
 - EQ-5D-5L
- PRO questionnaires should be completed by the participant in a quiet and private location.
- The participant should be given sufficient time to complete the PRO questionnaires at their own speed.
- The research nurse or appointed site staff should stress that the information is not routinely shared with study staff. Therefore, if the participant has any medical problems, he/she should discuss them with the doctor or research nurse separately from the ePRO assessment.
- The research nurse or appointed site staff must train the participant on how to use the ePRO device using the materials and training provided by the ePRO vendor.
- The research nurse or appointed site staff must provide guidance on whom to call if there are problems with the device when the participant is completing the ePRO at home.

- All PRO questionnaires are to be completed using an ePRO device. If technical or other device-related issues prohibit completion on the device, an appropriate back-up option may be considered with prior approval from AstraZeneca.
- The research nurse or appointed site staff must remind participants that there are no right or wrong answers and avoid introducing bias by not clarifying items.
- The participant must not receive help from relatives, friends, or clinic staff deciding on answers to the PRO questionnaires. The responses are the participant's alone.
- If a participant uses visual aids (eg, glasses or contact lenses) for reading, it is important that these are brought to the baseline visit so questionnaires can be completed before dosing. If the participant does not have them when he or she attends the site visit, the participant may be exempted from completing the PRO questionnaires at that site visit but should be encouraged to complete them as soon as possible at home via device or web.
- Site staff must not read or complete the PRO questionnaires on behalf of the participant. If the participant is unable to read the questionnaire (eg, is blind, illiterate or not fluent in the available language), that participant is exempted from completing PRO questionnaires but may still participate in the study. If the participant cannot complete the PRO questionnaires due to reasons other than being blind, illiterate, or fluent in language, the AstraZeneca study team must be contacted to determine if they can be exempted. Participants exempted in this regard should be flagged appropriately by the site staff in the source documents and in the designated eCRF.
- Questions must not be translated from an available language in the device into the language the participant speaks.
- Reminders should be provided to participants, as needed, to ensure compliance with the assessment schedules.
- The research nurse or appointed site staff must monitor compliance since minimising missing data is a key aspect of study success.
- Finally, the research nurse or appointed site staff will review the completion status of questionnaires during site visits and document the reason(s) why a participant could not complete assessments, in the source documents and in the designated eCRF. If the site receives an email notification regarding the participant's compliance, appropriate action will be taken (eg, discussion with participant to improve compliance, a check in call from the site to ask the participant if they have any difficulties in completing questionnaires on schedule, etc). A solution to enhance/resolve compliance should be discussed with the participant. Discussions and compliance review should be reflected in source documents.

8.2 Safety Assessments

Planned timepoints for all safety assessments are provided in the SoAs.

8.2.1 Physical Examinations

A complete physical examination will be performed and will include assessments of the following; general appearance, respiratory, cardiovascular, abdomen; skin, head and neck (including ears, eyes, nose and throat), lymph nodes, thyroid, musculoskeletal (including spine and extremities), urogenital, dermatological, gastrointestinal, endocrine, haematologic/lymphatic, and neurological systems. Targeted physical examinations are to be used by the investigator on the basis of clinical observations and symptomatology.

Physical examination, as well as assessment of height and weight, will be performed at timelines as specified in the SoA; investigators should pay special attention to clinical signs related to previous serious illnesses, new or worsening abnormalities may qualify as AEs, see Section 8.3.5 for details.

8.2.2 Vital Signs

Vital signs will be performed at timelines as specified in the SoA and will include the following:

- Temperature: oral, tympanic, rectal, axillary, skin, or temporal artery per the standard local practice
- Pulse rate
- Respiratory rate
- Blood pressure

Blood pressure and pulse measurements will be assessed with a completely automated device. Manual techniques will be used only if an automated device is not available.

Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the participant in a quiet setting without distractions (eg, television, cell phones).

Vital signs (to be taken before blood collection for laboratory tests) will consist of one pulse and three blood pressure measurements (3 consecutive blood pressure readings will be recorded at intervals of at least 1 minute). The average of the three blood pressure readings will be recorded on the eCRF.

Situations in which vital signs results should be reported as AEs are described in Section 8.3.5.

For any AEs of infusion reactions, the vital signs values should be entered into the eCRF.

8.2.2.1 Vital Signs at First Infusion (Group A and Group B)

On the first infusion day, participants in Group A (ceralasertib plus durvalumab combination

therapy group) and the Group B (docetaxel group) will be monitored, and vital signs collected and recorded in the eCRF prior to, during, and after infusion of study intervention as presented in the bulleted list below.

Blood pressure and pulse measured before, during, and after each infusion at the following times (based on a 60-minute infusion):

- Prior to the beginning of the infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion])
- Approximately 30 minutes during the infusion (halfway through infusion)
- At the end of the infusion (approximately 60 minutes $\pm$ 5 minutes)

If the infusion takes longer than 60 minutes, then blood pressure and pulse measurements should follow the principles as described above or be taken more frequently if clinically indicated.

- A 1-hour observation period is recommended after the first infusion of durvalumab of durvalumab. If no clinically significant infusion reactions are observed during or after the first cycle, subsequent infusion observation periods can be at the investigator's discretion (suggested 30 minutes after each durvalumab infusion).

Additional monitoring with assessment of vital signs will be at the discretion of the investigator per standard clinical practice or as clinically indicated.

8.2.2.2 Vital Signs at Subsequent Infusions (Group A and Group B)

Blood pressure, pulse and other vital signs should be measured, collected/recorded in eCRF prior to the start of the infusion. Participants should be carefully monitored, and blood pressure and other vital signs should be measured during and post infusion as per institution standard and as clinically indicated. Any clinically significant changes in vital signs should be entered onto an unscheduled vital signs eCRF page.

8.2.2.3 Vital Signs for Docetaxel (Group B)

Participants receiving the SoC, docetaxel, should have vital signs performed prior to the start of administration, as clinically indicated, and according to local practice.

8.2.3 Electrocardiograms

Triplicate 12-lead ECGs will be done at screening, and single ECG readings will be collected at other visits as clinically indicated. The participant should rest in a semi-supine position for at least 5 minutes before and also during the recording. The ECG machine will automatically calculate the heart rate and measures PR, QRS, QT, and QTcF intervals.

All ECGs should be assessed by the investigator as to whether they are clinically significantly abnormal. Any clinically significant abnormalities detected including a QTcF value > 470 ms require triplicate ECG results. For triplicate ECGs, 3 individual ECG tracings should be obtained in succession, no more than 2 minutes apart. The full set of triplicates should be completed within 5 minutes.

Situations in which ECG results should be reported as AEs are described in Section [8.3.5](#).

8.2.4 Clinical Safety Laboratory Assessments

Blood and urine samples for determination of clinical chemistry, haematology, coagulation, and urinalysis will be taken at the visits indicated in the SoA.

Additional safety samples may be collected if clinically indicated at the discretion of the investigator. The date, time of collection and results (values, units and reference ranges) will be recorded on the appropriate eCRF.

The clinical chemistry, haematology including coagulation and urinalysis will be performed at a local laboratory at or near to the investigator site. Sample tubes and sample sizes may vary depending on laboratory method used and routine practice at the site.

Other safety laboratory tests include assessment for pregnancy (serum at screening or urine at other timepoints). Pregnancy tests may be performed at the site using a licensed test (urine or serum pregnancy test). Female participants who are WOCBP are required to have a pregnancy test within 7 days prior to the first dose of study intervention and then as clinically indicated. Pregnancy test may occur on Day 1, but results must be available and reviewed by the treating physician or investigator prior to commencing an infusion. Abnormal clinically significant laboratory results should be repeated as soon as possible (preferably within 24 to 48 hours).

The following laboratory variables will be measured.

Table 9 Laboratory Safety Variables

Haematology/Haemostasis (whole blood)	Clinical chemistry (serum or plasma)
Leukocyte count (total white cell count) ^a	Albumin
Absolute lymphocyte count ^a	Alkaline phosphatase ^c
Absolute neutrophil count ^a	ALT ^c
Haemoglobin	Amylase
Platelet count	AST ^c
Coagulation	Bicarbonate ^b
PTT and INR ^b	Bilirubin, total ^c
	Calcium, total or calcium corrected (for albumin)
Urinalysis	Chloride ^b
Colour and appearance	CRP
pH and specific gravity	Creatinine
Blood ^e	CrCL ^{b, d}
Bilirubin	Gamma-glutamyl transferase ^b
Glucose	Glucose (random)
Ketones	Lactate dehydrogenase
Protein	Lipase ^f
	Magnesium ^b
Virology (serum)	Potassium
HBsAg ^h , HCV	Protein, total
HIV antibodies	Sodium
	TSH
	T3 free (reflex) ^g
Pregnancy	T4 free (reflex) ^g
β-hCG in serum (pre-/screening) or urine (all other visits); only in WOCBP	Urea nitrogen/blood urea nitrogen, depending on local practice

- a Can be recorded as absolute counts or as percentages. Therefore, total white cell count has to be provided.
- b Bicarbonate (where available), chloride, creatinine clearance, gamma glutamyltransferase, INR, magnesium and PTT testing are to be performed at baseline, on Day 1 (unless all screening laboratory clinical chemistry and haematology assessments are performed within 3 days prior to Day 1), and if clinically indicated.
- c Tests for ALT, AST, alkaline phosphatase, and total bilirubin must be conducted and assessed concurrently. If total bilirubin is $\geq 2 \times$ upper limit of normal (and no evidence of Gilbert's syndrome), then fractionate into direct and indirect bilirubin.
- d Creatinine clearance to be calculated using Cockcroft-Gault equation (Cockcroft and Gault 1976).
- e For urinalysis, microscopy is preferred to investigate white blood cells, with use of high-power field; dipstick can be used as well.
- f It is preferable that both amylase and lipase parameters are assessed. For sites where only one of these parameters is routinely measured, either lipase or amylase is acceptable.

- g Free T3 or free T4 will only be measured if TSH is abnormal or if there is a clinical suspicion of an AE related to the endocrine system.
- h If participant HCV-Ag positive, should be checked for HCV RNA.

The investigator should assess the available results with regard to clinically relevant abnormalities in documentation. Any clinically significant abnormal laboratory values should be repeated as clinically indicated and recorded on the eCRF. Situations in which laboratory safety results should be reported as AEs are described in Section 8.3.5.

All participants with Grade 3 or 4 laboratory values at the time of completion or discontinuation from study intervention must be followed and have further tests performed until the laboratory values have returned to Grade 1 or 2 unless these values are not likely to improve because of the underlying disease.

NB. In case a participant shows an AST or ALT $\geq 3 \times$ ULN together with TBL $\geq 2 \times$ ULN please refer to [Appendix E](#) for further instructions.

8.2.5 Other Safety Assessments

For Group A participants, if on Day 8 of any cycle when the participants visit for the durvalumab infusion, if the participants have haematological or GI toxicity of CTCAE Grade ≥ 3 then the participant should have additional visits for a safety blood sample on D15 of that cycle and on D1 of the next cycle.

8.2.5.1 Pneumonitis (ILD) Investigation

If new or worsening pulmonary symptoms (eg, dyspnoea) or radiological abnormality suggestive of pneumonitis/ILD is observed, toxicity management as described in detail in the Dosing Modification and TMGs (see the Annex document to this CSP) will be applied. The results of the full diagnostic workup (including high-resolution CT, blood and sputum culture, haematological parameters, etc.) will be captured in the eCRF. It is strongly recommended to perform a full diagnostic workup, to exclude alternative causes such as lymphangitic carcinomatosis, infection, allergy, cardiogenic oedema, or pulmonary haemorrhage. In the presence of confirmatory high-resolution CT scans where other causes of respiratory symptoms have been excluded, a diagnosis of pneumonitis (ILD) should be considered and the Dosing Modification and TMGs (see the Annex document to this CSP) should be followed.

The following assessments, and additional assessments if required, will be performed to enhance the investigation and diagnosis of potential cases of pneumonitis. The results of the assessment will be collected.

- Physical examination
 - Signs and symptoms (cough, shortness of breath, and pyrexia, etc) including auscultation for lung field will be assessed.
 - SpO₂.
- Other items
 - When pneumonitis (ILD) is suspected during study intervention, the following markers should be measured where possible:
 - ILD Markers (KL-6, SP-D) and β -D-glucan.
 - Tumour markers: Particular tumour markers that are related to disease progression.
 - Additional clinical chemistry: C-reactive protein and lactate dehydrogenase.

8.2.5.2 ECOG/WHO Performance Status

The participant's performance status will be assessed at screening using the ECOG/WHO Performance Status scale. Participants must have an ECOG/WHO performance status 0 to 1 to be eligible for enrolment.

These scales and criteria are used by doctors and researchers to assess how a participant's disease is progressing, assess how the disease affects the daily living abilities of the participant, and determine appropriate treatment and prognosis ([Oken et al 1982](#), see [Table 10](#)). Any significant change from baseline or screening must be reported as an AE.

Table 10 Eastern Cooperative Oncology Group Performance Status

Grade	Performance
0	Fully active; able to carry out all usual activities without restrictions.
1	Restricted in strenuous activity, but ambulatory and able to carry out light work or work of a sedentary nature (eg, light housework or office work).
2	Ambulatory and capable of self-care, but unable to carry out any work activities; up and about more than 50% of waking hours.
3	Capable of only limited self-care; confined to bed or chair more than 50% of waking hours.
4	Completely disabled; unable to carry out any self-care and totally confined to bed or chair.
5	Dead.

8.3 Adverse Events and Serious Adverse Events

The principal investigator is responsible for ensuring that all staff involved in the study are familiar with the content of this section.

The definitions of an AE or SAE can be found in [Appendix B](#).

Adverse events will be reported by the participant (or, when appropriate, by a caregiver, surrogate, or the participant's legally authorised representative).

The investigator and any designees are responsible for detecting, documenting, recording, and reporting events that meet the definition of an AE.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Adverse events and SAEs will be collected at every study visit from the time of signature of the screening ICF, throughout the treatment period and follow-up period (up to 90 days after the last dose of study intervention) or until initiation of another anti-cancer therapy, whichever is earlier. Collection and reporting of AEs and SAEs after the final DCO is described in Section [8.3.11](#).

If the investigator becomes aware of an SAE with a suspected causal relationship to the study intervention that occurs after the end of the clinical study in a participant treated by him or her, the investigator shall, without undue delay, report the SAE to the sponsor.

8.3.2 Follow-up of AEs and SAEs

Any AEs that are unresolved at the participant's last study visit will be followed up by the investigator for as long as medically indicated (this may be beyond the 90 days after the last dose of the study intervention), but without further recording in the eCRF. AstraZeneca retains the right to request additional information for any participant with ongoing AE(s)/SAE(s) at the end of the study, if judged necessary.

Adverse Event Variables

The following variables will be collected for each AE:

- AE (verbatim)
- The date and time when the AE started and stopped
- CTCAE grade and changes in CTCAE grade
- Whether the AE is serious or not ([Appendix B](#))
- Investigator causality rating against the study intervention(s) (Yes or No)
- Action taken with regard to study interventions
- AE caused participant's withdrawal from study (Yes or No)
- Administration of treatment for the AE
- Outcome

In addition, the following variables will be collected for SAEs:

- Date AE met criteria for SAE
- Date investigator became aware of SAE
- Seriousness criteria
- Date of hospitalisation
- Date of discharge
- Probable cause of death
- Date of death
- Autopsy performed
- Causality assessment in relation to study procedure(s)
- Causality assessment to other medication

The grading scales found in the NCI CTCAE version x will be utilised for all events with an assigned CTCAE grading. For those events without assigned CTCAE grades, the recommendation in the CTCAE criteria that converts mild, moderate, and severe events into CTCAE grades should be used. A copy of the CTCAE can be downloaded from the Cancer Therapy Evaluation Program website (<http://ctep.cancer.gov>).

8.3.3 Causality Collection

The investigator should assess causal relationship between study intervention and each AE, and answer “Yes” or “No” to the question “Do you consider that there is a reasonable possibility that the event may have been caused by the investigational product?”

For SAEs, causal relationship should also be assessed for other medication and study procedures. Note that for SAEs that could be associated with any study procedure the causal relationship is implied as “yes”.

A guide to the interpretation of the causality question is found in [Appendix B](#).

8.3.4 Adverse Events Based on Signs and Symptoms

All AEs spontaneously reported by the participant or care provider or reported in response to the open question from the study site staff, eg, “Have you had any health problems since the previous visit/you were last asked?” or revealed by observation will be collected and recorded in the eCRF. When collecting AEs, the recording of diagnoses is preferred (when possible) to recording a list of signs and symptoms. However, if a diagnosis is known and there are other signs or symptoms that are not generally part of the diagnosis, the diagnosis and each sign or symptom will be recorded separately.

8.3.5 Adverse Events Based on Examinations and Tests

The results from the CSP-mandated laboratory tests, vital signs, physical examinations, and ECGs will be summarised in the CSR.

Deterioration as compared with baseline in protocol-mandated laboratory values, vital signs and physical examinations should therefore only be reported as AEs if they fulfil any of the SAE criteria, are the reason for discontinuation of treatment with the study intervention or are considered to be clinically relevant as judged by the investigator (which may include but not limited to consideration as to whether treatment or non-planned visits were required or other action was taken with the study intervention, eg, dose adjustment or study intervention interruption).

If deterioration in a laboratory value/vital sign is associated with clinical signs and symptoms, the sign or symptom will be reported as an AE and the associated laboratory result/vital sign will be considered as additional information. Wherever possible the reporting investigator uses the clinical, rather than the laboratory term (eg, anaemia versus low haemoglobin value). In the absence of clinical signs or symptoms, clinically relevant deteriorations in non-mandated parameters should be reported as AE(s).

Deterioration of a laboratory value, which is unequivocally due to PD, should not be reported as an AE/SAE.

Any new or aggravated clinically relevant abnormal medical finding at a physical examination as compared with the baseline assessment will be reported as an AE unless unequivocally related to the disease under study.

8.3.6 Hy's Law

Cases where a participant shows elevations in liver biochemistry may require further evaluation and occurrences of AST or ALT $\geq 3 \times \text{ULN}$ together with TBL $\geq 2 \times \text{ULN}$ may need to be reported as SAEs. Please refer to [Appendix E](#) for further instruction on cases of increases in liver biochemistry and evaluation of HL.

8.3.7 Disease Progression

Disease progression can be considered as a worsening of a participant's condition attributable to the disease for which the study intervention is being studied. It may be an increase in the severity of the disease under study and/or increases in the symptoms of the disease. The development of new, or progression of existing metastasis to the primary cancer under study should be considered as PD and not an AE. Events, which are unequivocally due to PD, should not be reported as an AE during the study.

8.3.8 New Cancers

The development of a new cancer should be regarded as an AE and will generally meet at least one of the serious criteria. New primary cancers are those that are not the primary reason for the administration of study intervention and are identified after the participant's inclusion in this study. They do not include metastases of the original cancer.

8.3.9 Deaths

All deaths that occur during the study intervention period, or within the protocol-defined follow-up period after the administration of the last dose of study intervention, must be reported as follows:

- Death clearly resulting from PD should be documented in the eCRF in the Statement of Death page. It should not be reported as an SAE.
- Where death is not due (or not clearly due) to PD under study, the AE causing the death must be reported as an SAE within 24 hours. It should also be documented in the Statement of Death page in the eCRF. The report should contain a comment regarding the co-involvement of PD, if appropriate, and should assign the main and contributory causes of death.
- Deaths with an unknown cause should always be reported as an SAE and documented in the Statement of Death page in the eCRF, but every effort should be made to determine a cause of death. A post-mortem may be helpful in the assessment of the cause of death, and if performed, a copy of the post-mortem results should be forwarded to AstraZeneca Participant Safety or its representative within the usual time frames.

Deaths occurring after the protocol-defined follow-up period after the administration of the last dose of study intervention should be documented in the Statement of Death page. If the death occurred as a result of an event that started after the defined follow-up period and the event is considered to be due to a late-onset toxicity to study intervention, then it should also be reported as an SAE.

8.3.10 Adverse Events of Special Interest

Adverse events of special interest are events of scientific and medical interest specific to the further understanding of ceralasertib and durvalumab safety profile and require close monitoring and rapid communication by the investigators to AstraZeneca. An AESI can be serious or non-serious. All AESIs will be recorded in the eCRF. Serious AESIs will be recorded and reported as per Section [8.3.12](#).

8.3.10.1 Ceralasertib AESIs

At present there are no AESIs defined for ceralasertib. For specific MedDRA terms used to determine expectedness for SUSAR reporting refer to the ceralasertib IB Section 5.6.

8.3.10.2 Durvalumab AESIs

Adverse events of special interest are events of scientific and medical interest specific to the further understanding of durvalumab safety profile and require close monitoring and rapid communication by the investigators to AstraZeneca. An AESI can be serious or non-serious. All AESIs will be recorded in the eCRF. Serious AESIs will be recorded and reported as per Section [8.3.12](#).

Adverse events of special interest for durvalumab include but are not limited to events with a potential inflammatory or immune-mediated mechanism and which may require more frequent monitoring and/or interventions such as steroids, immunosuppressants and/or hormone replacement therapy. An imAE is defined as an AESI that is associated with drug exposure and is consistent with an immune-mediated mechanism of action and where there is no clear alternate aetiology.

If the investigator has any questions in regard to an event being an imAE, the investigator should promptly contact the study clinical lead.

Adverse events of special interest/imAEs observed with anti-PD-(L)1 agents such as durvalumab include:

- Diarrhoea/colitis and intestinal perforation
- Pneumonitis/ILD
- Endocrinopathies (ie, events of hypophysitis/hypopituitarism, adrenal insufficiency, hyper- and hypo-thyroidism and type I diabetes mellitus)
- Hepatitis/transaminase increases
- Nephritis/blood creatinine increases
- Pancreatitis/serum lipase and amylase increases
- Rash/dermatitis
- Myocarditis
- Myositis/polymyositis
- Neuropathy/neuromuscular toxicity (eg, Guillain-Barré, and myasthenia gravis)
- Other inflammatory responses that are rare/less frequent with a potential immune-mediated aetiology include, but are not limited to, pericarditis, sarcoidosis, uveitis, other events involving the eye, skin, haematological, and rheumatological events, vasculitis, non-infectious meningitis and non-infectious encephalitis

In addition, infusion-related reactions and hypersensitivity/anaphylactic reactions with a different underlying pharmacological aetiology are also considered AESIs.

More detailed guidelines for their evaluation and treatment are described in detail in the Dose Modification and TMGs (see the Annex document to this CSP). These guidelines have been prepared by the sponsor to assist the investigator in the exercise of his/her clinical judgement in treating these types of toxicities. These guidelines apply to AEs considered causally related to the study intervention/study regimen by the reporting investigator.

8.3.11 Safety Data to be Collected Following the Final Data Cut-off of the Study

For participants continuing to receive ceralasertib and/or durvalumab after the final DCO and database closure, AEs and SAEs will be collected, but only SAEs will be reported. In addition, it is recommended that participants continue the scheduled site visits and investigators monitor the participant's safety laboratory results periodically during this treatment in order to manage AEs, consistent with the ceralasertib and durvalumab dose modification guidelines for management of study intervention-related toxicities (see the Annex document to this CSP). All data after the final DCO and database closure will be recorded in the participant notes but, with the exception of SAEs, will not otherwise be reported for the purposes of this study.

All SAEs that occur in patients still receiving ceralasertib and/or durvalumab (or within the 90 days following the last dose) after the final DCO must be reported as detailed in Section [8.3.12](#).

8.3.12 Reporting of Serious Adverse Events

All SAEs have to be reported, whether or not considered causally related to the study intervention, or to the study procedure(s). All SAEs will be recorded in the eCRF.

If any SAE occurs in the course of the study, then investigators or other site personnel inform the appropriate AstraZeneca representatives within 1 day ie, immediately but no later than 24 hours of when he or she becomes aware of it.

The designated AstraZeneca representative works with the investigator to ensure that all the necessary information is provided to the AstraZeneca Patient Safety data entry site within 1 calendar day of initial receipt for fatal and life-threatening events and within 5 calendar days of initial receipt for all other SAEs.

For fatal or life-threatening AEs where important or relevant information is missing, active follow-up is undertaken immediately. Investigators or other site personnel inform AstraZeneca representatives of any follow-up information on a previously reported SAE within 1 calendar day ie, immediately but no later than 24 hours of when he or she becomes aware of it.

Once the investigators or other site personnel indicate an AE is serious in the EDC system, an automated email alert is sent to the designated AstraZeneca representative.

If the EDC system is not available, then the investigator or other study site staff reports an SAE to the appropriate AstraZeneca representative by telephone followed by completion of a paper SAE form.

The AstraZeneca representative will advise the investigator/study site staff how to proceed.

For further guidance on the definition of an SAE, see [Appendix B](#).

The reference documents for definition of expectedness/listedness of AEs are the IBs for ceralasertib and durvalumab; and the applicable document for docetaxel (EU SmPC, US Prescribing Information or other local labels).

8.3.13 Pregnancy

All pregnancies and outcomes of pregnancy with conception dates following the first date of study intervention, including pregnancy in the partner of male participants, should be reported to AstraZeneca.

8.3.13.1 Maternal Exposure

If a participant becomes pregnant during the course of the study, the study intervention should be discontinued immediately.

Pregnancy itself is not regarded as an AE unless there is a suspicion that the study intervention under study may have interfered with the effectiveness of a contraceptive medication.

Congenital abnormalities/birth defects and spontaneous miscarriages should be reported and handled as SAEs. Elective abortions without complications should not be handled as AEs. The outcome of all pregnancies (spontaneous miscarriage, elective termination, ectopic pregnancy, normal birth or congenital abnormality) should be followed up and documented even if the participant was discontinued from the study.

If any pregnancy occurs in the course of the study, then the investigator or other site personnel informs the appropriate AstraZeneca representatives within 1 day, ie, immediately but no later than 24 hours of when he or she becomes aware of it.

The designated AstraZeneca representative works with the investigator to ensure that all relevant information is provided to the AstraZeneca Patient Safety data entry site within 1 or 5 calendar days for SAEs (see Section [8.3.12](#)) and within 30 days for all other pregnancies.

The same timelines apply when outcome information is available.

The PREGREP module in the eCRF is used to report the pregnancy and the PREGOUT is used to report the outcome of the pregnancy.

8.3.13.2 Paternal Exposure

Non-sterilised male participants who intend to be sexually active with a WOCBP partner should refrain from fathering a child or donating or banking sperm for the duration of the study (from the time of screening) and for 6 months after the last dose of the study intervention.

Participants receiving docetaxel should follow the local Prescribing Information relating to contraception, the time limits for such precautions, and any additional restrictions.

Pregnancy of the participant's partner is not considered to be an AE. However, the outcome of all pregnancies (spontaneous miscarriage, elective termination, ectopic pregnancy, normal birth, or congenital abnormality) occurring from the date of the first dose of study intervention until 90 days after the last dose of study intervention should be followed up and documented in the medical record and provided to the AstraZeneca Patient Safety data entry site. Consent from the partner must be obtained before the information is collected and reported to AstraZeneca.

Where a report of pregnancy is received, prior to obtaining information about the pregnancy, the investigator must obtain the consent of the participant's partner. The local study team should adopt the Master Pregnant Partner Form in line with local procedures/requirements and submit it to the relevant Regulatory Authority/IRBs/IECs prior to use.

8.3.14 Medication Error

If a medication error occurs in the course of the study, then the investigator or other site personnel informs the appropriate AstraZeneca representatives within 1 day, ie, immediately but no later than 24 hours of when he or she becomes aware of it.

The designated AstraZeneca representative works with the investigator to ensure that all relevant information is completed within 1 (Initial Fatal/Life-threatening or follow-up Fatal/Life-threatening) or 5 (other serious initial and follow-up) calendar days if there is an SAE associated with the medication error (see Section 8.3.12) and within 30 days for all other medication errors.

The definition of a medication error can be found in [Appendix B](#).

8.4 Overdose

Use of any study intervention in doses exceeding that specified in the protocol is considered to be an overdose. There is currently no specific treatment in the event of overdose of ceralasertib or durvalumab, and possible symptoms of overdose are not established.

- An overdose with associated AEs is recorded as the AE diagnosis/symptoms on the relevant AE modules in the eCRF and on the Overdose eCRF module.
- An overdose without associated symptoms is only reported on the Overdose eCRF module.

If an overdose on an AstraZeneca study intervention occurs in the course of the study, the investigator or other site personnel inform appropriate AstraZeneca representatives immediately, but no later than 24 hours of when he or she becomes aware of it.

The designated AstraZeneca representative works with the investigator to ensure that all relevant information is provided to the AstraZeneca Patient Safety data entry site within 1 or 5 calendar days for overdoses associated with an SAE (see Section 8.3.12) and within 30 days for all other overdoses.

For participants receiving docetaxel, refer to the local Prescribing Information for treatment of cases of overdose. If any overdose is associated with an AE or SAE, record the AE/SAE diagnosis or symptoms in the relevant AE modules only of the eCRF.

8.5 Human Biological Samples for Pharmacokinetics and Pharmacodynamics

Instructions for the collection, handling, storage, and shipping of biological samples will be provided in the study-specific Laboratory Manual. Samples should be stored in a secure storage space with adequate measures to protect confidentiality.

Samples will be stored for a maximum of 15 years from the end of the study (as defined in Section 4.4) in line with consent and local requirements, after which they will be destroyed/repatriated.

- Samples will be disposed of after the Bioanalytical Report finalisation or 6 months after issuance of the draft Bioanalytical Report (whichever is earlier), unless consented for future analyses.
 - Samples may be disposed of or anonymised by pooling. Additional analyses may be conducted on the anonymised, pooled samples to further evaluate and validate the analytical method. Any results from such analyses may be reported separately from the CSR.
 - Samples collected in China will be stored and disposed of according to local laws and regulations. PK samples collected in China will be destroyed or repatriated maximally 5 years after study intervention is approved for marketing in China, or 5 years after study completion if this study not been used for registration.

- Remaining sample aliquots will be retained at AstraZeneca or its designee for a maximum of 15 years from the end of the study (as defined in Section 4.4). Additional use includes but is not limited to confirmation and/or requalification of the assay as well as additional assay development work. The results from future analysis will not be reported in the CSR.
 - Residual samples from those collected in China will be stored and disposed of according to local laws and regulations and will be destroyed or repatriated maximally 5 years after study intervention is approved for marketing in China, or 5 years after study completion if this study not been used for registration. This applies to all samples described in Sections 8.5.1 and 8.5.2.

For further details on Handling of Human Biological Samples, see [Appendix C](#).

8.5.1 Pharmacokinetics

Pharmacokinetic analysis will be done only for participants in Group A (ceralasertib plus durvalumab combination therapy). Whole blood samples will be collected for measurement of plasma concentrations of ceralasertib.

Sparse PK sampling is planned to analyse variables such as peak and trough concentrations, as data allow. Samples collected for PK analyses may also be used to evaluate safety or efficacy aspects related to concerns arising during or after the study.

A total of 8 plasma samples will be collected for ceralasertib PK analysis ([Table 11](#)). Note that on C1D1 and C2D1 participants should take the ceralasertib morning dose at the study centre.

Table 11 Pharmacokinetic Sampling Timepoints

Time relative to dose	C1D1	C1D8	C2D1	C2D8	C4D8	C8D8	C12D8
Ceralasertib plasma PK sampling							
Pre-dose	X ^a	X ^c		X ^c	X ^c	X ^c	X ^c
Post-dose	X ^b		X ^b				

a Pre-dose: within 1 hour ($\pm$ 10 min) before the first ceralasertib dose (to be taken at the study centre)

b Post-dose: within 1 hour ($\pm$ 10 min) after ceralasertib dose

c Pre-durvalumab dose: within 1 hour ($\pm$ 10 min) before durvalumab infusion; D8 sample will be approximately 24 hours after the D7 ceralasertib morning dose.

8.5.1.1 Determination of Drug Concentration

Samples collected for determination of ceralasertib plasma concentrations will be assayed by bioanalytical test sites operated by or on behalf of AstraZeneca, using an appropriately validated bioanalytical method. Full details of the analytical method used will be described in a separate Bioanalytical Report.

Incurred sample reproducibility analysis, if any, will be performed alongside the bioanalysis of the test samples. The results from the evaluation, if performed, will be reported in a

separate Bioanalytical Report.

8.5.2 Pharmacodynamics

Pharmacodynamic analysis will be performed using the mandatory blood samples collected.

8.6 Human Biological Samples for Biomarker Analysis

Participant consent to the study includes participation in the mandatory biomarker assessment components of the study. Samples for biomarker assessment are required and will be collected from all participants in this study as specified in the SoA and in compliance with applicable local regulations.

Residual samples will be used for future research, outside the scope of the protocol-defined objectives, only where optional consent allows.

In China, the mandatory tissue and blood samples will not be collected or analysed until after approval from relevant local authorities. This applies to samples described in the Sections [8.6.1](#).

Tumour samples collected from participants in China will be destroyed or repatriated maximally 5 years after study intervention is approved for marketing in China, or 5 years after study completion if this study not been used for registration. This applies to samples described in the Sections [8.6.1](#).

For details on Handling of Human Biological Samples, including storage, re-use and destruction, refer to [Appendix C](#) and the Laboratory Manual.

8.6.1 Collection of Mandatory Tumour Tissue Sample

Tumour samples at screening are mandatory in this study.

Where available, a tissue sample obtained after progression on prior anti-PD-(L)1 therapy and ≤ 3 months prior to screening consent should be provided. Where no such sample is available a tumour sample taken ≤ 24 months prior to screening is acceptable. Samples should be of sufficient quality to enable assessment of tumour cell PD-L1 expression. Tumour sample must be FFPE, with sufficient material for sectioning of preferably 16 slides (5 micron thick sections). If FFPE blocks cannot be provided, then a set of newly cut unstained slides that enable necessary testing should be provided (preferably a minimum of 16 slides). Further details are provided in the Laboratory Manual.

If a suitable archival tumour sample is not available, a fresh tumour sample should be taken via a biopsy. The tumour biopsy must not be taken from a previously irradiated lesion. Tumour lesions used for fresh biopsies should not be the same lesions used as RECIST 1.1 TLs unless there are no other lesions suitable for biopsy and in this instance only core needle

(not excisional/incisional) biopsy is allowed. For participants with a single TL, if screening biopsy is collected prior to screening imaging for baseline tumour assessment, allow approximately 2 weeks before imaging scans are acquired.

Any sample (ie, newly acquired or archived) with insufficient tumour content and/or those obtained through a fine needle aspirate are inadequate for defining tumour PD-L1 status. Collection of tumour cells from fluid such as ascites or pleural effusion is not permitted.

8.6.1.1 Eligibility and Stratification by PD-L1 Expression

Tumour PD-L1 expression status will be used to determine eligibility and stratification. All participants must have the qualifying PD-L1 expression result to be randomised. PD-L1 expression will be evaluated prospectively for all screened participants at a central laboratory using VENTANA PD-L1 (SP263) IHC assay. For the purposes of stratification, PD-L1 expression result will remain blinded to the site and investigator and will be transmitted directly to the IRT/IXRS system.

Data may be used to evaluate their association with the observed clinical responses to ceralasertib and durvalumab combination therapy versus doctaxel therapy. Data and samples collected as part of the study may be used for development of current or future diagnostic tests, as required.

8.6.1.2 Exploratory Tumour Biomarker Analysis

Following completion of PD-L1 testing, any residual mandatory tissue collected at screening may be used for the assessment of exploratory biomarkers. These markers may include, but not be limited to, sequencing of DNA in order to assess the presence of mutations in one or more genes (eg, ATM), tumour mutational burden and immune cell diversity within the tumour; sequencing of RNA in order to assess the expression level of one or more genes or gene signatures; assessment of expression of one or more proteins (eg, ATM).

Data may be used to evaluate the association of individual markers, or groups of markers, with each other, with participant demographics, with safety outcomes, and with clinical outcomes (eg, ORR, PFS, OS).

8.6.2 Collection of Mandatory Blood Samples

Whole blood samples will be collected and used for biomarker analyses as follows.

These samples will not be collected for participants in China (applies to all the subsections below).

8.6.2.1 Whole Blood Samples for ctDNA and DNA analysis

Whole blood samples will be collected as specified in the SoA. Blood samples will be processed to generate ctDNA and, in some instances, germline DNA (see Section [8.6.2.2](#)). The

ctDNA may be used for a range of analyses including but not limited to quantitation and profiling of somatic molecular defects through sequencing, assessment of methylation, or other appropriate analysis. Applications of ctDNA quantitation and profiling will include but may not be limited to assessing the association between ctDNA dynamics and response to treatment as well as interrogation of changes in the molecular profile of the tumour through the course of treatment.

8.6.2.2 Whole Blood Samples for RNA and DNA Analysis

Whole blood samples will be obtained and used to extract RNA and/or DNA as specified in the SoA.

RNA, where isolated, may be used for analysis including, but not limited to quantifying the expression of specific genes or gene signatures before and after treatment, as well as assessment of the relationship of the gene expression to the outcome following treatment.

DNA, where isolated, may be used for a range of analyses including, but not be limited to targeted sequencing of T-cell and B cell receptor sequences, in order to assess the diversity and clonality of baseline, and on-treatment immune responses, and targeted sequencing of the HLA loci, in order to assess the impact of HLA allotype and diversity on treatment outcome. Sequencing may also involve whole exome sequencing, which will be conducted in order to provide a control germline sequence for comparison to similar sequencing from tumour DNA. Such comparisons are critical for certain analyses, such as for highly sensitive detection of ctDNA as a means of assessing minimal residual disease.

8.6.2.3 Plasma Samples for Biomarker Analysis

Plasma will be collected as specified in the SoA. Plasma may be used for analysis of blood-based biomarkers including, but not limited to, quantification of circulating proteins.

8.6.3 Collection of Optional Biomarker Samples

Collection of a tumour sample at progression is encouraged. Participants will not be excluded from the study if these samples are not collected.

If tumour sample at progression is provided, it will be used to evaluate association of biomarkers with emerging resistance to the study intervention, in line with exploratory biomarker objectives described in [Section 3](#).

Biopsy on progression may be particularly valuable when there is a marked phenotypic change in a particular lesion, and investigators are encouraged to contact AstraZeneca in these cases. The provision of tumour tissue is encouraged only if clinically appropriate and not considered detrimental to patient care.

These samples will not be collected for participants in China.

For further details on Handling of Human Biological Samples, including storage, re-use and destruction, refer to [Appendix C](#) and the Laboratory Manual.

8.7 Optional Genomics Initiative Sample

Collection of optional samples for Genomics Initiative research is also part of this study as specified in the SoA and is subject to agreement in the ICF addendum.

This sample will not be collected in China.

Blood sample for DNA isolation will be collected from participants who have consented to participate in the genetic analysis component of the study. Participation is optional. Participants who do not wish to participate in the genetic research may still participate in the study.

See [Appendix D](#) for information regarding the storage and destruction of Genomics Initiative genetic sample. Details on processes for collection and shipment and destruction of these samples can be found either in the appendices or in the Laboratory Manual.

8.8 Medical Resource Utilisation and Health Economics

Medical resource utilisation and health economics data, associated with medical encounters, will be collected in the eCRF by the investigator and study site personnel for all participants throughout the study. Protocol-mandated procedures, tests, and encounters are excluded.

The assessment of health care resource use will increase the understanding regarding the relationship between treatment and tumour-related cancer symptoms on resource use, such as the need for palliative procedures to address obstruction or bleeding. This will be captured and analysed to inform submissions to payers.

The data collected may be used to conduct exploratory economic analyses of treatment and will include:

- Number and duration of medical care encounters, including surgeries, and other selected procedures (inpatient and outpatient)
- Duration of hospitalisation (total days or length of stay, including duration by wards eg, intensive care unit)
- Number and type of diagnostic and therapeutic tests and procedures
- Outpatient medical encounters and interventions (including physician or emergency room visits, tests and procedures, and medications)
- Treatment-related to AEs (including the method of delivery of the treatment)

9 STATISTICAL CONSIDERATIONS

Statistical analyses will be performed by AstraZeneca or its representatives.

A comprehensive SAP will be prepared and signed off within 3 months of the first participant randomised.

9.1 Statistical Hypotheses

The hypothesis of interest with regard to the efficacy is:

- H0: No difference between ceralasertib plus durvalumab and docetaxel in terms of OS
- H1: Difference between ceralasertib plus durvalumab and docetaxel in terms of OS

To preserve the overall type 1 error (familywise error rate) in the strong sense, an MTP including the primary and key secondary endpoints will be implemented. The MTP will be fully specified in the SAP. The primary endpoint of OS will be tested using overall 2-sided alpha level 5% (split between the interim and final analysis by using the Lan DeMets spending function). The significance level applied at the interim analysis depends on the information fraction (ie the proportion of events included in the analysis data set). For example, if approximately 71% of the target OS events are available at the time of the interim efficacy analysis (ie, 290/406), the 2-sided significance level to be applied for the interim and final OS analysis would be 1.6% and 4.6% respectively, based on the stratified log-rank test.

Hypotheses will be tested using an MTP with an alpha-exhaustive recycling strategy ([Burman et al 2009](#)). With this approach, hypotheses will be tested in a pre-defined order, where OS for ceralasertib plus durvalumab combination therapy versus docetaxel treatment (FAS) is tested first, followed by test of secondary endpoints of PFS and ORR in the same way. The other hypotheses will then be tested in the MTP using alpha (test mass) splitting and alpha recycling, where the test mass that becomes available after each rejected hypothesis is recycled to secondary hypotheses not yet rejected. This testing procedure stops when the entire test mass is allocated to non-rejected hypotheses. Implementation of this pre-defined ordered testing procedure, including recycling, will strongly control type I error at 5% (2-sided), among all key hypotheses.

The MTP will be fully specified in the SAP.

9.2 Sample Size Determination

Approximately 726 patients will be screened/enrolled to achieve 580 participants randomised 1:1 to ceralasertib plus durvalumab combination therapy or docetaxel (290 participants per arm).

Enrolled/screened	Estimated 760 patients
Randomised/assigned	Estimated 580 participants

Note: “Enrolled” means a participant’s, or their legally acceptable representative’s, agreement to participate in a clinical study following completion of the informed consent process. Potential patients who are screened for the purpose of determining eligibility for the study, but are not randomised/assigned in the study, are considered “screen failures”, unless otherwise specified by the protocol.

The study is sized to characterise the OS benefit of ceralasertib plus durvalumab combination treatment versus SoC, docetaxel, in patients with advanced or metastatic NSCLC without actionable genomic alterations and refractory to anti-PD-(L)1 therapy and platinum-based chemotherapy. The sizing assumes no treatment effect for the first 2 months, followed by separation of the OS curves with a HR < 1. The SoC control group OS is assumed to follow an exponential distribution, whereas the experimental group OS follows a piece-wise exponential distribution.

The primary OS analysis will be performed when approximately 406 OS events have occurred across the total study population (70% maturity), which will provide 90% power to demonstrate statistical significance at the 4.6% level (using a 2-sided test, significance level determined by Lan DeMets). The smallest treatment difference that could be statistically significant at the final analysis is an average HR of 0.82.

Assuming an 18-month recruitment period and a minimum follow-up of 13.5 months, it is anticipated that the final OS analysis will be performed approximately 32 months after the first participant is randomised.

Other assumptions are median OS of 9 months for docetaxel and 12.5 months for ceralasertib plus durvalumab combination treatment with 2 months delay of separation of the OS curves (assumed HR = 0.72). The critical HR is 0.82 which translates to an approximate median OS of 10.9 months in ceralasertib plus durvalumab treatment group compared to 9 months in docetaxel group.

9.3 Populations for Analyses

The populations defined are prespecified in [Table 12](#).

Table 12 Populations for Analysis

Population/Analysis set	Description
Enrolled	All participants who sign the ICF.
FAS	All participants who are randomised in the study. The FAS will be used for all the efficacy analyses. Treatment groups will be compared on the basis of randomised study intervention, regardless of the treatment actually received. Participants who were randomised but did not subsequently receive study intervention are included in the analysis in the treatment group to which they were randomised.
Safety analysis set	All participants who have received any amount of the study intervention. Safety data will be summarised using the safety analysis set according to the treatment received, that is, erroneously treated participants (eg, those randomised to treatment A but actually given treatment B) will be summarised according to the treatment they actually received.
PK analysis set	All participants who received at least 1 dose of study intervention (ceralasertib) and for whom at least 1 reportable post-dose PK sample data are available.

9.4 Statistical Analyses

The SAP will be finalised prior to database lock, and it will include a more technical and detailed description of the statistical analyses described in this section. This section is a summary of the planned statistical analyses of the most important endpoints including primary and key secondary endpoints.

9.4.1 General Considerations

Data will be presented by treatment group. Descriptive statistics will be used for all variables. Continuous variables will be summarised by the number of observations, mean, standard deviation, median, minimum, and maximum.

Categorical variables will be summarised by frequency counts and percentages for each category. Unless otherwise stated, percentages will be calculated based on the population total, by dose regimen and by timepoint as appropriate.

Time-to-event variables will be presented using the Kaplan-Meier methodology, including median time calculated from the Kaplan-Meier curves.

In general, the last observed measurement prior to first dose of study intervention will be considered the baseline measurement. For efficacy variables, baseline is defined as the last visit within the protocol-allowed window prior to randomisation.

Depending on the extent of impact, if any, of COVID-19 on-study conduct (in particular missed visits, delayed or discontinued study treatment, and other protocol deviations), summaries of data relating to participants diagnosed with COVID-19 may be generated. Further details will be provided in the SAP.

9.4.2 Efficacy

9.4.2.1 Primary Endpoint: Overall Survival

OS is defined as the time from the date of randomisation until death due to any cause. The comparison will include all randomised participants, as randomised, regardless of whether the participant withdraws from randomised therapy or receives another anti-cancer therapy. Any participant not known to have died at the time of analysis will be censored based on the last recorded date on which the participant was known to be alive.

The final analysis of OS will be performed at approximately 70% of total possible survival events (406 deaths) in the FAS population, assuming median OS of 9 months for docetaxel and 12.5 months for ceralasertib plus durvalumab combination treatment. Assuming an 18 month recruitment period and a minimum follow-up of 13.5 months, this analysis is expected to be performed approximately 32 months after first participant is randomised.

Note: During the Survival Follow-up period, calls will be made to reach the participants following the date of DCO for the analysis (these contacts should generally occur within 7 days of the DCO). If participants are confirmed to be alive or if the death date is after the DCO date, then these participants will be censored at the date of DCO.

9.4.2.1.1 Analysis Method for OS

OS will be analysed using a log-rank test stratified by histology (squamous versus non-squamous), resistance to prior anti-PD-(L)1 therapy (primary versus acquired), tumour cell PD-L1 expression level ($<1\%$ versus $\geq 1\%$), and geographical region (North America and Western Europe versus RoW). Kaplan-Meier plots of OS will be presented by treatment group and the same stratification variables, where appropriate. The stratification variables will be defined according to data from the IXRS/IRT at randomisation.

The effect of ceralasertib plus durvalumab combination treatment versus docetaxel treatment will be estimated based on the HR together with its 95% CI and p-value. The HR and CI will be estimated from a stratified Cox Proportional Hazards model (with ties = Efron and stratification variables: histology, anti-PD-(L)1 resistance, PD-L1 expression status, and geographical region; and the CI calculated using a profile likelihood approach. An HR < 1 will be considered to favour ceralasertib plus durvalumab treatment over docetaxel treatment.

The assumption of proportionality will be assessed. In the event of non-proportionality, the HR will be interpreted as an average HR over the observed extent of follow-up. Proportionality will be tested firstly by examining the plots of complementary log-log(event times) versus log(time) and, if necessary, a time-dependent covariate will be fitted to assess the extent to which this represents random variation. In case of the violation of proportionality assumption further analysis may be performed and detailed in SAP.

Stratified analysis of OS will allow a pooling strategy, to be pre-specified in the SAP, in the

event of insufficient events per stratum.

9.4.2.1.2 Subgroup Analysis of OS

Subgroup analyses will be conducted to compare OS between ceralasertib plus durvalumab combination treatment and docetaxel treatment in the following subgroups of the FAS (but not limited to):

- Histology: squamous versus non-squamous
- Previous anti-PD-(L)1 therapy (primary versus acquired)
- PD-L1 (IVRS) expression status ($< 1\%$ versus $\geq 1\%$)
- Sex: male versus female
- Age at randomisation: < 65 years versus ≥ 65 years
- Smoking: current, former or never
- Geographical region: North America and Western Europe versus RoW

Other baseline variables, for eg, race (Asian versus non-Asian), liver metastasis (presence versus absence), ATM mutation(s) (presence versus absence), or ATM protein (normal versus loss), may also be assessed if there is clinical justification or an imbalance is observed between the treatment groups. The purpose of the subgroup analyses is to assess the consistency of treatment effect across expected prognostic and/or predictive factors.

The subgroup analyses for the stratification factors will be based on the values entered into the IXRS/IRT; all other factors will be based on values recorded on the eCRF, or from the TPV data.

Additional subgroups of interest and analysis methods will be outlined in the SAP.

9.4.2.2 Secondary Endpoints

9.4.2.2.1 Derivation of Tumour Response Variables

All RECIST 1.1 assessments, whether scheduled or unscheduled, will be included in the calculations. This is also regardless of whether a participant discontinues study intervention or receives another anti-cancer therapy.

At each visit, participants will be programmatically assigned a RECIST 1.1 visit response of CR, PR, SD, PD, or NE depending on the status of their disease compared with baseline and previous assessments. Baseline will be assessed within the 28 days prior to randomisation. The tumour response endpoints (PFS, ORR, and DoR) will then be derived from the scan dates and overall visit responses.

9.4.2.2.2 Overall Survival at 12 Months

The OS12 will be defined as the Kaplan-Meier estimate of OS at 12 months after randomisation and will be estimated as described for the primary endpoint in Section [9.4.2.1](#).

9.4.2.2.3 Progression-Free Survival

PFS will be defined as the time from the date of randomisation until the date of objective PD per RECIST 1.1 as assessed by the investigator, and derived from the raw tumour data or death (by any cause in the absence of progression), (ie, date of event or censoring – date of randomisation + 1). The comparison will include all randomised participants, as randomised, regardless of whether the participant withdraws from randomised therapy, receives another anti-cancer therapy or clinically progresses prior to RECIST 1.1 progression. Participants who have not progressed or died at the time of analysis will be censored at the time of the latest date of assessment from their last evaluable RECIST 1.1 assessment. However, if the participant progresses or dies after 2 or more consecutive missed visits, the participant will be censored at the time of the latest evaluable RECIST 1.1 assessment prior to the 2 missed visits. (Note: NE visit is not considered as missed visit).

If the participant has no evaluable visits or does not have baseline data, they will be censored at Day 1, unless they die within 2 visits of baseline (plus 1 week allowing for a late assessment within the visit window).

The PFS time will always be derived based on tumour assessment dates and not visit dates.

9.4.2.2.3.1 Analysis Methods for PFS

The PFS analysis will be performed by the same methods as described for OS in Section [9.4.2.1.1](#).

Summaries of the number and percentage of participants experiencing a PFS event and the type of event (RECIST 1.1 or death) will be provided along with median PFS for each treatment. The proportion of participants alive and progression-free at 12 monthly intervals from randomisation will be summarised by treatment group.

9.4.2.2.3.2 Subgroup Analysis for PFS

Subgroup analyses of PFS will be performed using the same methods as described for OS in Section [9.4.2.1.2](#). Additional subgroups of interest and analysis methods will be outlined in the SAP.

9.4.2.2.4 Objective Response Rate

ORR is defined as the proportion of participants who have a CR or PR as assessed by the investigator per RECIST 1.1. The analysis will include all randomised participants with measurable disease at baseline.

Data obtained from date of randomisation until date of progression, or of the last evaluable

assessment in the absence of progression, will be included in the assessment of ORR, regardless of whether the participant withdraws from study treatment. Participants who discontinue the study intervention without a response or progression, receive a subsequent therapy, and then respond will not be included as responders in the ORR analysis.

The analysis will be performed on the subset of FAS who have measurable disease at baseline using a logistic regression model adjusted for the same stratification factors as for the primary analysis (Section 9.4.2.1.1). The results of the analysis will be presented in terms of an OR together with its corresponding profile likelihood 95% CI and p-value.

Summaries will be produced that present the number and percentage of participants with a tumour response (CR/PR). Overall visit response data will be listed for all participants (ie, the FAS). For each treatment arm, best overall response will be summarised by n (%) for each category (CR, PR, SD, PD, and NE). No formal statistical analyses are planned for best overall response.

9.4.2.2.5 Duration of Response

DoR will be defined as the time from the date of first documented response until date of documented progression as assessed by the investigator per RECIST 1.1; or death due to any cause (ie, date of PFS event – date of first response + 1).

The end of response coincides with the date of progression or death from any cause used for the PFS endpoint. The time of the initial response is defined as the latest of the dates contributing towards the first visit response of PR or CR.

The analysis will include all randomised participants as randomised with measurable disease at baseline who have a response, regardless of whether the participant withdraws from study treatment, receives another anti-cancer therapy or clinically progresses prior to RECIST 1.1 progression.

Kaplan-Meier estimates for median DoR and 25th and 75th percentiles for DoR are summarised. Kaplan-Meier estimates for percentage remaining in response at 12 monthly intervals are presented for each treatment group.

Additionally, median, 25th and 75th percentiles for time from randomisation to onset of response calculated using standard descriptive statistics are summarised for participants who have a response.

9.4.2.2.6 Time to Response

Time to response is defined as the time from the date of randomisation until the date of first documented objective response, as assessed by the investigator per RECIST 1.1. The date of first documented response should coincide with that used for the DoR endpoint.

Time to response will not be defined for those participants who do not have a documented response. The analysis will include all randomised participants as randomised with measurable disease at baseline.

Data obtained from randomisation up until progression, or the last evaluable assessment in the absence of progression, will be included in the assessment of TTR. Participants who discontinue study intervention without a response or progression, receive a subsequent therapy, and then respond will not be included in the TTR analysis.

The TTR will be summarised as number (%) of participants based upon the number responders for each treatment by the scheduled assessment timepoint at which the response was first observed. Additionally, descriptive summary statistics (ie, minimum, maximum, median, Q1 and Q3) will also be presented. Other (supportive) secondary endpoints need not be described with the same level of detail as the key secondary endpoints.

Additional analyses will be described in the SAP.

9.4.2.2.7 Disease Control Rate

DCR at 18 weeks is defined as the percentage of participants who have at least 1 visit response of CR or PR or who have demonstrated SD for at least 17 weeks (ie, 18 weeks – 1 week window), as assessed by the investigator per RECIST 1.1.

Data obtained from date of randomisation until date of progression, or of the last evaluable assessment in the absence of progression, will be included in the assessment of DCR, regardless of whether the participant withdraws from treatment.

The analysis will include all randomised participants with measurable disease at baseline.

The measure of interest is the OR of the DCR at 18 weeks. The analysis will be performed on the subset of FAS for participants who have measurable disease at baseline using a logistic regression model adjusted for the same stratification factors as the primary objective (Section 9.4.2.1.1). The results of the analysis will be presented in terms of an OR together with its corresponding profile likelihood 95% CI and p-value.

9.4.2.2.8 Second Progression-Free Survival

Time from randomisation to second progression or death (PFS2) will be defined as the time from date of randomisation to date of either the earliest progression event following the initial objective progression subsequent to first subsequent therapy, or death. The date of second progression will be recorded by the investigator in the eCRF and defined according to local standard clinical practice.

The comparison will include all randomised participants (FAS) regardless of whether the

participant withdraws from subsequent therapy and regardless of missed visits. The censoring rules will be detailed out in the SAP.

PFS2 will be analysed using identical methods as outlined for PFS and adjusting for the same stratification factors as the primary objective (Section 9.4.2.1.1). The HR for the treatment effect together with its 95% CI will be presented. Medians and Kaplan-Meier plots will be presented to support the analysis. Descriptive summaries will also be presented and detailed out in SAP.

9.4.2.2.9 Clinical Outcome Assessments

All COA, including the PROs, will be based on FAS, if not otherwise stated. Tolerability-related analyses will use the safety analysis set. Compliance per assessment timepoint will be reported for all COA measures.

Secondary endpoints include:

- TTD of health-related QoL
- TTD of physical function
- Proportion of participants experiencing each level of symptomatic AEs and descriptive summary of proportion of participants reporting each level of overall side effect both while on treatment

The analysis of selected PRO scales may include change from baseline using a mixed-model for repeated measurements. Adjusted mean change from baseline estimates per treatment arm and corresponding 95% CIs may be presented, along with an overall estimate of the treatment difference, 95% CI, and p-value.

Details of all the above COA statistical analyses will be described in full in the SAP.

9.4.2.2.9.1 EORTC QLQ-C30

Analysis of TTD of health-related QoL will be performed by using the two global health status items from the EORTC QLQ-C30, and TTD of physical functioning will be performed using the physical functioning subscale score of the EORTC QLQ-C30. The TTD PRO endpoints will be analysed using the same time-to-event analysis methodology as applied to the PFS endpoint.

Meaningful change thresholds will be derived and selected as part of a separate, pre-specified, psychometric SAP. Anchor-based methods using the participant-based anchors PGI-S and PGI-C will be considered to define thresholds for clinically meaningful within-participant change used in the TTD endpoints. Other methods including distribution-based methods, cumulative distribution function, and probability density function curves, and methods using other anchors may also be considered.

9.4.2.2.9.2 PRO-CTCAE

The proportion of participants experiencing each level of symptomatic AEs and overall side effect bother while on treatment will be measured by the PRO-CTCAE and PGI-TT, respectively.

Descriptive analyses will be performed using methods appropriate for the scale of measurement.

9.4.2.2.10 WHO/ECOG Performance Status

WHO/ECOG performance status will be summarised by visit and by treatment group.

9.4.2.2.11 Medical Resource Utilisation and Health Economics

To investigate the impact of treatment and disease on health care resource use, the following variables will be captured:

- Planned and unplanned hospital attendances beyond protocol-mandated visits (including physician visits, emergency room visits, day cases, and admissions)
- Primary sign or symptom the participant presents with
- Length of hospital stay
- Length of any time spent in an intensive care unit

Where admitted overnight, the length of hospital stay will be calculated as the difference between the date of hospital discharge (or death date) and the start date of hospitalisation or start of study intervention if the start of study intervention is after start date of hospitalisation.

Length of hospital stay = end date of hospitalisation – start date of hospitalisation + 1

Participants with missing discharge dates will be calculated as the difference between the last day with available data and the start date of hospitalisation. The length of intensive care unit stay will be calculated using the same method.

Analysis Methods

The potential impact of the disease and treatment on health care resource use will be analysed for the purposes of submissions to payers. Descriptive statistics (as appropriate, including means, median, ranges or frequencies, and percentages) will be provided for each treatment group on the different types of hospital admissions, the length of stay for participants admitted to hospital for at least 1 overnight stay, and the length of stay for participants admitted to intensive care/high dependency units, as well as the primary sign or symptom the participant presents with.

9.4.2.3 Tertiary/Exploratory Endpoints

9.4.2.3.1 ATM Deficiency

The relationship of tumoural expression and/or mutation status of ATM relative to efficacy outcomes (OS, PFS and ORR) will be presented for ceralasertib plus durvalumab relative to docetaxel. Analysis methods described above will be presented for subgroups of participants by ATM alteration. Refer to [9.4.4](#) for other assessments.

9.4.3 Safety

9.4.3.1 Adverse Events

Adverse events will be coded using the most recent version of the MedDRA that will have been released for execution at AstraZeneca/designee.

Adverse events will be presented for each treatment group and by MedDRA SOC/PT, covering number and percentage of participants reporting at least one event and number of events where appropriate.

Adverse events occurring prior to the start of study interventions, TEAEs, and post-treatment AEs will be presented separately.

Treatment-emergent Adverse Events

The following AEs will be considered as TEAEs:

- Adverse events with an onset date on or after first dose of study intervention and within 90 days after last dose of study intervention or up to the day prior to start of subsequent anti-cancer therapy, whichever comes first.
- Worsening of pre-existing events on or after first dose of study intervention and within 90 days after last dose of study intervention, or up to the day prior to start of subsequent anti-cancer therapy, whichever comes first.

An overview of AEs will be presented for each treatment group with the number and percentage of participants and the number of individual occurrences for the following categories of AEs:

- any AE
- AEs with outcome of death
- serious AEs
- AEs leading to discontinuation of study intervention
- AEs leading to study intervention dose interruptions
- AEs leading to dose reduction

- AEs leading to treatment or study withdrawal

Separate AE tables will be provided taken into consideration relationship as assessed by the investigator, maximum CTC grading, seriousness, death, and events leading to discontinuation of study interventions as well as other action taken related to study interventions, events of special interest, other significant adverse events and timing of events.

An additional table will present the number and percentage of participants with most common AEs. Most common will be defined in the SAP.

Key participant information will be presented for participants with AEs with outcome of death, serious AEs, and AEs leading to discontinuation of study interventions.

An AE listing for the safety analysis set will cover details for each individual AE. Full details of AE analyses will be provided in the SAP.

9.4.3.2 Vital Signs

For each scheduled post-baseline visit, descriptive statistics for all vital signs will be presented for observed values and for change from baseline.

Details of vital signs analyses, including definition of abnormality criteria (eg, definition of low, normal, high) and study-specific criteria for treatment-emergent changes in vital signs parameters will be provided in the SAP.

9.4.3.3 Laboratory Parameters

For each scheduled post-baseline visit, descriptive statistics for all clinical chemistry, haematology, and TSH parameters will be presented for observed values and change from baseline.

Details of analyses of laboratory safety variables will be provided in the SAP.

9.4.3.4 Electrocardiogram

For each scheduled post-baseline assessment, descriptive statistics for all ECG parameters will be presented for observed values and change from baseline. Details of ECG analyses, including definition of reference values and abnormalities will be provided in the SAP.

9.4.3.5 Other Safety Endpoints

9.4.3.5.1 ECOG/WHO Performance Status

The ECOG/WHO performance status will be summarised by visit and by treatment group.

9.4.4 Other Analyses

Other analyses may include analyses of assessments, which are not defined as endpoints, that

need to be pre-specified and not necessarily be reported in the CSR such as, but not limited to, biomarkers, population PK, health care utilisation endpoints and health technology assessment related endpoints.

9.4.4.1 Biomarkers

Biomarker status will be assessed for participants in each treatment group according to pre-specified criteria that may be detailed in the SAP. The relationship of exploratory biomarkers to clinical outcomes (including but not restricted to) of PFS, ORR, and OS may be presented.

Biomarker exploratory analyses may be described in a separate analysis plan and may be reported outside the CSR in a separate report. The results of this biomarker assessment will be reported either in the CSR itself or as an addendum, or separately in a scientific report or publication. The results of this biomarker assessment may be pooled with biomarker data from other studies with the study intervention to generate hypotheses to be tested in future research.

9.4.4.2 Optional Exploratory Genetic Sample

Data will be reported outside the CSR (please see [Appendix D](#)).

9.5 Interim Analyses

Two interim analyses, IA1 and IA2, will be performed for analysing OS.

- IA1 will be performed after reaching approximately 32% information fraction, ie, 32% of total estimated survival events (130 deaths) in the FAS population (22% maturity based on a total of 580 participants). This analysis is aimed at assessing the futility of continuing the study. If the $HR \geq 1.05$ then the study will be stopped (non-binding boundary). The probability of crossing the HR threshold if the true $HR = 0.72$ (“false stop”) is 2%, and if the true $HR = 1$ the probability is 39% (“correct stop”). The DCO for this analysis is expected to be performed approximately 16 months after first participant is randomised.
- IA2 will be performed after reaching approximately 71% information fraction, ie, 71% of total estimated survival events (290 deaths) in the FAS population (50% maturity based on a total of 580 participants). This analysis is aimed at assessing the superiority of the experimental treatment over SoC. It is considered that 290 events will have 53% power at the 1.6% significance level to find a statistically significant benefit in OS for ceralasertib plus durvalumab combination treatment over docetaxel (critical $HR < 0.75$). This analysis is expected to be performed approximately 23 months after first participant is randomised.

The interim analyses will be performed by an IDMC. For the interim analyses, the IDMC will review unblinded interim data and inform the sponsor whether the interim boundaries specified in Section [9.6](#) are met.

The 5.0% alpha (2-sided) for the study will be controlled across IA2 and final OS analysis using the Lan DeMets spending function that approximates the O'Brien-Fleming approach where the significance level applied depends upon the actual proportion of information (ie, information fraction) available at the time of analysis ([Lan and DeMets 1983](#)).

The planned interim analyses will be described in greater detail in the SAP.

9.6 Independent Data Monitoring Committee

For details on IDMC refer to Appendix [A 5](#).

An IDMC comprised of independent experts will be convened and will meet approximately 6 months after first participant enrolled to review safety data and make recommendations to continue, amend, or stop the study based on safety findings. The committee will meet approximately every 6 months thereafter. The IDMC will review unblinded safety data.

In addition, the IDMC may be requested to review efficacy data. For the interim futility and interim superiority analyses, the IDMC will review unblinded interim data and inform the sponsor whether the interim boundaries specified in this section are met.

An additional safety review for Japanese participants will take place when the first 9 participants in Japan have completed from C1D1 to C1D28. This review will be carried out by the IDMC in an unblinded manner. After review of the unblinded data, the IDMC will make a recommendation on whether the study should continue recruitment as planned or hold recruitment in Japan.

Full details of the IDMC procedures, processes, and interim analyses can be found in the IDMC charter.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Appendix A Regulatory, Ethical, and Study Oversight Considerations

A 1 Regulatory and Ethical Considerations

This study will be conducted in accordance with the protocol and with the following:

- Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organisations of Medical Sciences International Ethical Guidelines.
- Applicable ICH GCP Guidelines.
- Applicable laws and regulations.

The protocol, protocol amendments, ICF, IB, and other relevant documents (eg, advertisements) must be submitted to an IRB/IEC by the investigator and reviewed and approved by the IRB/IEC before the study is initiated.

Any amendments to the protocol will require IRB/IEC and applicable Regulatory Authority approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants.

AstraZeneca will be responsible for obtaining the required authorisations to conduct the study from the concerned Regulatory Authority. This responsibility may be delegated to a CRO, but the accountability remains with AstraZeneca.

Regulatory Reporting Requirements for Serious Adverse Events

- Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities towards the safety of participants and the safety of a study intervention under clinical investigation are met.
- The sponsor has a legal responsibility to notify both the local Regulatory Authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The sponsor will comply with country-specific regulatory requirements relating to safety reporting to the Regulatory Authority, IRB/IEC, and investigators.
- For all studies except those utilising medical devices investigator safety reports must be prepared for suspected unexpected serious adverse reactions according to local regulatory requirements and sponsor policy and forwarded to investigators as necessary.
- An investigator who receives an investigator safety report describing an SAE or other specific safety information (eg, summary or listing of SAEs) from the sponsor will review and then file it along with the IB and will notify the IRB/IEC, if appropriate according to local requirements.

A 2 Financial Disclosure

Investigators and sub-investigators will provide the sponsor with sufficient, accurate financial information as requested to allow the sponsor to submit complete and accurate financial certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

A 3 Informed Consent Process

The investigator or his/her representative will explain the nature of the study to the participant or his/her legally authorised representative and answer all questions regarding the study.

Participants must be informed that their participation is voluntary, and they are free to refuse to participate and may withdraw their consent at any time and for any reason during the study. Participants or their legally authorised representative will be required to sign a statement of informed consent that meets the requirements of 21 Code of Federal Regulations 50, local regulations, ICH guidelines, Health Insurance Portability and Accountability Act requirements, where applicable, and the IRB/IEC or study centre.

The medical record must include a statement that written informed consent was obtained before the participant was enrolled in the study and the date the written consent was obtained. The authorised person obtaining the informed consent must also sign the ICF.

Participants must be re-consented to the most current version of the ICF(s) during their participation in the study.

A copy of the ICF(s) must be provided to the participant or the participant's legally authorised representative.

Participants who are rescreened are required to sign a new ICF.

The ICF will contain a separate section that addresses and documents the collection and use of any mandatory and/or optional human biological samples. The investigator or authorised designee will explain to each participant the objectives of the analysis to be done on the samples and any potential future use. Participants will be told that they are free to refuse to participate in any optional samples or the future use and may withdraw their consent at any time and for any reason during the retention period.

A 4 Data Protection

Participants will be assigned a unique identifier by the sponsor. Any participant records or datasets that are transferred to the sponsor will contain the identifier only; participant names or any information, which would make the participant identifiable, will not be transferred.

The participant must be informed that his/her personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure and use of their data must also be explained to the participant in the informed consent

The participant must be informed that his/her medical records may be examined by Clinical Quality Assurance auditors or other authorised personnel appointed by the sponsor, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.

Unless previously specified, the biomarker data will have unknown clinical significance and AstraZeneca will not provide biomarker assessment results to participants, their family members, any insurance company, any employer, a clinical study investigator, a general physician, or any other third-party, unless required to do so by law.

The participant's samples will not be used for any purpose other than those described in the study protocol.

A 5 Committees Structure

The safety of all AstraZeneca clinical studies is closely monitored on an ongoing basis by AstraZeneca representatives in consultation with Participant Safety. Issues identified will be addressed; for instance, this could involve amendments to the CSP and letters to investigators.

An IDMC is to be established to provide an independent, external, systematic, and unbiased assessment of safety for this study. The role and the primary responsibilities of the IDMC members, as well as the purpose and timing of IDMC meetings are described in the IDMC charter. The IDMC will provide review independent of AstraZeneca, the study clinical team and all other individuals associated with the conduct of the study.

A Steering Committee has been established to review relevant research (completed, ongoing, and pending) which may impact upon the study and to support the study team with interpretation of study outcomes. A Steering Committee Charter will define the primary responsibilities of the steering committee, its members, and the purpose and timing of meetings.

A 6 Dissemination of Clinical Study Data

A description of this clinical study will be available on <http://astrazenecaclinicaltrials.com> and <http://www.clinicaltrials.gov>, as will the summary of the study results when they are available. The clinical study and/or summary of study results may also be available on other websites according to the regulations of the countries in which the study is conducted.

A 7 Data Quality Assurance

All participant data relating to the study will be recorded on the eCRF unless transmitted to

the sponsor or designee electronically (eg, laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the eCRF.

The investigator must maintain accurate documentation (source data) that supports the information entered in the eCRF.

The investigator must permit study-related monitoring, audits, IRB/IEC review, and Regulatory Authority inspections and provide direct access to source data documents.

Monitoring details describing strategy (eg, risk-based initiatives in operations and quality such as Risk Management and Mitigation Strategies and Analytical Risk-Based Monitoring), methods, responsibilities and requirements, including handling of non-compliance issues and monitoring techniques (central, remote, or on-site monitoring) are provided in the Monitoring Plan.

The sponsor or designee is responsible for the data management of this study including quality checking of the data.

The sponsor assumes accountability for actions delegated to other individuals (eg, CRO).

Study monitors will perform ongoing source data verification to confirm that data entered into the eCRF by authorised site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the investigator for 15 years from the end of the study (as defined in the protocol) unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor.

A 8 Source Documents

Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator's site.

Data reported on the eCRF or entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.

All information in original records and certified copies of original records of clinical findings, observations, or other activities in a clinical study necessary for the reconstruction and evaluation of the study are defined as source documents. Source data include but are not limited to medical history and physical examination notes, hospital discharge summary, autopsy report when available, results of relevant diagnostic tests completed.

A 9 Study and Site Start and Closure

The study start date is the date on which the clinical study will be open for recruitment of participants. The first act of recruitment is the first site open.

The sponsor designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of the sponsor. Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study site closure visit has been performed.

The investigator may initiate study site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to:

- Failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines.
- Inadequate recruitment of participants by the investigator.
- Discontinuation of further study intervention development.

If the study is prematurely terminated or suspended, the sponsor shall promptly inform the investigators, the IECs/IRBs, the regulatory authorities, and any CRO(s) used in the study of the reason for termination or suspension, as specified by the applicable regulatory requirements. The investigator shall promptly inform the participant and should assure appropriate participant therapy and/or follow-up.

Participants from terminated sites will have the opportunity to be transferred to another site to continue the study.

A 10 Publication Policy

The results of this study may be published or presented at scientific meetings. If this is foreseen, the investigator agrees to submit all manuscripts or abstracts to the sponsor before submission. This allows the sponsor to protect proprietary information and to provide comments.

The sponsor will comply with the requirements for publication of study results. In accordance with standard editorial and ethical practice, the sponsor will generally support publication of multicentre studies only in their entirety and not as individual site data. In this case, a co-ordinating investigator will be designated by mutual agreement.

Authorship will be determined by mutual agreement and in line with International Committee of Medical Journal Editors authorship requirements.

Appendix B Adverse Events: Definitions and Procedures for Recording, Evaluating, Follow-up, and Reporting

B 1 Definition of Adverse Events

An AE is the development of any untoward medical occurrence (other than progression of the malignancy under evaluation) in a participant or clinical study participant administered a study intervention and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavourable and unintended sign (eg, an abnormal laboratory finding), symptom (for example nausea, chest pain), or disease temporally associated with the use of a medicinal product, whether or not considered related to the study intervention.

The term AE is used to include both serious and non-serious AEs and can include a deterioration of a pre-existing medical occurrence. An AE may occur at any time, including run-in or washout periods, even if no study intervention has been administered.

B 2 Definitions of Serious Adverse Event

An SAE is an AE occurring during any study phase (ie, run-in, treatment, washout, follow-up), that fulfils one or more of the following criteria:

- Results in death.
- Is immediately life-threatening.
- Requires inpatient hospitalisation or prolongation of existing hospitalisation.
- Results in persistent or significant disability or incapacity.
- Is a congenital abnormality or birth defect.
- Is an Important Medical Event that may jeopardise the participant or may require medical treatment to prevent one of the outcomes listed above.

Adverse events for malignant tumours reported during a study should generally be assessed as SAEs. If no other seriousness criteria apply, the “Important Medical Event” criterion should be used. In certain situations, however, medical judgement on an individual event basis should be applied to clarify that the malignant tumour event should be assessed and reported as a **Non-Serious AE**. For example, if the tumour is included as medical history and progression occurs during the study, but the progression does not change treatment and/or prognosis of the malignant tumour, the AE may not fulfil the attributes for being assessed as serious, although reporting of the progression of the malignant tumour as an AE is valid and should occur. Also, some types of malignant tumours, which do not spread remotely after a routine treatment that does not require hospitalisation, may be assessed as non-serious; examples in adults include Stage 1 basal cell carcinoma and Stage 1A1 cervical cancer removed via cone biopsy.

The above instruction applies only when the malignant tumour event in question is a new malignant tumour (ie, it is not the tumour for which entry into the study is a criterion and that is being treated by the study intervention under study and is not the development of new or progression of existing metastasis to the tumour under study). Malignant tumours that – as part of normal, if rare, progression – undergo transformation (eg, Richter’s transformation of B cell chronic lymphocytic leukaemia into diffuse large B cell lymphoma) should not be considered a new malignant tumour.

Life-threatening

“Life-threatening” means that the participant was at immediate risk of death from the AE as it occurred, or it is suspected that use or continued use of the product would result in the participant’s death. “Life-threatening” does not mean that had an AE occurred in a more severe form it might have caused death (eg, hepatitis that resolved without hepatic failure).

Hospitalisation

Outpatient treatment in an emergency room is not in itself an SAE, although the reasons for it may be (eg, bronchospasm, laryngeal oedema). Hospital admissions and/or surgical operations planned before or during a study are not considered AEs if the illness or disease existed before the participant was enrolled in the study, provided that it did not deteriorate in an unexpected way during the study.

Important Medical Event or Medical Treatment

Medical and scientific judgement should be exercised in deciding whether a case is serious in situations where important medical events may not be immediately life-threatening or result in death, hospitalisation, disability or incapacity but may jeopardise the participant or may require medical treatment to prevent one or more outcomes listed in the definition of serious. These should usually be considered as serious.

Simply stopping the suspect drug does not mean that it is an Important Medical Event; medical judgement must be used.

- Angioedema not severe enough to require intubation but requiring IV hydrocortisone treatment.
- Hepatotoxicity caused by paracetamol (acetaminophen) overdose requiring treatment with N-acetylcysteine.
- Intensive treatment in an emergency room or at home for allergic bronchospasm.
- Blood dyscrasias (eg, neutropenia or anaemia requiring blood transfusion, etc.) or convulsions that do not result in hospitalisation.
- Development of drug dependency or drug abuse.

Intensity Rating Scale:

The grading scales found in the revised National Cancer Institute CTCAE latest version will be utilised for all events with an assigned CTCAE grading. For those events without assigned CTCAE grades, the recommendation in the CTCAE criteria that converts mild, moderate and severe events into CTCAE grades should be used. A copy of the CTCAE can be downloaded from the Cancer Therapy Evaluation Program website (<http://ctep.cancer.gov>). The applicable version of CTCAE should be described clearly.

It is important to distinguish between serious and severe AEs. Severity is a measure of intensity whereas seriousness is defined by the criteria in Appendix B 2. An AE of severe intensity need not necessarily be considered serious. For example, nausea that persists for several hours may be considered severe nausea, but not an SAE unless it meets the criteria shown in Appendix B 2. On the other hand, a stroke that results in only a limited degree of disability may be considered a mild stroke but would be an SAE when it satisfies the criteria shown in Appendix B 2.

B 3 A Guide to Interpreting the Causality Question

When making an assessment of causality consider the following factors when deciding if there is a “reasonable possibility” that an AE may have been caused by the drug.

- Time Course. Exposure to suspect drug. Has the participant actually received the suspect drug? Did the AE occur in a reasonable temporal relationship to the administration of the suspect drug?
- Consistency with known drug profile. Was the AE consistent with the previous knowledge of the suspect drug (pharmacology and toxicology) or drugs of the same pharmacological class? Or could the AE be anticipated from its pharmacological properties?
- De-challenge experience. Did the AE resolve or improve on stopping or reducing the dose of the suspect drug?
- No alternative cause. The AE cannot be reasonably explained by another aetiology such as the underlying disease, other drugs, other host or environmental factors.
- Re-challenge experience. Did the AE reoccur if the suspected drug was reintroduced after having been stopped? AstraZeneca would not normally recommend or support a re-challenge.
- Laboratory tests. A specific laboratory investigation (if performed) has confirmed the relationship.

In difficult cases, other factors could be considered such as:

- Is this a recognised feature of overdose of the drug?
- Is there a known mechanism?

Causality of “related” is made if following a review of the relevant data, there is evidence for a “reasonable possibility” of a causal relationship for the individual case. The expression “reasonable possibility” of a causal relationship is meant to convey, in general, that there are facts (evidence) or arguments to suggest a causal relationship.

The causality assessment is performed based on the available data including enough information to make an informed judgement. With no available facts or arguments to suggest a causal relationship, the event(s) will be assessed as “not related”.

Causal relationship in cases where the disease under study has deteriorated due to lack of effect should be classified as no reasonable possibility.

B 4 Medication Error

For the purposes of this clinical study a medication error is an unintended failure or mistake in the treatment process for an AstraZeneca study intervention that either causes harm to the participant or has the potential to cause harm to the participant.

A medication error is not lack of efficacy of the drug, but rather a human or process related failure while the drug is in control of the study site staff or participant.

Medication error includes situations where an error:

- Occurred.
- Was identified and intercepted before the participant received the drug.
- Did not occur, but circumstances were recognised that could have led to an error.

Examples of events to be reported in clinical studies as medication errors:

- Drug name confusion.
- Dispensing error eg, medication prepared incorrectly, even if it was not actually given to the participant.
- Drug not administered as indicated, for example, wrong route or wrong site of administration.
- Drug not taken as indicated eg, tablet dissolved in water when it should be taken as a solid tablet.

- Drug not stored as instructed eg, kept in the fridge when it should be at room temperature.
- Wrong participant received the medication (excluding IXRS/IRT errors).
- Wrong drug administered to participant (excluding IXRS/IRT errors).

Examples of events that **do not** require reporting as medication errors in clinical studies:

- Errors related to or resulting from IXRS/IRT - including those which lead to one of the above listed events that would otherwise have been a medication error.
- Participant accidentally missed drug dose(s) eg, forgot to take medication.
- Accidental overdose (will be captured as an overdose).
- Participant failed to return unused medication or empty packaging.
- Errors related to background and rescue medication, or SoC medication in open-label studies, even if an AstraZeneca product.

Medication errors are not regarded as AEs, but AEs may occur as a consequence of the medication error.

Appendix C Handling of Human Biological Samples

C 1 Chain of Custody

A full chain of custody is maintained for all samples throughout their lifecycle.

The investigator at each centre keeps full traceability of collected biological samples from the participants while in storage at the centre until shipment or disposal (where appropriate) and records relevant processing information related to the samples whilst at site.

The sample receiver keeps full traceability of the samples while in storage and during use until used or disposed of or until further shipment and keeps record of receipt of arrival and onward shipment or disposal.

AstraZeneca or delegated representatives will keep oversight of the entire life cycle through internal procedures, monitoring of study sites, auditing or process checks, and contractual requirements of external laboratory providers

Samples retained for further use will be stored in the AstraZeneca-assigned biobanks or other sample archive facilities and will be tracked by the appropriate AstraZeneca Team during for the remainder of the sample life cycle.

If required, AstraZeneca will ensure that remaining biological samples are returned to the site according to local regulations or at the end of the retention period, whichever is the sooner.

C 2 Withdrawal of Informed Consent for Donated Biological Samples

AstraZeneca ensures that biological samples are returned to the source or destroyed at the end of a specified period as described in the informed consent.

If a participant withdraws consent specifically to the subsequent use of donated biological samples, the samples will be disposed of/destroyed/repatriated, and the action documented. If samples are already analysed, AstraZeneca is not obliged to destroy the results of this research. The participant will be presented with the option to opt out of the subsequent use of the donated samples during the withdrawal process. If the participant decides to opt out, then the donated samples will be disposed of. If the participant withdraws consent without opting out for the subsequent use of the donated samples, then the samples will be used as per protocol.

Following withdrawal of consent for biological samples, further study participation should be considered in relation to the withdrawal processes outlined in the informed consent.

The investigator:

- Ensures participant's withdrawal of informed consent to the use of donated samples is highlighted immediately to AstraZeneca or delegate.
- Ensures that relevant human biological samples from that participant, if stored at the study site, are immediately identified, disposed of as appropriate, and the action documented.
- Ensures that the participant and AstraZeneca are informed about the sample disposal.

AstraZeneca ensures the organisation(s) holding the samples is/are informed about the withdrawn consent immediately and that samples are disposed of or repatriated as appropriate, and the action documented, and study site notified.

C 3 International Airline Transportation Association (IATA) 6.2 Guidance Document

LABELLING AND SHIPMENT OF BIOHAZARD SAMPLES

The IATA (<https://www.iata.org/whatwedo/cargo/dgr/Pages/download.aspx>) classifies infectious substances into three categories: Category A, Category B or Exempt

Category A Infectious Substances are infectious substances in a form that, when exposure to it occurs, is capable of causing permanent disability, life-threatening or fatal disease in otherwise healthy humans or animals.

Category A pathogens are, for example, Ebola, Lassa fever virus. Infectious substances meeting these criteria which cause disease in humans or both in humans and animals must be assigned to UN 2814. Infectious substances which cause disease only in animals must be assigned to UN 2900.

Category B Infectious Substances are infectious Substances that do not meet the criteria for inclusion in Category A. Category B pathogens are, for example, Hepatitis A, C, D, and E viruses. They are assigned the following UN number and proper shipping name:

- UN 3373 – Biological Substance, Category B.
- Are to be packed in accordance with UN3373 and IATA 650.

Exempt - Substances which do not contain infectious substances or substances which are unlikely to cause disease in humans or animals are not subject to these regulations unless they meet the criteria for inclusion in another class.

- Clinical study samples will fall into Category B or exempt under IATA regulations.
- Clinical study samples will routinely be packed and transported at ambient temperature in IATA 650 compliant packaging.
(<https://www.iata.org/whatwedo/cargo/dgr/Documents/DGR-60-EN-PI650.pdf>).

- Biological samples transported in dry-ice require additional dangerous goods specification for the dry-ice content.

Appendix D Optional Genomics Initiative Sample

D 1 Use/Analysis of DNA

- AstraZeneca intends to collect and store DNA for genetic research to explore how genetic variations may affect clinical parameters, risk and prognosis of diseases, and the response to medications. This genetic research may lead to better understanding of diseases, better diagnosis of diseases or other improvements in health care and to the discovery of new diagnostics, treatments or medications. Therefore, where local regulations and IRB/IEC allow, tumour tissue and/or blood sample will be collected for DNA analysis from consenting participants.
- This optional genetic research may consist of the analysis of the structure of the participant's DNA, ie, the entire genome.
- The results of genetic analyses may be reported in a separate study summary.
- The sponsor will store the DNA samples in a secure storage space with adequate measures to protect confidentiality.
- The samples will be retained while research on ceralasertib and durvalumab continues but no longer than 15 years from the end of the study (as defined in the protocol) or other period as per local requirements.

D 2 Genetic Research Plan and Procedures

Selection of Genetic Research Population

- All participants will be asked to participate in this genetic research. Participation is voluntary and if a participant declines to participate there will be no penalty or loss of benefit. The participant will not be excluded from any aspect of the main study.

Inclusion Criteria

For inclusion in this genetic research, participants must fulfil all of the inclusion criteria described in the main body of this CSP and provide informed consent for the Genomics Initiative sampling and analyses.

Exclusion Criteria

- Exclusion from this genetic research may be for any of the exclusion criteria specified in the main study or any of the following:
 - Previous allogeneic bone marrow transplant.
 - Transfusion of non-leukocyte depleted blood or blood component within 120 days of genetic sample collection.

- Healthy volunteers and paediatric participant samples will not be collected for the Genomics Initiative.

Withdrawal of Consent for Genetic Research

Participants may withdraw from this genetic research at any time, independent of any decision concerning participation in other aspects of the main study. Voluntary withdrawal will not prejudice further treatment. Procedures for withdrawal are outlined in Section 7.2 of the main CSP.

Collection of Samples for Genetic Research

The blood sample for this genetic research will be obtained from the participants pre-dose at the first dosing visit. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding participants who may withdraw due to an AE. If for any reason the sample is not drawn at the first dosing visit, it may be taken at any visit until the last study visit. Only one sample should be collected per participant for genetics during the study.

Coding and Storage of DNA Samples

- The processes adopted for the coding and storage of samples for genetic analysis are important to maintain participant confidentiality. Samples may be stored for a maximum of 15 years from the end of the study (as defined in the protocol), after which they will be destroyed. DNA is a finite resource that is used up during analyses. Samples will be stored and used until no further analyses are possible or the maximum storage time has been reached.
- An additional second code will be assigned to the sample either before or at the time of DNA extraction replacing the information on the sample tube. Thereafter, the sample will be identifiable only by the second, unique number. This number is used to identify the sample and corresponding data at the AstraZeneca genetics laboratories, or at the designated organisation. No personal details identifying the individual will be available to any person (AstraZeneca employee or designated organisations working with the DNA).
- The link between the participant enrolment/randomisation code and the second number will be maintained and stored in a secure environment, with restricted access at AstraZeneca or designated organisations. The link will be used to identify the relevant DNA samples for analysis, facilitate correlation of genotypic results with clinical data, allow regulatory audit, and permit tracing of samples for destruction in the case of withdrawal of consent.

Ethical and Regulatory Requirements

The principles for ethical and regulatory requirements for the study, including this genetics

research component, are outlined in [Appendix A](#).

Informed Consent

The genetic component of this study is optional, and the participant may participate in other components of the main study without participating in this genetic component. To participate in the genetic component of the study the participant must sign and date both the consent form for the main study and the addendum for the Genomics Initiative component of the study. Copies of both signed and dated consent forms must be given to the participant and the original filed at the study centre. The principal investigator is responsible for ensuring that consent is given freely, and that the participant understands that they may freely withdraw from the genetic aspect of the study at any time.

Participant Data Protection

AstraZeneca will not provide individual genotype results to participants, any insurance company, any employer, their family members, general physician unless required to do so by law.

Extra precautions are taken to preserve confidentiality and prevent genetic data being linked to the identity of the participant. In exceptional circumstances, however, certain individuals might see both the genetic data and the personal identifiers of a participant. For example, in the case of a medical emergency, an AstraZeneca physician or an investigator might know a participant's identity and also have access to his or her genetic data. Regulatory authorities may require access to the relevant files, though the participant's medical information and the genetic files would remain physically separate.

Data Management

Any genetic data generated in this study will be stored at a secure system at AstraZeneca and/or designated organisations to analyse the samples.

AstraZeneca and its designated organisations may share summary results (such as genetic differences from groups of individuals with a disease) from this genetic research with other researchers, such as hospitals, academic organisations or health insurance companies. This can be done by placing the results in scientific databases, where they can be combined with the results of similar studies to learn even more about health and disease. The researchers can only use this information for health-related research purposes. Researchers may see summary results, but they will not be able to see individual participant data or any personal identifiers.

Some or all of the clinical datasets from the main study may be merged with the genetic data in a suitable secure environment separate from the clinical database.

Statistical Methods

The number of participants that will agree to participate in the genetic research is unknown. It is therefore not possible to establish whether sufficient data will be collected to allow a formal statistical evaluation or whether only descriptive statistics will be generated. A SAP may be prepared where appropriate.

Appendix E Actions Required in Cases of Increases in Liver Biochemistry and Evaluation of Hy's Law

E 1 Introduction

This appendix describes the process to be followed in order to identify and appropriately report PHL cases and HL cases. It is not intended to be a comprehensive guide to the management of elevated liver biochemistries.

During the course of the study, the investigator will remain vigilant for increases in liver biochemistry. The investigator is responsible for determining whether a participant meets potential PHL criteria at any point during the study.

All sources of laboratory data are appropriate for the determination of PHL and HL events; this includes samples taken at scheduled study visits and other visits including central and all local laboratory evaluations even if collected outside of the study visits; for example, PHL criteria could be met by an elevated ALT from a central laboratory and/or elevated TBL from a local laboratory.

The investigator will also review AE data (for example, for AEs that may indicate elevations in liver biochemistry) for possible PHL events.

The investigator participates, together with AstraZeneca clinical project representatives, in review and assessment of cases meeting PHL criteria to agree whether HL criteria are met. HL criteria are met if there is no alternative explanation for the elevations in liver biochemistry other than drug-induced liver injury caused by the study intervention.

The investigator is responsible for recording data pertaining to PHL/HL cases and for reporting SAEs and AEs according to the outcome of the review and assessment in line with standard safety reporting processes.

E 2 Definitions

PHL

Levels of AST or ALT $\geq 3 \times \text{ULN}$ **together with** TBL $\geq 2 \times \text{ULN}$ at any point during the study following the start of study intervention irrespective of an increase in alkaline phosphatase.

HL

Levels of AST or ALT $\geq 3 \times \text{ULN}$ **together with** TBL $\geq 2 \times \text{ULN}$, where no other reason, other than the study intervention, can be found to explain the combination of increases, eg, elevated alkaline phosphatase indicating cholestasis, viral hepatitis, another drug.

For PHL and HL the elevation in transaminases must precede or be coincident with (ie, on the same day) the elevation in TBL, but there is no specified timeframe within which the elevations in transaminases and TBL must occur.

E 3 Identification of Potential Hy's Law Cases

In order to identify cases of PHL it is important to perform a comprehensive review of laboratory data for any participant who meets any of the following identification criteria in isolation or in combination:

- $ALT \geq 3 \times ULN$.
- $AST \geq 3 \times ULN$.
- $TBL \geq 2 \times ULN$.

Local laboratories being used

The investigator will, without delay, review each new laboratory report and if the identification criteria are met will:

- Notify the AstraZeneca representative
- Determine whether the participant meets PHL criteria (see Section [E 2](#) Definitions within this Appendix for definition) by reviewing laboratory reports from all previous visits.
- Promptly enter the laboratory data into the laboratory eCRF.

E 4 Follow-up

E 4.1 Potential Hy's Law Criteria Not Met

If the participant does not meet PHL criteria, the investigator will:

- Inform the AstraZeneca representative that the participant has not met PHL criteria.
- Perform follow-up on subsequent laboratory results according to the guidance provided in the CSP.

E 4.2 Potential Hy's Law Criteria Met

If the participant does meet PHL criteria the investigator will:

- Determine whether PHL criteria were met at any study visit prior to starting study intervention (See Section [E 6](#)).
- Notify the AstraZeneca representative who will then inform the central Study Team.

- Within 1 day of PHL criteria being met, the investigator will report the case as an SAE of PHL; serious criterion “Important Medical Event” and causality assessment “yes/related” according to CSP process for SAE reporting.
- For participants that met PHL criteria prior to starting study intervention, the investigator is not required to submit a PHL SAE unless there is a significant change# in the participant’s condition.
- The study clinical lead contacts the investigator, to provide guidance, discuss and agree an approach for the study participants’ follow-up (including any further laboratory testing) and the continuous review of data.
- Subsequent to this contact the investigator will:
 - Monitor the participant until liver biochemistry parameters and appropriate clinical symptoms and signs return to normal or baseline levels, or as long as medically indicated. Completes follow-up SAE Form as required.
 - Investigate the aetiology of the event and perform diagnostic investigations as discussed with the study clinical lead. For studies using a central laboratory, this includes deciding which the tests available in the HL laboratory kit should be used.
 - Complete the three Liver eCRF Modules as information becomes available.

#A “**significant**” change in the participant’s condition refers to a clinically relevant change in any of the individual liver biochemistry parameters (ALT, AST or TBL) in isolation or in combination, or a clinically relevant change in associated symptoms. The determination of whether there has been a significant change will be at the discretion of the investigator, this may be in consultation with the study clinical lead if there is any uncertainty.

E 5 Review and Assessment of Potential Hy’s Law Cases

The instructions in this section should be followed for all cases where PHL criteria are met.

As soon as possible after the biochemistry abnormality was initially detected, the study clinical lead contacts the investigator in order to review available data and agree on whether there is an alternative explanation for meeting PHL criteria other than drug-induced liver injury caused by the study intervention, to ensure timely analysis and reporting to health authorities within 15 calendar days from date PHL criteria was met. The AstraZeneca global clinical lead or equivalent and global safety physician will also be involved in this review together with other subject matter experts as appropriate.

According to the outcome of the review and assessment, the investigator will follow the instructions below.

Where there is an agreed alternative explanation for the ALT or AST and TBL elevations, a

determination of whether the alternative explanation is an AE will be made and subsequently whether the AE meets the criteria for an SAE:

- If the alternative explanation is **not** an AE, record the alternative explanation on the appropriate eCRF.
- If the alternative explanation is an AE/SAE: update the previously submitted PHL SAE and AE eCRFs accordingly with the new information (reassessing event term; causality and seriousness criteria) following the AstraZeneca standard processes.

If it is agreed that there is **no** explanation that would explain the ALT or AST and TBL elevations other than the study intervention:

- Send updated SAE report (report term “Hy’s Law”) according to AstraZeneca standard processes.
 - The “Medically Important” serious criterion should be used if no other serious criteria apply.
 - As there is no alternative explanation for the HL case, a causality assessment of “related” should be assigned.

If, there is an unavoidable delay, of over 15 calendar days in obtaining the information necessary to assess whether or not the case meets the criteria for HL, then it is assumed that there is no alternative explanation until such time as an informed decision can be made:

- Provides any further update to the previously submitted SAE report of PHL, (report term now “Hy’s Law case”) ensuring causality assessment is related to study intervention and seriousness criterion is medically important, according to CSP process for SAE reporting.
- Continue follow-up and review according to agreed plan. Once the necessary supplementary information is obtained, repeat the review and assessment to determine whether HL criteria are still met. Update the previously submitted PHL SAE report following CSP process for SAE reporting, according to the outcome of the review and amending the reported term if an alternative explanation for the liver biochemistry elevations is determined.

E 6 Actions Required when Potential Hy’s Law Criteria are Met Before and After Starting Study Intervention

This section is applicable to participants with liver metastases who meet PHL criteria on study intervention, having previously met PHL criteria at a study visit prior to starting study intervention.

At the first on-study intervention occurrence of PHL criteria being met, the investigator will determine if there has been a **significant change** in the participant's condition compared with the last visit where PHL criteria were met.

- If there is no significant change, no action is required.
- If there is a significant change, notify the AstraZeneca representative, who will inform the central Study Team, then follow the subsequent process described in Section [E 4.2](#).

E 7 Actions Required for Repeat Episodes of Potential Hy's Law

This section is applicable when a participant meets PHL criteria on study intervention and has already met PHL criteria at a previous on study intervention visit.

The requirement to conduct follow-up, review and assessment of a repeat occurrence(s) of PHL is based on the nature of the alternative cause identified for the previous occurrence.

The investigator should determine the cause for the previous occurrence of PHL criteria being met and answer the following question:

Was the alternative cause for the previous occurrence of PHL criteria being met found to be the disease under study eg, chronic or progressing malignant disease, severe infection or liver disease or did the participant meet PHL criteria prior to starting study intervention and at their first on-study intervention visit as described in Section [E 6](#) of this Appendix.

If **No**: follow the process described in Section [E 4.2](#) for reporting PHL as an SAE.

If **Yes**: Determine if there has been a significant change in the participant's condition compared with when PHL criteria were previously met.

- If there is no significant change, no action is required.
- If there is a significant change, follow the process described in Section [E 4.2](#) for reporting PHL as an SAE.

E 8 Laboratory Tests

The list below represents the standard, comprehensive list of follow-up tests, which are recommended but not mandatory. The list may be modified based on clinical judgement. Any test results need to be recorded.

Hy's Law Laboratory Kit

Additional standard chemistry and coagulation tests	GGT LDH Prothrombin time INR
Viral hepatitis	IgM anti-HAV IgM and IgG anti-HBc HBsAg HCV DNA ^a IgM and IgG anti-HCV HCV RNA ^a IgM anti-HEV HEV RNA
Other viral infections	IgM and IgG anti-CMV IgM and IgG anti-HSV IgM and IgG anti-EBV
Alcoholic hepatitis	Carbohydrate-deficient transferrin ^b
Autoimmune hepatitis	Antinuclear antibody Anti-liver/kidney microsomal antibody Anti-smooth muscle antibody
Metabolic diseases	Alpha-1-antitrypsin Ceruloplasmin Iron Ferritin Transferrin ^b Transferrin saturation

CMV, cytomegalovirus; DNA, deoxyribonucleic acid; EBV, Epstein-Barr virus; GGT, gamma glutamyl transferase; HAV, hepatitis A virus; HBc, hepatitis B core antigen; HBsAg, hepatitis B surface antigen; HCV, hepatitis C virus; HEV, hepatitis E virus; HSV, herpes simplex virus; IgG, immuno-globulin G; IgM, immuno-globulin M; INR, international normalised ratio; LDH, lactate dehydrogenase; RNA, ribonucleic acid.

^a HCV RNA; HCV DNA are only tested when IgG anti-HCV is positive or inconclusive.

^b Carbohydrate-deficient transferrin (CD-transferrin) is not available in China. Study teams should amend this list accordingly.

E 9 References

Aithal et al 2011

Aithal GP, Watkins PB, Andrade RJ, Larrey D, Molokhia M, Takikawa H, et al. Case definition and phenotype standardization in drug-induced liver injury. Clin Pharmacol Ther 2011;89(6):806-15.

FDA Guidance 2009

Food and Drug Administration. Guidance for industry: Drug-induced liver injury: premarketing clinical evaluation. July 2009. Available from: URL: <https://www.fda.gov/downloads/guidances/UCM174090.pdf>. Accessed 29 November 2021.

Appendix F Guidelines for Evaluation of Objective Tumour Response Using RECIST Version 1.1 Criteria (Response Evaluation Criteria in Solid Tumours)

Introduction

This appendix details the implementation of RECIST 1.1 guidelines ([Eisenhauer et al 2009](#)). Investigator assessments will use the RECIST 1.1 guidelines described in this appendix.

Imaging Modalities and Acquisition Specifications for RECIST 1.1

A summary of the imaging modalities that can be used for tumour assessment of TLs, NTLs and NLs is provided in [Table 13](#).

Table 13 Summary of Imaging Modalities for Tumour Assessment

Target lesions	Non-target lesions	New lesions
CT MRI	CT MRI Plain X-ray Chest X-ray	CT MRI Plain X-ray Chest X-ray Bone scan (Scintigraphy) ¹⁸ F-fluoro-deoxyglucose-PET/CT

CT, computed tomography; PET/CT, positron emission tomography/CT; MRI, magnetic resonance imaging.

Computed Tomography and Magnetic Resonance Imaging

CT with IV contrast is the preferred imaging modality (although MRI with IV contrast is acceptable if CT is contraindicated) to generate reproducible anatomical images for tumour assessments (ie, for measurement of TLs, assessment of NTLs, and identification of NLs). It is essential that the same correct imaging modality, image acquisition parameters (eg, anatomic coverage, imaging sequences, etc.), imaging facility, tumour assessor (eg, radiologist), and method of tumour assessment (eg, RECIST 1.1) are used consistently for each participant throughout the study. The use of the same scanner for serial scans is recommended, if possible. It is important to follow the image collection/tumour assessment schedule as closely as possible (refer to the SoA), and this on-study imaging schedule MUST be followed regardless of any delays in dosing or missed imaging visits. If an unscheduled assessment is performed (eg, to investigate clinical signs/symptoms of progression) and the participant has not progressed, every attempt should be made to perform the subsequent scan acquisitions at the next scheduled imaging visit.

Due to its inherent rapid acquisition (seconds), CT is the imaging modality of choice. Body scans should be performed with breath-hold scanning techniques, if possible. Therefore, CT of the chest is recommended over MRI due to significant motion artefacts (eg, heart, major blood vessels, breathing) associated with MRI. MRI has excellent contrast and spatial and temporal

resolutions; however, there are many image acquisition variables involved in MRI, which greatly impact image quality, lesion conspicuity, and measurement. Furthermore, the availability of MRI is variable globally. The modality used at follow-up should be the same as was used at baseline, and the lesions should be measured/assessed on the same pulse sequence. In general, local oncology diagnostic imaging parameters are applied for scan acquisition. It is beyond the scope of this appendix to prescribe specific MRI pulse sequence parameters for all scanners, body parts, and diseases.

The most critical CT and MRI image acquisition parameters for optimal tumour evaluation are anatomic coverage, contrast administration, slice thickness, and reconstruction interval.

a. Anatomic coverage: Optimal anatomic coverage for most solid tumours is the chest-abdomen (-pelvis). Coverage should encompass all areas of known predilection for metastases in the disease under evaluation and should additionally investigate areas that may be involved based on signs and symptoms of individual participants. Because a lesion later identified in a body part not scanned at baseline would be considered as a NL representing PD, careful consideration should be given to the extent of imaging coverage at baseline and at subsequent follow-up timepoints. This will enable better consistency not only of tumour measurements but also identification of new disease.

Required anatomical regions to be imaged for assessment of tumour burden (TLs and/or NTLs) at baseline and follow-up visits vary according to the study, and these timepoints are specified in the SoA. Examples include the following:

- IV contrast-enhanced CT of chest-abdomen (including the entire liver and both adrenal glands) (-pelvis).
- Non-contrast CT of chest and IV contrast-enhanced abdomen (including the entire liver and both adrenal glands) (-pelvis).
- IV contrast-enhanced CT or MRI of the head and neck.
- IV contrast-enhanced MRI (preferred) or CT of the brain.

For chest-abdomen (-pelvis) imaging, the following are scanning options in decreasing order of preference, with additional options (2 to 4) for consideration when participants have

sensitivity to IV contrast or have compromised renal function:

- 1 Chest-abdomen (-pelvis) CT with IV CT contrast (most preferred).
- 2 Chest CT without IV contrast + abdomen (-pelvis) MRI with IV MRI contrast, if CT IV contrast (iodine based) is medically contraindicated at any time during the study.
- 3 Chest-abdomen (-pelvis) CT without IV contrast, if both IV CT and MRI contrast are medically contraindicated or the participant has compromised renal function.
- 4 Chest-abdomen (-pelvis) MRI with IV MRI contrast, if CT cannot be performed at any time during the study.

b. IV contrast administration: Optimal visualisation and measurement of metastases in solid tumours require consistent administration (dose and rate) of IV contrast as well as timing of scanning. An adequate volume of a suitable contrast agent should be given so that the tumour lesions are demonstrated to best effect and a consistent method is used on subsequent examinations for any given participant. Oral contrast is recommended to help visualise and differentiate structures in the abdomen and pelvis.

c. Slice thickness and reconstruction interval: It is recommended that CT or MRI scans be acquired/reconstructed as contiguous (no gap) slices with ≤ 5 mm thickness throughout the entire anatomic region of interest for optimal lesion measurements. Exceptionally, particular institutions may perform medically acceptable scans at slice thicknesses > 5 mm. If this occurs, the minimum size of measurable lesions at baseline should be twice the slice thickness of the baseline scans.

For CT scans, all window settings should be included in the assessment, particularly in the thorax where lung and soft tissue windows should be considered. When measuring lesions, the TL should be measured on the same window setting for repeated examinations throughout the study.

Chest X-ray

Chest X-ray assessment will not be used for the assessment of TLs. Chest X-ray can, however, be used to assess NTLs and to identify the presence of NLs. However, there is preference that a higher resolution modality, such as CT, be used to confirm the presence of NLs.

Plain X-ray

Plain X-ray may be used as a method of assessment for bone NTLs and to identify the presence of new bone lesions.

Isotopic bone scan

Bone lesions identified on an isotopic bone scan at baseline and confirmed by CT, MRI, or X-ray at baseline should be recorded as NTLs and followed by the same method per baseline

assessment (CT, MRI, or X-ray).

Isotopic bone scans may be used as a method of assessment to identify the presence of new bone lesions at follow-up visits. NLs may be recorded in case positive hot-spots appear on a bone scan that were not present on a previous bone scan; however, a newly observed equivocal hot-spot on a bone scan that cannot be verified with correlative imaging (CT, MRI, or X-ray) of the same anatomical region shall not be the only trigger for a PD assessment at that timepoint.

¹⁸F-Fluoro-deoxyglucose-PET/CT

¹⁸F-fluoro-deoxyglucose PET/CT scans may be used as a method for identifying new extrahepatic lesions (but not intrahepatic lesions) for RECIST 1.1 assessments according to the following algorithm: NLs will be recorded where there is positive ¹⁸F-Fluoro-deoxyglucose uptake¹ not present on baseline or prior ¹⁸F-fluoro-deoxyglucose-PET scan or in a location corresponding to a NL on a companion CT/MRI collected close in time to the ¹⁸F-fluoro-deoxyglucose-PET scan. The PET portion of the PET/CT introduces additional data that may bias an investigator if it is not routinely or serially performed. Therefore, if there is no baseline or prior ¹⁸F-fluoro-deoxyglucose-PET scan available for comparison, and no evidence of NLs on companion CT/MRI scans, then follow-up CT/MRI assessments should continue as per the regular imaging schedule to verify the unequivocal presence of NLs.

At present, low dose or attenuation correction CT portions of a combined ¹⁸F-fluoro-deoxyglucose-PET/CT scan are of limited use in anatomically based efficacy assessments, and it is therefore suggested that they should not substitute for dedicated diagnostic contrast-enhanced CT scans for tumour measurements by RECIST 1.1. In exceptional situations, if a site can document that the CT performed, as part of a PET/CT examination, is of identical diagnostic quality (with IV contrast) to a dedicated diagnostic CT scan, then the CT portion of the PET/CT can be used for RECIST 1.1 tumour assessments. Caution that this is not recommended because the PET portion of the CT introduces additional (PET) data that may bias an investigator if it is not routinely or serially performed.

Ultrasound

Ultrasound examination will not be used for RECIST 1.1 assessment of tumours as it is not a reproducible acquisition method (operator dependent), is subjective in interpretation, and may not provide an accurate assessment of the true tumour size. Tumours identified by ultrasound will need to be assessed by correlative CT or MRI anatomical scan.

1 A positive ¹⁸F-fluoro-deoxyglucose-PET scan lesion should be reported only when an uptake (eg, standard uptake value) greater than twice that of the surrounding tissue or liver is observed.

Other Tumour Assessments

Clinical examination

Clinical examination of skin/surface lesions (by visual inspection or manual palpation) will not be used for RECIST 1.1 assessments. Tumours identified by clinical examination will need to be assessed by correlative CT or MRI anatomical scans.

Endoscopy and laparoscopy

Endoscopy and laparoscopy will not be used for tumour assessments as they are not validated in the context of tumour assessment.

Histology and cytology

Histology or tumour markers on tumour biopsy samples will not be used as part of the tumour response assessment as per RECIST 1.1.

Results of cytological examination for the neoplastic origin of any effusion (eg, ascites, pericardial effusion, and pleural effusion) that appears or worsens during the study will not be used as part of the tumour response assessment as per RECIST 1.1.

Furthermore, an overall assessment of CR (all other disease disappears/reverts to normal) would be changed to PR if an effusion remains present radiologically.

Measurability of Tumour Lesions at Baseline

RECIST 1.1 measurable lesions at baseline

A tumour lesion that can be accurately measured at baseline as ≥ 10 mm in the longest diameter for non-nodal lesions or ≥ 15 mm in short axis² diameter for lymph node lesions with IV contrast-enhanced CT or MRI and that is suitable for accurate repeated measurements. Please see additional RECIST 1.1 guidance below on measurability of intrahepatic hepatocellular carcinoma lesions and porta hepatis lymph nodes.

2 The short axis is defined as the longest in-plane axis perpendicular to the long axis.

Non-measurable lesions at baseline

- Truly non-measurable lesions include the following:
 - Bone lesions (see exception below for soft tissue component).
 - Leptomeningeal disease.
 - Ascites, pleural effusion, or pericardial effusion.
 - Inflammatory breast disease.
 - Lymphangitic involvement of skin or lung.
- All other lesions, including small lesions (longest diameter < 10 mm or pathological lymph nodes with ≥ 10 mm to < 15 mm short axis diameter at baseline).³
- Previously irradiated lesions.⁴
- Brain metastasis.

Special considerations regarding lesion measurability at baseline

- Bone lesions:
 - Bone scan, PET scan, or plain X-ray are not considered adequate imaging techniques to measure bone lesions; however, these techniques can be used to confirm the presence or disappearance of bone lesions.
 - Lytic bone lesions or mixed lytic-blastic lesions, with identifiable soft tissue components, can be considered measurable if the soft tissue component meets the definition of measurability.
 - Blastic lesions are considered non-measurable.
- Cystic lesions thought to represent cystic metastases can be considered measurable lesions if they meet the criteria for measurability from a radiological point of view, but if non-cystic lesions are present in the same participant, these should be selected over cystic lesions as TLs.

RECIST 1.1 TL selection at baseline

A maximum of 5 measurable lesions, with a maximum of 2 lesions per organ (including lymph nodes collectively considered as a single organ), representative of all lesions involved should be identified as TLs at baseline. TLs should be selected on the basis of their size (longest diameter for non-nodal lesions or short axis diameter for nodal lesions), but in addition should be those that lend themselves to reproducible repeated measurements. It may

3 Lymph nodes with < 10 mm short axis diameter are considered non-pathological and should not be recorded or followed as NTLs.

4 Localised post-radiation changes that affect lesion size may occur. Therefore, lesions that have been previously irradiated are typically considered non-measurable and as NTL at baseline and followed up as part of the NTL assessment.

be the case that, on occasion, the largest lesion does not lend itself to reproducible measurement, in which circumstance the next largest lesion that can be measured reproducibly should be selected.

Lymph nodes, in any location (local/regional and distant), are collectively considered as a single organ, with a maximum of 2 lymph nodes as TLs. A bilateral organ (eg, adrenal glands), a segmented organ (eg, liver), or a multilobed organ (eg, lung) is each considered as a single organ.

The site and location of each TL should be documented, as well as the longest axis diameter for non-nodal lesions (or short axis diameter for lymph nodes). All measurements should be recorded in millimetres. At baseline, the sum of the diameters for all TLs will be calculated and reported as the baseline sum of diameters. At follow-up visits, the sum of diameters for all TLs will be calculated and reported as the follow-up sum of diameters.

Special cases for TL assessment at baseline

- For TLs measurable in 2 or 3 dimensions, always report the longest diameter. For pathological lymph nodes measurable in 2 or 3 dimensions, always report the short axis diameter.
- When lymph nodes are coalesced and no longer separable in a conglomerate mass, the vector of the longest diameter should be used to determine the perpendicular vector for the maximal short axis diameter of the coalesced mass. Non-nodal lesions that coalesce should similarly be assessed by the longest axis diameter.
- Tumour lesions selected for newly acquired screening biopsy should not be selected as TLs, unless imaging occurred at least approximately 2 weeks after biopsy, allowing time for healing.
- If the CT/MRI slice thickness used is > 5 mm, the minimum size of measurable disease at baseline should be twice the slice thickness of the baseline scan.
- If a lesion has completely disappeared, the diameter should be recorded as 0 mm. If a lesion appears in the same location on a subsequent scan, it will be recorded as a NL.

RECIST 1.1 NTL selection at baseline

All other lesions, including non-measurable lesions and surplus measurable lesions, not recorded as TLs should be identified as NTLs at baseline. Measurements of these lesions are not required, but the presence or absence of each should be noted throughout follow-up.

Evaluation of Tumour Response and Progression

RECIST 1.1 TL assessment at follow-up

This section defines the criteria used to determine objective tumour visit response for

RECIST 1.1-defined TLs. The imaging modality, location, and scan date of each TL identified previously at baseline should be documented at follow-up visits with the long axis diameter for non-nodal lesions or short axis diameter for lymph node lesions. All measurements should be recorded in millimetres. The sum of the diameters for all TLs at each follow-up visit will be compared with the baseline sum of diameters (for response or SD) or to the smallest prior (nadir) sum of diameters (for progression).

Special cases for TL assessment at follow-up:

- If a lesion has completely disappeared, the diameter should be recorded as 0 mm. If a lesion appears in the same location on a subsequent scan, it will be recorded as an NL.
- If a TL splits into 2 or more parts, the sum of the diameters of those parts should be recorded.
- If 2 or more TLs merge, then the sum of the diameters of the combined lesion should be recorded for 1 of the lesions and 0 mm recorded for the other lesion(s). If the merged TLs are non-nodal lesions, record the long axis diameter of the merged lesion. If pathologic lymph nodes coalesce and are no longer individually separable within a conglomerate mass, the vector of the longest diameter of the coalesced mass should be used to determine the perpendicular vector for the maximal short axis diameter.
- If a TL is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned. If an accurate measure can be given, this should be recorded, even if it is below 5 mm.
- If a TL cannot be measured accurately due to it being too large, provide an estimate of the size of the lesion. The choice of “Too large to measure” in the eCRF will trigger an overall visit response of PD.
- When a TL has had any intervention (eg, definitive radiotherapy, embolisation, surgery, transarterial chemoembolisation, etc.) during the study, the size of the TL should still be provided where possible and the intervention recorded in the RECIST 1.1 CRF for the current imaging visit and all subsequent visits. If a TL has been completely removed (surgery) or disappears, the longest diameter should be recorded as 0 mm.

Table 14 **RECIST 1.1 Evaluation of Target Lesions**

CR	Disappearance of all TLs since baseline. Any pathological lymph nodes selected as TLs must have a reduction in short axis diameter to < 10 mm.
PR	At least a 30% decrease in the sum of the diameters of TL, taking as reference the baseline sum of diameters.
SD	Neither sufficient decrease in the sum of diameters to qualify for PR nor sufficient increase to qualify for PD.
PD	At least a 20% increase in the sum of diameters of TLs, taking as reference the smallest previous sum of diameters (nadir). This includes the baseline sum if that is the smallest on study. In addition to the relative increase of 20%, the sum must demonstrate an absolute increase of at least 5 mm from nadir.
NE	Only relevant if any of the TLs at follow-up were not assessed or NE (eg, missing anatomy) or had a lesion intervention at this visit. Note: If the sum of diameters meets the PD criteria, PD overrides NE as a TL response.
Not applicable	Only relevant if no TLs present at baseline.

CR, complete response; NE, not evaluable; PD, progression of disease; PR, partial response; SD, stable disease; TL, target lesion.

RECIST 1.1 NTL assessment at follow-up

All other lesions (or sites of disease) not recorded as TLs should be identified as NTLs at baseline. Measurements are not required for these lesions, but their status should be followed at subsequent visits. At each visit, an overall assessment of the NTL response should be recorded by the investigator.

To achieve “unequivocal progression” on the basis of NTLs, there must be an overall level of substantial worsening in non-target disease such that, even in presence of SD or PR in TLs, the overall tumour burden has increased sufficiently to merit unequivocal progression by NTLs. A modest “increase” in the size of 1 or more NTLs is usually not sufficient to qualify for unequivocal progression status. The designation of overall progression solely on the basis of change in non-target disease in the face of SD or PD of target disease will therefore be extremely rare.

Table 15 **RECIST 1.1 Evaluation of Non-Target Lesions**

CR	Disappearance of all NTLs since baseline. All lymph nodes must be non-pathological in size (<10 mm short axis).
Non-CR/non-PD	Persistence of 1 or more NTLs.
PD	Unequivocal progression of existing NTLs. Unequivocal progression may be due to an important progression in 1 lesion only or in several lesions. In all cases, the progression MUST be clinically significant for the physician to consider changing (or stopping) therapy.
NE	Only relevant when 1 or some of the NTLs were not assessed and, in the investigator's opinion, they are not able to provide an evaluable overall NTL assessment at this visit. Note: For participants without TLs at baseline, this is relevant if any of the NTLs were not assessed at this visit and the progression criteria have not been met.
Not applicable	Only relevant if no NTLs present at baseline.

CR, complete response; NE, not evaluable; NTL, non-target lesion; PD, progression of disease; TL, target lesion.

RECIST 1.1 NL identification at follow-up

Details, including the imaging modality, the date of scan, and the location of any NLs will also be recorded in the eCRF. The presence of 1 or more NLs is assessed as progression. The finding of a NL should be unequivocal, ie, not attributable to differences in scanning technique, change in imaging modality, or findings thought to represent something other than tumour. If a NL is equivocal, for example because of its small size, the treatment and tumour assessments should be continued until the previously (pre-existing) NL has been assessed as unequivocal at a follow-up visit, and then the progression date should be declared using the date of the initial scan when the NL first appeared.

A lesion identified at a follow-up assessment in an anatomical location that was not scanned at baseline is considered a NL and will indicate PD.

RECIST 1.1 evaluation of overall visit response at follow-up

Derivation of overall visit response as a result of the combined assessment of TLs, NTLs, and NLs uses the algorithm shown in [Table 16](#).

Table 16 **RECIST 1.1 Overall Visit Response**

Target lesions	Non-target lesions	New lesions	Overall visit response
CR	CR	No	CR
CR	NA	No	CR
NA	CR	No	CR
CR	Non-CR/Non-PD	No	PR
CR	NE	No	PR
PR	Non-PD or NE or NA	No	PR
SD	Non-PD or NE or NA	No	SD
NA	Non-CR/Non-PD	No	SD (non-CR/non-PD)
NE	Non-PD or NE	No	NE
NA	NE	No	NE
NA	NA	No	NED
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

Non-CR/Non-PD for overall response if only NTL (no TLs) are present at baseline.

Note: An overall assessment of CR (all other disease disappears/reverts to normal) would be changed to PR if ascites remains present radiologically.

CR, complete response; NA, not applicable (only relevant if there were no TLs at baseline or NTLs at baseline), NE, not evaluable; NED, no evidence of diseases (only relevant if there were neither TLs nor NTLs at baseline); NTL, non-target lesion; PD, progression of disease; PR, partial response; SD, stable disease; TL, target lesion.

The following overall visit responses are possible depending on the extent of tumour disease at baseline:

- For participants with TLs (at baseline): CR, PR, SD, PD, or NE.
- For participants with NTLs only (at baseline): CR, Non-CR/Non-PD, PD, or NE.
- For participants with no disease at baseline: no evidence of disease (available as an option in the eCRF), PD, or NE.

Evaluation of scans subsequent to RECIST 1.1-defined progression

A follow-up scan is requested at least 4 weeks after a RECIST 1.1-defined radiological progression and no longer than the next regularly scheduled imaging visit. The follow-up scans provide additional information to the investigator for patient management and further treatment decisions, and since the published RECIST 1.1 criteria ([Eisenhauer et al 2009](#)) do not provide guidance on how to assess scans acquired after RECIST 1.1-defined PD, supplemental instructions for investigators on how to evaluate these follow-up scans are provided below. An immediate prior RECIST 1.1-defined radiologic PD would be considered

confirmed if *any* of the following criteria are met in the subsequent follow-up scan:

- $\geq 20\%$ increase and at least a 5 mm increase in the sum of diameters of TLs compared with the nadir sum of diameters at 2 consecutive visits, and a further increase of ≥ 5 mm in the sum of diameters at the follow-up scan timepoint compared with the immediate prior timepoint.
- Significant progression (worsening) of NTLs at the follow-up scan timepoint compared with the immediate prior timepoint.
- Significant progression (worsening) of previously NLs (pre-existing NLs) at the follow-up scan timepoint compared with the immediate prior timepoint.
- Additional brand-new unequivocal lesions at the follow-up scan timepoint.

Central imaging

Images, including unscheduled visit scans, will be collected on an ongoing basis and sent to an AstraZeneca-appointed iCRO for quality control and storage. Digital copies of all original scans should be stored at the investigator site as source documents. Electronic image transfer from the sites to the iCRO is strongly encouraged.

The management of patients will be based upon the results of the tumour assessments conducted by the investigator.

References

Eisenhauer et al 2009

Eisenhauer EA, Therasse P, Bogaerts J, Schwartz LH, Sargent D, Ford R, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer* 2009;45(2):228-47.

Appendix G Contraception Requirements

Contraception requirements for this study are as follows.

G 1 Female Participants

Women not of childbearing potential are defined as those who are surgically sterile (ie, bilateral salpingectomy, bilateral oophorectomy, or complete hysterectomy) or who are post-menopausal.

Women will be considered post-menopausal if they have been amenorrhoeic for 12 months without an alternative medical cause. The following age-specific requirements apply:

- Women < 50 years of age would be considered post-menopausal if they have been amenorrhoeic for 12 months or more following cessation of all hormonal replacement therapy and if they have luteinising hormone and follicle-stimulating hormone levels in the post-menopausal range for the institution.
- Women $\geq$ 50 years of age would be considered post-menopausal if they have been amenorrhoeic for 12 months or more following cessation of all hormonal replacement therapy or had radiation-induced menopause with last menses > 1 year ago or had chemotherapy-induced menopause with last menses > 1 year ago.

Women of childbearing potential who are not totally sexually abstinent (ie, refraining from heterosexual intercourse during the entire period of risk associated with study interventions) and intend to be sexually active with a non-sterilised male partner must use at least one highly effective method of contraception ([Table 17](#)). They should have been stable on their chosen method of birth control for a minimum of 3 months before entering the study and continue to use it throughout the total duration of the study treatment and for 3 months (Group A) or 6 months (Group B) after the last dose of study intervention.

Non-sterilised male partners of a WOCBP must use a male condom plus spermicide (condom alone in countries where spermicides are not approved) throughout this period. Cessation of birth control after this point should be discussed with a responsible physician. Periodic abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception. Total sexual abstinence is an acceptable method provided it is the usual lifestyle of the participant. Female participants should refrain from breastfeeding throughout this period.

G 2 Male Participants

Male participants in Group A should use condom with spermicide from the time of screening, throughout the total duration of the study, and up to 1 week post last dose with all sexual partners.

Non-sterilised male participants (including males sterilised by a method other than bilateral orchidectomy, eg, vasectomy) who intend to be sexually active with a WOCBP partner must be using an acceptable method of contraception such as male condom plus spermicide (condom alone in countries where spermicides are not approved) from the time of screening throughout the total duration of the study and for 6 months after the last dose of ceralasertib plus durvalumab combination therapy to prevent pregnancy in a partner. Periodic abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception.

Vasectomised males are considered fertile and should still use a male condom plus spermicide as indicated above during the clinical study.

Even if the female partner is pregnant, male participants should still use a condom plus spermicide (where approved), as indicated above during the clinical study, if there is a concern about damaging the developing foetus from drug in ejaculate.

Male participants should refrain from sperm donation or banking from the time of screening throughout the total duration of the study and for 6 months after the last dose of ceralasertib plus durvalumab combination therapy.

WOCBP partners of male participants with must also use a highly effective method of contraception throughout this period ([Table 17](#)).

G 3 Highly Effective Methods of Contraception

Highly effective methods of contraception, defined as one that results in a low failure rate (ie, less than 1% per year) when used consistently and correctly, are described in [Table 17](#). Note that some contraception methods are not considered highly effective (eg, male or female condom with or without spermicide; female cap, diaphragm, or sponge with or without spermicide; non-copper containing intrauterine device; progestogen-only oral hormonal contraceptive pills where inhibition of ovulation is not the primary mode of action [excluding Cerazette/desogestrel which is considered highly effective]; and triphasic combined oral contraceptive pills).

Participants receiving docetaxel should follow the local Prescribing Information relating to contraception, the time limits for such precautions, and any additional restrictions.

Table 17 Highly Effective Methods of Contraception (< 1% Failure Rate)

Non-hormonal methods	Hormonal methods
• Total sexual abstinence (evaluate in relation to the duration of the clinical study and the preferred and usual lifestyle choice of the participant) • Vasectomised sexual partner (with participant assurance that partner received post-vasectomy confirmation of azoospermia) • Tubal occlusion • Intrauterine device (provided coils are copper-banded)	• Injection: Medroxyprogesterone injection (eg, Depo-Provera[®])^{a,b} • Levonorgestrel-releasing intrauterine system (eg, Mirena[®])^{a,b} • Implants: Etonogestrel-releasing implants (eg, Implanon[®] or Norplant[®])^b • Intravaginal devices: Ethinylestradiol/etonogestrel-releasing intravaginal devices (eg, NuvaRing[®])^b • Combined pill: Normal and low dose combined oral contraceptive pill • Patch: Norelgestromin/ethinylestradiol-releasing transdermal system (eg, Ortho Evra[®])^b • Mini pill: Progesterone-based oral contraceptive pill using desogestrel: Cerazette[®]^b is currently the only highly effective progesterone-based pill

a Hormonal methods not prone to drug-drug interactions.

b Not approved for use in Japan.

Appendix H Clinical Outcomes Assessment

H 1.1 Non-Small Cell Lung Cancer Symptom Assessment Questionnaire (NSCLC-SAQ)

Items of the *Non-Small Cell Lung Cancer Symptom Assessment Questionnaire (NSCLC-SAQ)*

1. How would you rate your coughing at its worst over the last 7 days?
Response scale for item 1: Verbal rating scale with the following descriptors: No coughing at all/Mild coughing/Moderate coughing/Severe coughing/Very severe coughing
2. How would you rate the worst pain in your chest over the last 7 days?
3. How would you rate the worst pain in areas other than your chest over the last 7 days?
Response scale for items 2-3: Verbal rating scale with the following descriptors: No pain at all/Mild pain/Moderate pain/Severe pain/Very severe pain
4. How often did you feel short of breath during usual activities over the last 7 days?
5. How often did you have low energy over the last 7 days?
6. How often did you tire easily over the last 7 days?
7. How often did you have a poor appetite over the last 7 days?
Response scale for items 4-7: Verbal rating scale with the following descriptors: Never/Rarely/Sometimes/Often/Always

H 1.2 Physical Function Scale (EORTC IL 171)

ENGLISH



EORTC IL171

We are interested in some things about you and your health. Please answer all of the questions yourself by circling the number that best applies to you. There are no "right" or "wrong" answers. The information that you provide will remain strictly confidential.

	Not at All	A Little	Quite a Bit	Very Much
1. Do you have any trouble doing strenuous activities, like carrying a heavy shopping bag or a suitcase?	1	2	3	4
2. Do you have any trouble taking a <u>long</u> walk?	1	2	3	4
3. Do you have any trouble taking a <u>short</u> walk outside of the house?	1	2	3	4
4. Do you need to stay in bed or a chair during the day?	1	2	3	4
5. Do you need help with eating, dressing, washing yourself or using the toilet?	1	2	3	4
6. Do you have any trouble walking up a flight of stairs?	1	2	3	4
7. Do you have any trouble running a short distance, such as to catch the bus?	1	2	3	4

During the past week:

	Not at All	A Little	Quite a Bit	Very Much
8. Have you felt slowed down?	1	2	3	4

H 1.3 Patient-Reported Outcomes Common Terminology Criteria for Adverse Events (PRO-CTCAE)

NCI PRO-CTCAE™ ITEMS

Item Library Version 1.0

English

Form Created on 25 January 2022

As individuals go through treatment for their cancer they sometimes experience different symptoms and side effects. For each question, please select the one response that best describes your experiences over the past 7 days...

1a. In the last 7 days, what was the SEVERITY of your MOUTH OR THROAT SORES at their WORST?				
☐ None	☐ Mild	☐ Moderate	☐ Severe	☐ Very severe
1b. In the last 7 days, how much did MOUTH OR THROAT SORES INTERFERE with your usual or daily activities?				
☐ Not at all	☐ A little bit	☐ Somewhat	☐ Quite a bit	☐ Very much

2a. In the last 7 days, how OFTEN did you have NAUSEA?				
☐ Never	☐ Rarely	☐ Occasionally	☐ Frequently	☐ Almost constantly
2b. In the last 7 days, what was the SEVERITY of your NAUSEA at its WORST?				
☐ None	☐ Mild	☐ Moderate	☐ Severe	☐ Very severe

3a. In the last 7 days, how OFTEN did you have VOMITING?				
☐ Never	☐ Rarely	☐ Occasionally	☐ Frequently	☐ Almost constantly
3b. In the last 7 days, what was the SEVERITY of your VOMITING at its WORST?				
☐ None	☐ Mild	☐ Moderate	☐ Severe	☐ Very severe

4a. In the last 7 days, what was the SEVERITY of your CONSTIPATION at its WORST?				
☐ None	☐ Mild	☐ Moderate	☐ Severe	☐ Very severe

5a. In the last 7 days, how OFTEN did you have LOOSE OR WATERY STOOLS (DIARRHEA/DIARRHOEA)?				
☐ Never	☐ Rarely	☐ Occasionally	☐ Frequently	☐ Almost constantly

The PRO-CTCAE™ items and information herein were developed by the Division of Cancer Control and Population Sciences in the NATIONAL CANCER INSTITUTE at the NATIONAL INSTITUTES OF HEALTH, in Bethesda, Maryland, U.S.A. Use of the PRO-CTCAE™ is subject to NCI's Terms of Use.

6a. In the last 7 days, did you have any RASH?	
☐ Yes	☐ No

7a. In the last 7 days, did you have any HAIR LOSS?				
☐ Not at all	☐ A little bit	☐ Somewhat	☐ Quite a bit	☐ Very much

8a. In the last 7 days, what was the SEVERITY of your NUMBNESS OR TINGLING IN YOUR HANDS OR FEET at its WORST?				
☐ None	☐ Mild	☐ Moderate	☐ Severe	☐ Very severe
8b. In the last 7 days, how much did NUMBNESS OR TINGLING IN YOUR HANDS OR FEET INTERFERE with your usual or daily activities?				
☐ Not at all	☐ A little bit	☐ Somewhat	☐ Quite a bit	☐ Very much

The PRO-CTCAE™ items and information herein were developed by the Division of Cancer Control and Population Sciences in the NATIONAL CANCER INSTITUTE at the NATIONAL INSTITUTES OF HEALTH, in Bethesda, Maryland, U.S.A. Use of the PRO-CTCAE™ is subject to NCI's Terms of Use.

H 1.4 Patient Global Impression of Treatment Tolerability (PGI-TT)

PGI-TT

PATIENT GLOBAL IMPRESSION OF TREATMENT TOLERABILITY (PGI-TT)

In the last 7 days, how bothered were you by the side effects of your cancer treatment?

- ☐ Not at all
- ☐ A little bit
- ☐ Somewhat
- ☐ Quite a bit
- ☐ Very much

H 1.5 Patient Global Impression of Treatment Tolerability (PGI-C) and Patient Global Impression of Severity (PGI-S)

Patient Global Impression of Severity - Cancer (PGIS-Cancer-4-Item)

Please select the response below that best describes the severity of your overall cancer symptoms over the past 7 days.

Please select one response only

- ☐ None
- ☐ Mild
- ☐ Moderate
- ☐ Severe

PATIENT GLOBAL IMPRESSION OF CHANGE - GENERIC (PGIC-GENERIC)

Overall, how would you rate the change in your health status since starting this study?

Please select one response only

- ☐ Much better
- ☐ Moderately better
- ☐ A little better
- ☐ About the same
- ☐ A little worse
- ☐ Moderately worse
- ☐ Much worse

H 1.6 Modified EORTC QLQ-C30 (EORTC IL 172)

ENGLISH



EORTC IL172

We are interested in some things about you and your health. Please answer all of the questions yourself by circling the number that best applies to you. There are no "right" or "wrong" answers. The information that you provide will remain strictly confidential.

During the past week:

	Not at All	A Little	Quite a Bit	Very Much
1. Were you limited in doing either your work or other daily activities?	1	2	3	4
2. Were you limited in pursuing your hobbies or other leisure time activities?	1	2	3	4
3. Have you had difficulty in concentrating on things, like reading a newspaper or watching television?	1	2	3	4
4. Did you feel tense?	1	2	3	4
5. Did you worry?	1	2	3	4
6. Did you feel irritable?	1	2	3	4
7. Did you feel depressed?				
8. Have you had difficulty remembering things?	1	2	3	4
9. Has your physical condition or medical treatment interfered with your <u>family</u> life?	1	2	3	4
10. Has your physical condition or medical treatment interfered with your <u>social</u> activities?	1	2	3	4

For the following questions please circle the number between 1 and 7 that best applies to you

11. How would you rate your overall health during the past week?

1 2 3 4 5 6 7

Very poor

Excellent

12. How would you rate your overall quality of life during the past week?

1 2 3 4 5 6 7

Very poor

Excellent

H 1.7 EuroQoL 5-dimension, 5-level Health State Utility Index (EQ-5D-5L)

Appendix I Concomitant Medications

I 1 Guidance Regarding Potential Interactions with Concomitant Medications

The use of any natural/herbal products or other “folk remedies” should be discouraged, but use of these products, as well as use of all vitamins, nutritional supplements, and all other concomitant medications must be recorded in the eCRF.

I 1.1 Guidance Regarding Potential Interactions of Ceralasertib with Concomitant Medications

There are currently no data confirming that there is a pharmacokinetic (PK) interaction between these agents and ceralasertib; a potential interaction is considered on the basis of preclinical and in vitro data only. Ceralasertib is predominantly eliminated via CYP3A metabolism, therefore CYP3A inhibitors or inducers may increase or decrease exposure to ceralasertib, respectively. Potent inhibitors or inducers of CYP3A should not be combined with ceralasertib. In vitro data also suggest that ceralasertib may be metabolised by CYP2C8 but a lesser extent, therefore caution should be applied with co-administration of potent inhibitors or inducers of CYP2C8.

These lists are not intended to be exhaustive, and similar restrictions will apply to other agents that are known to modulate CYP3A or CYP2C8 activity. Please contact AstraZeneca with any queries you have on this issue. Please refer to full Prescribing Information for all drugs prior to co-administration with ceralasertib.

Drugs Known to be Inhibitors of CYP3A4

Potent CYP3A inhibitors	Potent CYP3A inducers
boceprevir ceritinib clarithromycin cobicistat (GS-9350) conivaptan danoprevir / RIT elvitegravir / RIT grapefruit juice ^a idelalisib indinavir indinavir /RIT itraconazole ketoconazole LCL161 lopinavir / RIT mibefradil mifepristone nefazodone nelfinavir posaconazole ribociclib ritonavir saquinavir saquinavir / RIT telaprevir telithromycin tipranavir/RIT troleandomycin VIEKIRA PAK2 ^b voriconazole	apalutamide avasimibe carbamazepine enzalutamide ivosidenib lumacaftor mitotane phenobarbital phenytoin rifampin rifapentine St John's Wort extract

- c Double-strength grapefruit juice. Patients should abstain from eating large amounts of grapefruit and Seville oranges (and other products containing these fruits eg, grapefruit juice or marmalade) during the study (eg, no more than a small glass of grapefruit juice [120 mL] or half a grapefruit or 1-2 teaspoons [15 g] of Seville orange marmalade daily).
- d VIEKIRA PAK = 150/100 mg paritaprevir/ritonavir + 25 mg ombitasvir + 800 mg dasabuvir for 28 days.

List created using the University of Washington Drug-Drug Interaction Database July 2019.

RIT = Ritonavir.

Ritonavir has dual effects of simultaneous CYP3A inhibition and induction, and the net pharmacokinetic outcome during chronic ritonavir therapy is inhibition of CYP3A activity

Drugs Known to be Inhibitors and Inducers of CYP2C8

Potent CYP2C8 inhibitors	Potent CYP2C8 inducers
Gemfibrozil clopidogrel	None identified

List created using the University of Washington Drug-Drug Interaction Database Jan 2020.

Drugs known to be inhibitors or inducers of P-gp and/or BCRP, undertake appropriate monitoring if co-administration is necessary

Ceralasertib is a substrate of P-gp and BCRP. Co-administration of P-gp inhibitors/inducers or BCRP inhibitors/inducers may affect exposure to ceralasertib therefore it is recommended that these are not co-administered with ceralasertib.

These lists are not intended to be exhaustive, and similar restrictions will apply to other agents that are known to modulate P-gp activity or BCRP activity. Please contact AstraZeneca with any queries you have on this issue. Please refer to full Prescribing Information for all drugs prior to co-administration with ceralasertib.

Drugs Known to be Inhibitors or Inducers of P-gp

Drugs Known to be Inhibitors of P-gp	Drugs Known to be Inducers of P-gp
alogliptin amiodarone asian ginseng (Panax ginseng) asunaprevir AZD5672 azithromycin canagliflozin captopril carvedilol clarithromycin clopidogrel cobicstat conivaptan cremophor EL cremophor RH curcumin daclatasvir daclatasvir/asunaprevir/beclabuvir diltiazem diosmin	apalutamide avasimibe carbamazepine danshen (Salvia miltiorrhiza) efavirenz genistein green tea phenytoin quercetin rifabutin rifampin ritonavir St. John's wort extract tivantinib

Drugs Known to be Inhibitors of P-gp	Drugs Known to be Inducers of P-gp
dronedarone elagolix eliglustat erythromycin felodipine five-flavour berry (schisandra chinensis) flibanserin fluvoxamine fostamatinib ginkgo glecaprevir/pibrentasvir indinavir indinavir/ritonavir isavuconazole itraconazole ivacaftor ketoconazole lapatinib lopinavir/ritonavir mibefradil mifepristone milk thistle mirabegron nelfinavir neratinib nifedipine nitrendipine osimertinib paritaprevir/ritonavir/ombitasvir paroxetine piperine propafenone quercetin quinidine quinine ranolazine rifampin ritonavir rolapitant rucaparib saquinavir/ritonavir	

Drugs Known to be Inhibitors of P-gp	Drugs Known to be Inducers of P-gp
sarecycline simeprevir sofosbuvir/velpatasvir/voxilaprevir St. John's wort extract surfactant TPGS suvorexant talinolol telithromycin telaprevir telmisartan tezacaftor/ivacaftor ticagrelor tipranavir/ritonavir tolvaptan valbenazine valspodar (PSC 833) vandetanib velpatasvir vemurafenib verapamil voclosporin vorapaxar	

Drugs Known to be Inhibitors or Inducers of BCRP

Drugs Known to be Inhibitors of BCRP	Drugs Known to Be Inducers of BCRP
Afatinib Aripiprazole Curcumin Cyclosporine Elacridar Erlotinib Fluvastatin Fumitremorgin Gefitinib Ivermectin Lapatinib Nilotinib Novobiocin Pantoprazole	Please check individual drugs on a case-by-case basis

Drugs Known to be Inhibitors of BCRP	Drugs Known to Be Inducers of BCRP
Pitavastatin Ponatinib Quercetin Quizartinib Rabeprazole Regorafenib Rilpivirine Sulfasalazine Sunitinib Tacrolimus Teriflunomide Trametinib Trifluoperazine Vismodegib eltrombopag Atazanavir Lopinavir Ritonavir Tipranavir Omeprazole Estrone 17b-estradiol Imatinib mesylate	

List created using <http://dmd.aspetjournals.org/content/dmd/43/4/490.full.pdf>

Note: Although BCRP is involved in a number of clinically relevant drug-drug interaction, none of the cited inhibitors above is truly specific for this transporter

Drugs known to be substrates of CYP3A4 undertake appropriate monitoring if co-administration is necessary

Ceralasertib is a potential inducer of CYP3A4. Therefore, caution should be applied with co-administration of drugs that are either completely metabolised by CYP3A4, or that are substrates of CYP3A4, and also have a narrow therapeutic index. Investigators should be aware that the exposure of other drugs metabolised by CYP3A4 may be reduced.

Drugs Known to be Metabolised by CYP3A4 Have a Narrow Therapeutic Index

Metabolised by CYP3A4 and with a narrow therapeutic index
alfentanil astemizole cisapride cyclosporine diergotamine

Metabolised by CYP3A4 and with a narrow therapeutic index
ergotamine fentanyl pimozide quinidine sirolimus tacrolimus terfenadine

List created using the University of Washington Drug-Drug Interaction Database October 2019.

Note: This is not an exhaustive list.

Drugs known to be substrates of OATP1B1 and BCRP, undertake appropriate monitoring if co-administration is necessary

Ceralasertib is also an inhibitor of OATP1B1 and BCRP. Caution should be applied with co-administration of substrates of OATP1B1 and/or BCRP as ceralasertib may increase their exposure.

These lists are not intended to be exhaustive and appropriate medical judgement is required. Please contact AstraZeneca with any queries you have on this issue. Please refer to full Prescribing Information for all drugs prior to co-administration with ceralasertib.

Drugs Known to be Substrates of OATP1B1

docetaxel enalapril olmesartan phalloidin repaglinide statins* temocaprilat valsartan
--

* all statins

List created using <https://www.solvobiotech.com/transporters/OATP1B1>, latest access Nov 2019

Drugs known to be substrates of BCRP

anthracyclines chlorothiazide daunorubicin doxorubicin imatinib irinotecan methotrexate

mitoxantrone
nucleoside analogues
pantoprazole
prazosin
SN-38
topotecan
teriflunomide
rosuvastatin

List created using <https://www.solvobiotech.com/transporters/bcrp>, latest access Nov 2019

DRUGS INDUCING OR INHIBITING CYP3A4/5 METABOLISM THAT ASTRAZENECA STRONGLY RECOMMENDS ARE NOT COMBINED WITH STUDY INTERVENTION

Ceralasertib is metabolised by CYP3A4 and CYP3A5 enzymes. Therefore, inhibitors or inducers of CYP3A4/5 may increase or decrease exposure, respectively, to ceralasertib.

Strong inhibitors and strong inducers of CYP3A4/5 should not be combined with ceralasertib intervention and should be stopped at least 2 weeks before the first dose of ceralasertib (3 weeks for St John's Wort).

Moderate inhibitors and inducers of CYP3A4/5 are permitted, but caution should be exercised and participants monitored closely for possible drug interactions.

The following are drugs known to be inducers of CYP3A4/5 ([Table 18](#)).

Table 18 **Drugs Known to be Inducers of CYP3A4/5**

Strong CYP3A4/5 inducers (should not be combined)	Moderate CYP3A4/5 inducers (permitted with caution)
avasimibe carbamazepine enzalutamide mitotane nevirapine phenobarbital phenytoin rifabutin rifampin rifapentine St John's Wort	semagacestat ^a talviraline ^a bosentan efavirenz etravirine genistein lersivirine lopinavir modafinil nafcillin thioridazine tipranavir and ritonavir ^b

a Not available on the United States market.

b Ritonavir has dual effects of simultaneous CYP3A inhibition and induction; the net pharmacokinetic outcome during chronic ritonavir therapy is inhibition of CYP3A activity.

The following are drugs known to be inhibitors of CYP3A4/5 ([Table 19](#)).

Table 19 Drugs Known to be Inhibitors of CYP3A4/5

Strong CYP3A4/5 inhibitors (should not be combined)	Moderate CYP3A4/5 inhibitors (permitted with caution)
boceprevir clarithromycin conivaptan elvitegravir/ritonavir fluconazole grapefruit juice ^{a,b} indinavir itraconazole ketoconazole lopinavir/RIT mibefradil nefazodone nelfinavir posaconazole ritonavir saquinavir	amprenavir aprepitant atazanavir casopitant cimetidine ciprofloxacin crizotinib cyclosporine darunavir diltiazem dronedarone erythromycin grapefruit juice ^b imatinib schisandra sphenanthera tofisopam
telaprevir telithromycin tipranavir/ritonavir troleandomycin voriconazole	verapamil

a Double-strength grapefruit juice

b Participants should abstain from eating large amounts of grapefruit and Seville oranges (and other products containing these fruits eg, grapefruit juice or marmalade) during the study (eg, no more than a small glass of grapefruit juice [120 mL] or half a grapefruit or 1 to 2 teaspoons [15 g] of Seville orange marmalade daily)

This list is not intended to be exhaustive, and a similar restriction will apply to other agents that are known to strongly modulate CYP3A4/5 activity. Appropriate medical judgement is required. Please contact AstraZeneca with any queries you have on this issue.

If the investigator feels that concomitant administration of medications or herbal supplements that strongly modulate CYP3A4/5 is essential (eg, to treat AEs) ceralasertib treatment should be discontinued.

I 2 Restricted, Prohibited, and Permitted Concomitant Medications/Therapies

Restricted, prohibited, and permitted concomitant medications/therapies are described in [Table 20](#) and [Table 21](#). Refer also to the dose modification guidelines for management of study intervention-related toxicities in the Annex document to this CSP.

Table 20 Prohibited Medications/Therapies

Prohibited medication/class of drug/therapy	Usage
For all treatment arms	
Any investigational anticancer therapy other than those under investigation in this study	Must not be given concomitantly while the participant is on study intervention.
mAbs against CTLA-4, PD-1, or PD-L1 other than those under investigation in this study	Must not be given concomitantly while the participant is on study intervention.
Any concurrent treatment, including chemotherapy, radiotherapy, immunotherapy, or biologic or hormonal therapy for cancer treatment other than those under investigation in this study, except for those medications identified as “restricted,” as listed above	Must not be given concomitantly whilst the participant is on study intervention. Local treatment of isolated lesions, excluding target lesions, for palliative intent is acceptable [eg, by local surgery or radiotherapy]).
Live attenuated vaccines	Must not be given while receiving study intervention and up to 30 days after the last dose of study intervention. Local guidance should be consulted to determine the acceptable timeframe for vaccine administration following chemotherapy study intervention.
Herbal and natural remedies that may have immune-modulating effects and interfere with interpretation of study results	Must not be given concomitantly unless agreed by the sponsor.
Foods or medications that are strong inducers or inhibitors of CYP3A4/5, which are deemed to impact the pharmacokinetics of study medications.	Must not be given concomitantly unless agreed by the sponsor (See Appendix I 1).
For Group A (ceralasertib plus durvalumab combination treatment) only	
Immunosuppressive medications including, but not limited to, systemic corticosteroids at doses exceeding 10 mg/day of prednisone or equivalent, methotrexate, azathioprine, and TNF- α blockers	Must not be given concomitantly, or used for premedication prior to the IO infusions. The following are allowed exceptions: • Use of immunosuppressive medications for the management of study intervention-related AEs. • Use in participants with contrast allergies. • In addition, use of inhaled, topical, and intranasal corticosteroids is permitted. A temporary period of steroids will be allowed if clinically indicated and considered to be essential for the management

Table 20 Prohibited Medications/Therapies

Prohibited medication/class of drug/therapy	Usage
	of non-immunotherapy related events experienced by the participant (eg, chronic obstructive pulmonary disease, radiation, nausea, etc). Consult with study clinical lead for prolonged therapy or need for a slow taper.
EGFR TKIs	Must not be given concomitantly. Must be used with caution in the 90 days post last dose of durvalumab. Increased incidences of pneumonitis (with third generation EGFR TKIs) and increased incidence of transaminase increases (with first generation EGFR TKIs) has been reported when durvalumab has been given concomitantly.

AE, adverse event; CTLA-4, cytotoxic T-lymphocyte-associated antigen-4; CYP3A4/5, cytochrome P450 3A4/5; EGFR, epidermal growth factor receptor; IO, immune-oncology; PD-1, programmed cell death protein-1; PD-L1, programmed death-ligand 1; TKI, tyrosine kinase inhibitor.

Table 21 Supportive Medications/Therapies

Supportive medication/class of drug/therapy	Usage
Concomitant medications or treatments (eg, acetaminophen or diphenhydramine) deemed necessary to provide adequate AE management, except for those medications identified as “prohibited,” as listed above	To be administered as prescribed by the investigator except for those medications identified as “prohibited,” as listed in Table 20 .
Best supportive care (including antibiotics, nutritional support, correction of metabolic disorders, optimal symptom control, and pain management [including palliative radiotherapy, etc]) except for those medications identified as “prohibited,” as listed above	Should be used, when necessary, for all participants except for those medications identified as “prohibited,” as listed in Table 20 .
Vaccines limited only to non-live attenuated preparations (eg, influenza vaccine)	Permitted.
Required for management of other medical conditions	As required except for those identified as “prohibited,” as listed in Table 20 .

Appendix J Changes Related to Mitigation of Study Disruptions Due to Cases of Civil Crisis, Natural Disaster, or Public Health Crisis, including COVID-19 Outbreak

Note: Changes below should be implemented only during study disruptions due to any of or a combination of civil crisis, natural disaster, or public health crisis (eg, during quarantines and resulting site closures, regional travel restrictions and considerations if site personnel or study participants become infected with SARS-CoV-2 or similar pandemic infection) during which participants may not wish to or may be unable to visit the study site for study visits. These changes should only be implemented if allowable by local/regional guidelines and following agreement from the sponsor.

J 1 Reconsent of Study Participants During Study Interruptions

During study interruptions, it may not be possible for the participants to complete study visits and assessments on site and alternative means for carrying out the visits and assessments may be necessary, eg, remote visits. Reconsent should be obtained for the alternative means of carrying out visits and assessments and should be obtained prior to performing the procedures described in Sections J 2 to J 5. Local and regional regulations and/or guidelines regarding reconsent of study participants should be checked and followed. Reconsent may be verbal if allowed by local and regional guidelines (note, in the case of verbal reconsent the ICF should be signed at the participant's next contact with the study site). Visiting the study sites for the sole purpose of obtaining reconsent should be avoided.

J 2 Rescreening of Participants to Reconfirm Study Eligibility

Additional rescreening for screen failure due to study disruption can be performed in previously screened participants. The investigator should confirm this with the designated study clinical lead.

In addition, during study disruption there may be a delay between confirming eligibility of a participants and either enrolment into the study or commencing of dosing with study intervention. If this delay is outside the screening window specified in SoAs the participant will need to be rescreened to reconfirm eligibility before commencing study procedures. This will provide another opportunity to rescreen a participant in addition to that detailed in SoAs. The procedures detailed in SoAs must be undertaken to confirm eligibility using the same randomisation number as for the participant.

J 3 Home or Remote Visit to Replace On-site Visit (Where Applicable)

A qualified HCP from the study site or TPV service may visit the participant's home/or other remote location as per local SOPs, as applicable. Supplies will be provided for a safe and

efficient visit. The qualified HCP will be expected to collect information per the CSP.

J 4 Telemedicine Visit to Replace On-site Visit (Where Applicable)

In this appendix the term telemedicine visit refers to remote contact with the participants using telecommunications technology including phone calls, virtual or video visits, and mobile health devices.

During a civil crisis, natural disaster, or public health crisis, on-site visits may be replaced by a telemedicine visit if allowed by local/regional guidelines. Having a telemedicine contact with the participants will allow AEs, concomitant medication to be reported and documented.

J 5 Data Capture During Telemedicine or Home/Remote Visits

Data collected during telemedicine or home/remote visits will be captured by the qualified HCP from the study site or TPV service in the source documents, or by the participant themselves.

J 6 COVID-19 Risk Assessment

The safety of participants is of primary importance. Any potential risks of participating in the study, particularly with the added challenges due to COVID-19 outbreak, should be weighed against the anticipated benefit (see also principle 2.2 of ICH GCP). Investigators are advised to use clinical judgement in determining infection prevention precautions for study participants.

The emergence of SARS-CoV-2 presents a potential safety risk for cancer patients. Participants enrolling in this study may require more frequent visits to the site for study treatment administration and for study assessments compared to patients receiving standard of care. Therefore, several risk mitigation factors have been implemented related to study conduct during the COVID-19 outbreak, for patient management in an event of COVID-19, and actions to be taken on study treatment (see Section J 9). With these measures in place, it is considered that the anticipated potential benefits for the participants enrolled in this study outweigh the potential risks. All implemented measures prioritise trial participant safety and data validity; in case these two conflict with each other, trial participant safety should always prevail (see also European Medicines Agency Guidance on the management of clinical trials during the COVID-19 [coronavirus] pandemic [EMA 2020]).

Notably, participants with active COVID-19 infection confirmed by local laboratory testing will not be eligible for study enrolment (see CSP Section 5.2, Exclusion Criterion 1).

J 7 Potential Risks During COVID-19

Every effort should be made to follow the CSP. Section J 10 provides a dose modification and

management plan for participants with confirmed or suspected COVID-19 who are being treated with study intervention. The risk-benefit assessment should be carefully considered for each participant enrolling in the study based on the known safety risks related to COVID-19, individual needs, and local guidelines and restrictions. Investigators must continue to use their best clinical judgement in determining the most optimal care for participants and utmost diligence in determining their eligibility for study participation, continued study treatment, and overall assessment of benefit/risk of study treatment or participation.

The sponsor must be promptly notified of a site's inability to perform study activities due to COVID-19 outbreak in order to minimise any potential risks.

J 8 New Participant Enrolment

Study sites may continue to recruit new participants into the study provided the following activities to preserve study integrity can be met:

- Upon discussion with the site monitor, the study site has confirmed the ability to enrol and manage new participants effectively and in compliance with the protocol.
- Data will continue to be entered into the eCRF and queries resolved in a timely manner.

Per CSP Exclusion Criterion 1 (see CSP Section [5.2](#)), participants with uncontrolled intercurrent illness, including but not limited to, ongoing or active infection are not eligible for the study participation and hence such participants (including those who have confirmed COVID 19) should not be included for study participation.

J 9 Study Treatment Administration

If an AE or SAE is associated with COVID-19, the investigator should determine whether the participants' treatment with investigational product should continue, be interrupted, or be discontinued in accordance with the CSP.

AEs, SAEs, cycle delays and/or treatment suspensions associated with COVID-19 along with logistical issues should be reported according to the eCRF Completion Guidelines.

For dosing discontinuations, where applicable, the dosing discontinuation guidelines should be followed, and the End-of-Treatment Form(s) completed

J 10 Ongoing Participants

Participants receiving study intervention should continue to undergo safety assessments prior to dosing in accordance with the CSP. In case it is not feasible to perform safety assessments, study intervention should be interrupted until such assessments can be completed.

J 10.1 If a Participant has an Event Suspected to be COVID-19

Delay or omit study intervention as appropriate and test for COVID-19 per local health authority or institutional guidance.

- Signs and symptoms of COVID-19 include but are not limited to new onset of fever, new or worsening cough, shortness of breath, difficulty breathing and sometimes abnormal chest imaging and may be similar to those of an imAE.
- In accordance with the CSP and the TMGs for imAEs, thorough evaluation should be performed to accurately identify the underlying pathology in case an AE is encountered for a participant.
- If COVID-19 is ruled out, study intervention may be resumed per the CSP.
- If COVID-19 is **confirmed or diagnosis still suspected after evaluation**, manage COVID-19 per local guidance until full recovery.

J 10.2 Participants with Confirmed COVID-19

Participants with confirmed COVID-19 (by local laboratory testing and/or combination of key symptoms) should have study intervention withheld and COVID-19 managed per local guidance.

In case of confirmed COVID-19 and a simultaneous imAE requiring treatment, investigators are advised to apply clinical judgement regarding the use of corticosteroids as per the durvalumab/tremelimumab TMGs. This also includes the consideration of alternate immunosuppressive agents other than corticosteroids for imAE management, depending on the individual participant's presentation ([Curigliano et al 2020](#)).

J 10.3 Restarting Study Intervention

Study intervention must not be resumed until recovery from COVID-19 (eg, confirmed by imaging, laboratory testing and/or absence of symptoms) and COVID-19-specific treatment has been completed per local guidance.

The study clinical lead should be contacted if any additional guidance or clarification is needed.

J 10.4 Vaccination Against COVID-19

Protocol restrictions applying to live attenuated vaccines are relevant for live attenuated COVID-19 vaccines as well. Investigators should apply their discretion assessing the risk-benefit of other types of COVID-19 vaccines for participants in clinical trials. Ideally, administration of the vaccine should be done on a different day other than the day of study intervention administration to differentiate any potential AEs seen from the vaccine and study

intervention. The administration of the vaccine and any potential AEs associated with the vaccine are to be documented on the concomitant medication and AE eCRFs, respectively.

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Appendix K Abbreviations

Abbreviation or special term	Explanation
ADA	antidrug antibody
AE	adverse event
AESI	adverse event of special interest
ALK	anaplastic lymphoma kinase
ALP	alkaline phosphatase
ALT	alanine aminotransferase
anti-HBc	hepatitis B virus core antibody
AST	aspartate aminotransferase
ATM	ataxia telangiectasia mutated
ATR	ataxia telangiectasia and Rad3-related protein
BCRP	breast cancer resistance protein
BID	twice a day
BRCA	breast cancer susceptibility gene
BSA	body surface area
C#D#	Cycle number, Day number
CD	cluster of differentiation
CI	confidence interval
C _{max}	maximum observed concentration
CMC	Chemistry, Manufacturing and Controls
COA	clinical outcome assessment
CR	complete response
CrCL	creatinine clearance
CRF	case report form
CRO	Contract Research Organisation
CSP	Clinical Study Protocol
CSR	Clinical Study Report
CT	computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
ctDNA	circulating tumour DNA
CTLA-4	cytotoxic T-lymphocyte-associated antigen-4
CYP	cytochrome P450
DCO	data cut-off
DCR	disease control rate
DDR	DNA Damage Response

Abbreviation or special term	Explanation
DNA	deoxyribonucleic acid
DoR	duration of response
DSB	double strand break
DSUR	Development Safety Update Report
EC	Ethics Committee, synonymous to Institutional Review Board (IRB) and Independent Ethics Committee (IEC)
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	electronic case report form
EDC	electronic data capture
EGFR	epidermal growth factor receptor
EORTC	European Organisation for Research and Treatment of Cancer
EORTC IL	Modified EORTC Item Library
EORTC QLQ-C30	EORTC 30-item core quality of life questionnaire
ePRO	electronic patient-reported outcome
EQ-5D-5L	EuroQoL 5-dimension, 5-level health state utility index
EU	European Union
FAS	full analysis set
FDA	Food and Drug Administration
FFPE	formalin-fixed and paraffin-embedded
GCP	Good Clinical Practice
G-CSF	granulocyte-colony stimulating factor
GMP	Good Manufacturing Practice
HBsAg	hepatitis B virus surface antigen
HBV	hepatitis B virus
HCP	health care professional
HCV	hepatitis C virus
HIV	human immunodeficiency virus
HL	Hy's Law
HLA	Human Leukocyte Antigen
HOSPAD	Hospital Admission (form)
HR	hazard ratio
IATA	International Airline Transportation Association
IB	Investigator's Brochure
ICF	informed consent form

Abbreviation or special term	Explanation
ICH	International Council for Harmonisation
iCRO	imaging Contract Research Organisation
IDMC	Independent Data Monitoring Committee
IEC	Independent Ethics Committee
IFN	interferon
IFN- γ	interferon-gamma
Ig	immunoglobulin
IHC	immunohistochemistry
ILD	interstitial lung disease
imAE	immune-mediated AE
IND	Investigational New Drug
INR	international normalised ratio
International co-ordinating investigator	If a study is conducted in several countries the international co-ordinating investigator is the investigator co-ordinating the investigators and/or activities internationally
IO	immune-oncology
IRB	Institutional Review Board
IRT	interactive response technology
IV	intravenous
IVD	in vitro diagnostic
IXRS	interactive voice/web response system
LFT	liver function tests
LVEF	left ventricular ejection fraction
mAb	monoclonal antibody
MedDRA	Medical Dictionary for Regulatory Activities
MRI	magnetic resonance imaging
MTP	multiple testing procedure
NCI	National Cancer Institutes
NE	not evaluable
NIMP	non-investigational medicinal product
NK	natural killer
NL	new lesion
NSCLC	non-small cell lung cancer
NSCLC-SAQ	Non-Small Cell Lung Cancer Symptom Assessment Questionnaire
NTL	non-target lesion
OD	once a day

Abbreviation or special term	Explanation
OR	Odds ratio
ORR	objective response rate
OS	overall survival
OS12	overall survival at 12 months
PD	progression of disease
PD-1	programmed cell death protein-1
PD-L1	programmed death-ligand 1
PD-L2	programmed death-ligand 2
PET	positron emission tomography
PFS	progression-free survival
PFS2	time to second progression or death
PGI-C	Patient Global Impression of Change
PGI-C	Patient's Global Impression of Change
PGI-S	Patient Global Impression of Severity
PGI-S	Patient's Global Impression of Severity
PGI-TT	Patient Global Impression of Treatment Tolerability
PGI-TT	Patient's Global Impression of Treatment Tolerability
P-gp	P glycoprotein
PHL	potential Hy's Law
PIKK	phosphatidylinositol 3-kinase-related kinase
PK	pharmacokinetic
PR	partial response
PRO	patient-reported outcome
PSSR	Project Specific Safety Requirements
PTT	Partial thromboplastin time
QLQ-C30	30-item core quality of life questionnaire
QoL	quality of life
q2w	every 2 weeks
qXd	every X days
RECIST 1.1	Response Evaluation Criteria in Solid Tumours, Version 1.1
RNA	ribonucleic acid
RP2D	recommended phase 2 dose
RoW	Rest of the World
SAE	serious adverse event
SAP	statistical analysis plan
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2

Abbreviation or special term	Explanation
SD	stable disease
SmPC	Summary of Product Characteristics
SoA	Schedule of Activities
SoC	standard of care
SOC	System Organ Class
SOP	Standard Operating Procedure
SpO ₂	Saturation of peripheral oxygen
SUSAR	suspected unexpected serious adverse reaction
TBL	Total bilirubin
TCR	T-cell receptor
TdP	Torsades de Pointes
TEAE	treatment-emergent adverse event
TL	target lesion
TMG	Toxicity Management Guideline
TNF α	Tumour Necrosis Factor α
TPV	third-party vendor
TSH	thyroid-stimulating hormone
TTD	Time to deterioration
TTR	time to response
UK	United Kingdom
ULN	upper limit of normal
US	United States
VAS	visual analogue scale
w/v	weight/volume
WHO	World Health Organisation
WOBCP	women of childbearing potential

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